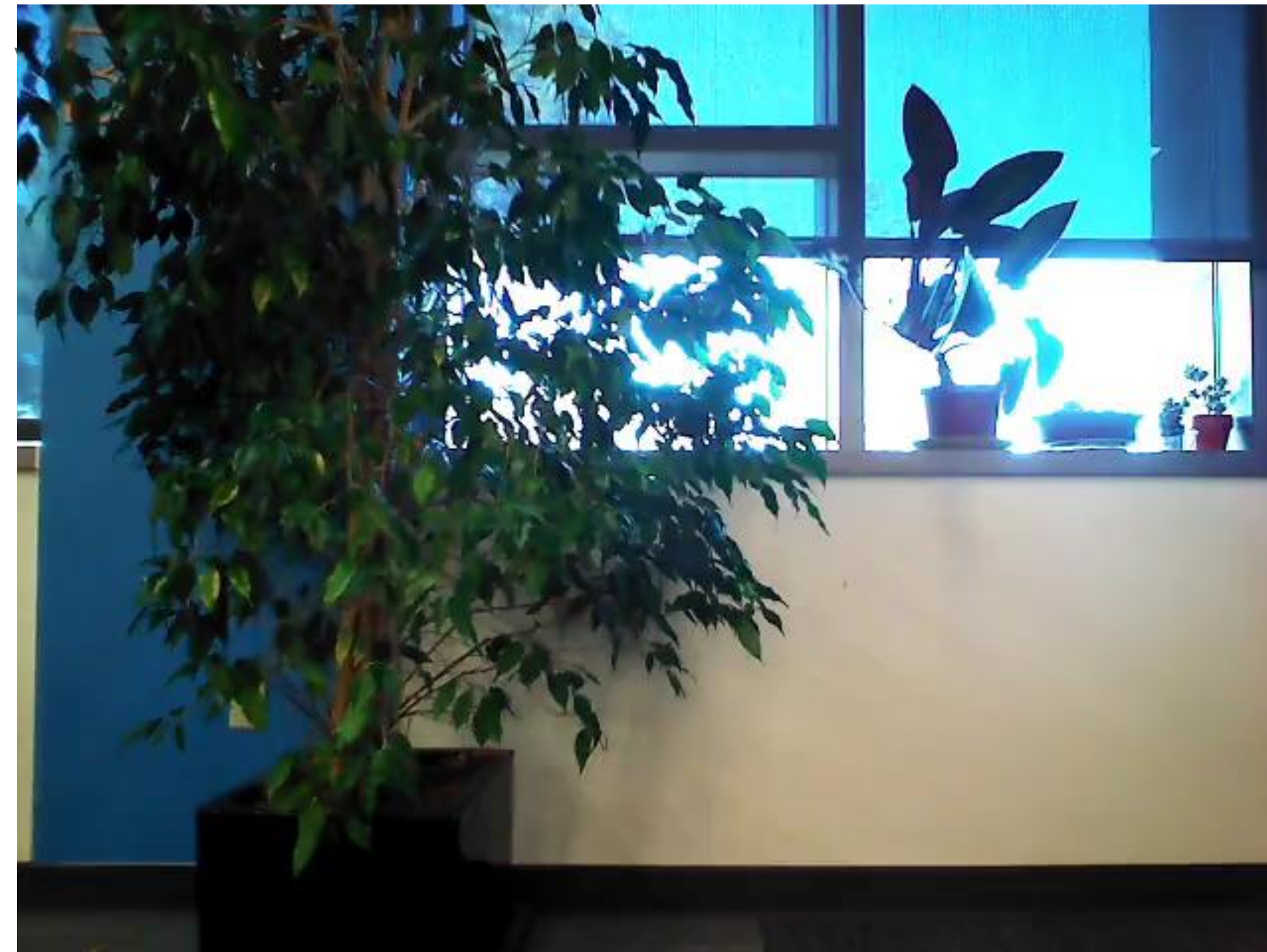
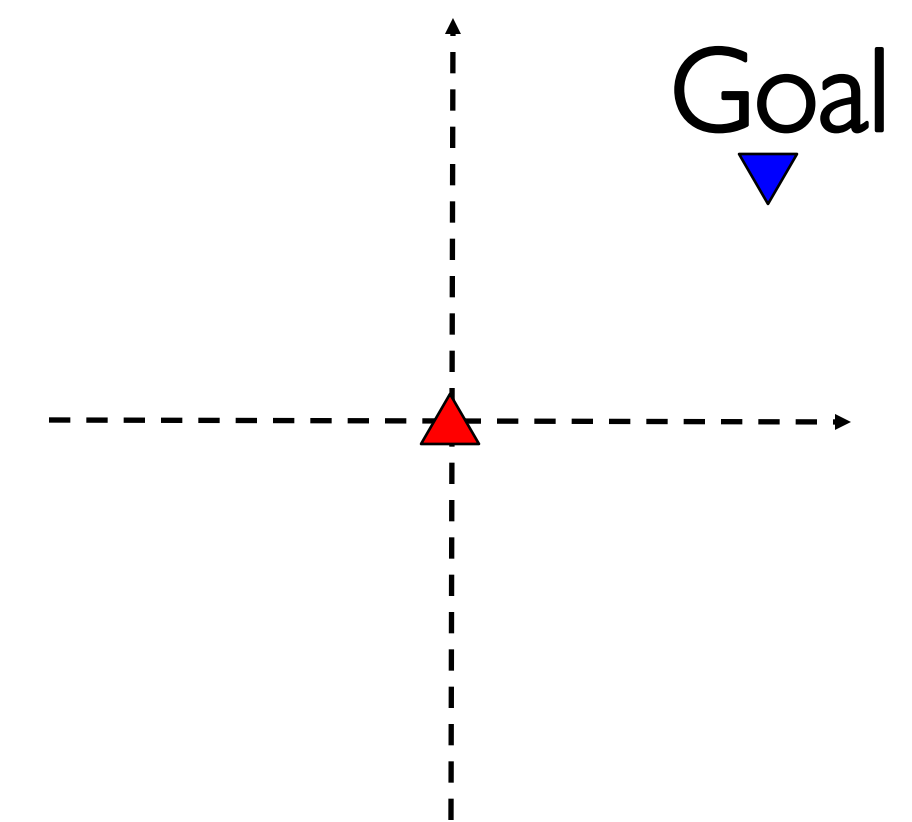




Robot with a first
person camera



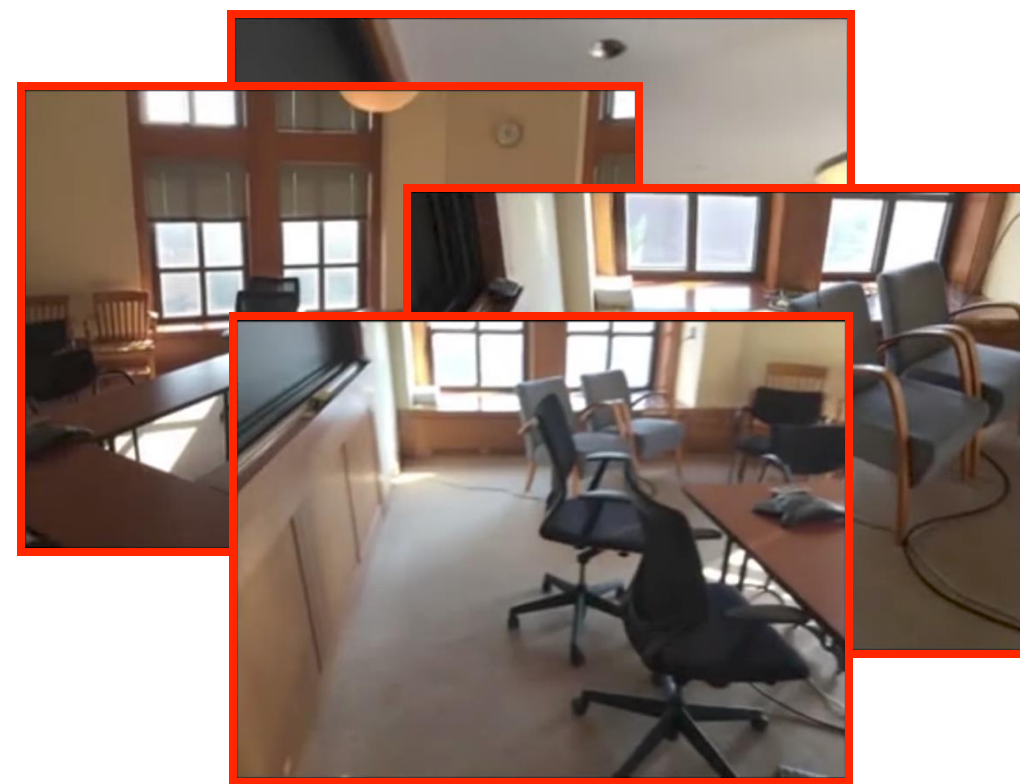
Dropped into a novel
environment



**“Go
300 feet North,
400 feet East”**

**“Go Find a
Chair”**

Navigate around



Observed Images

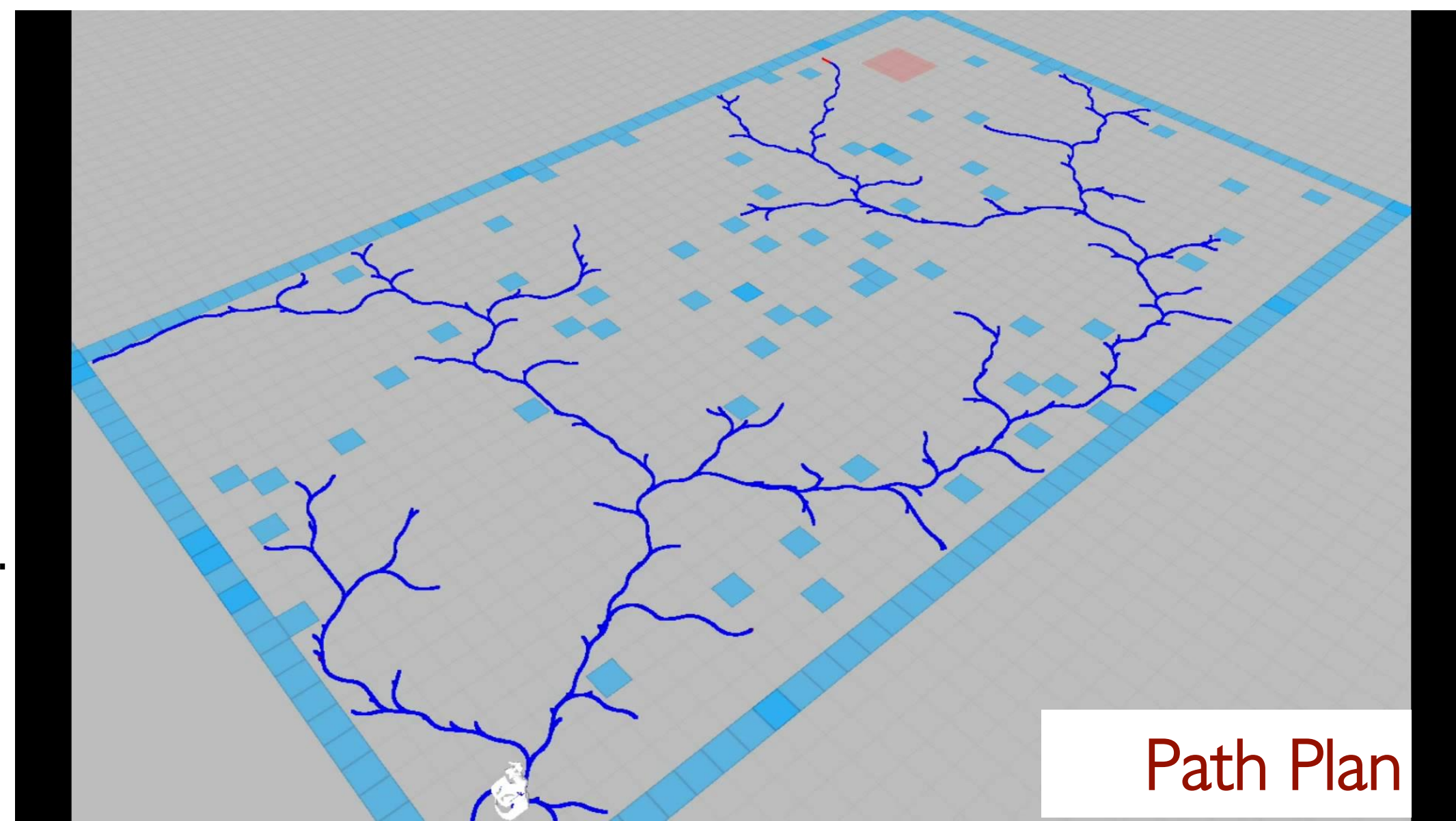
Mapping

Planning

Classical Solution



Geometric Reconstruction

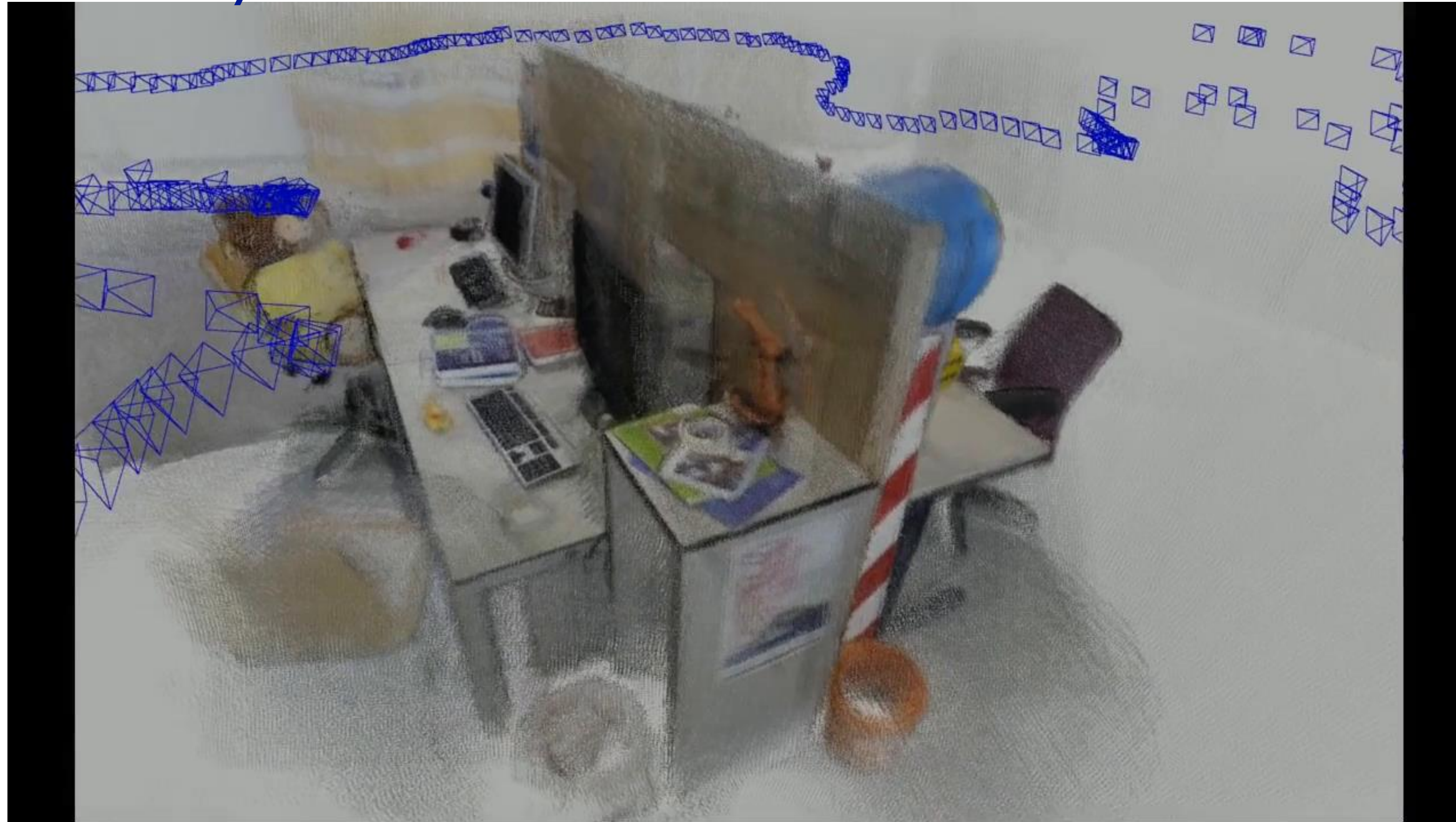


Path Plan

- Hartley and Zisserman. 2000.** Multiple View Geometry in Computer Vision
- Thrun, Burgard, Fox. 2005.** Probabilistic Robotics
- Canny. 1988.** The complexity of robot motion planning.
- Kavraki et al. RA1996.** Probabilistic roadmaps for path planning in high-dimensional configuration spaces.
- Lavalle and Kuffner. 2000.** Rapidly-exploring random trees: Progress and prospects.

Video Credits: Mur-Artal et al., Palmieri et al.

Unnecessary



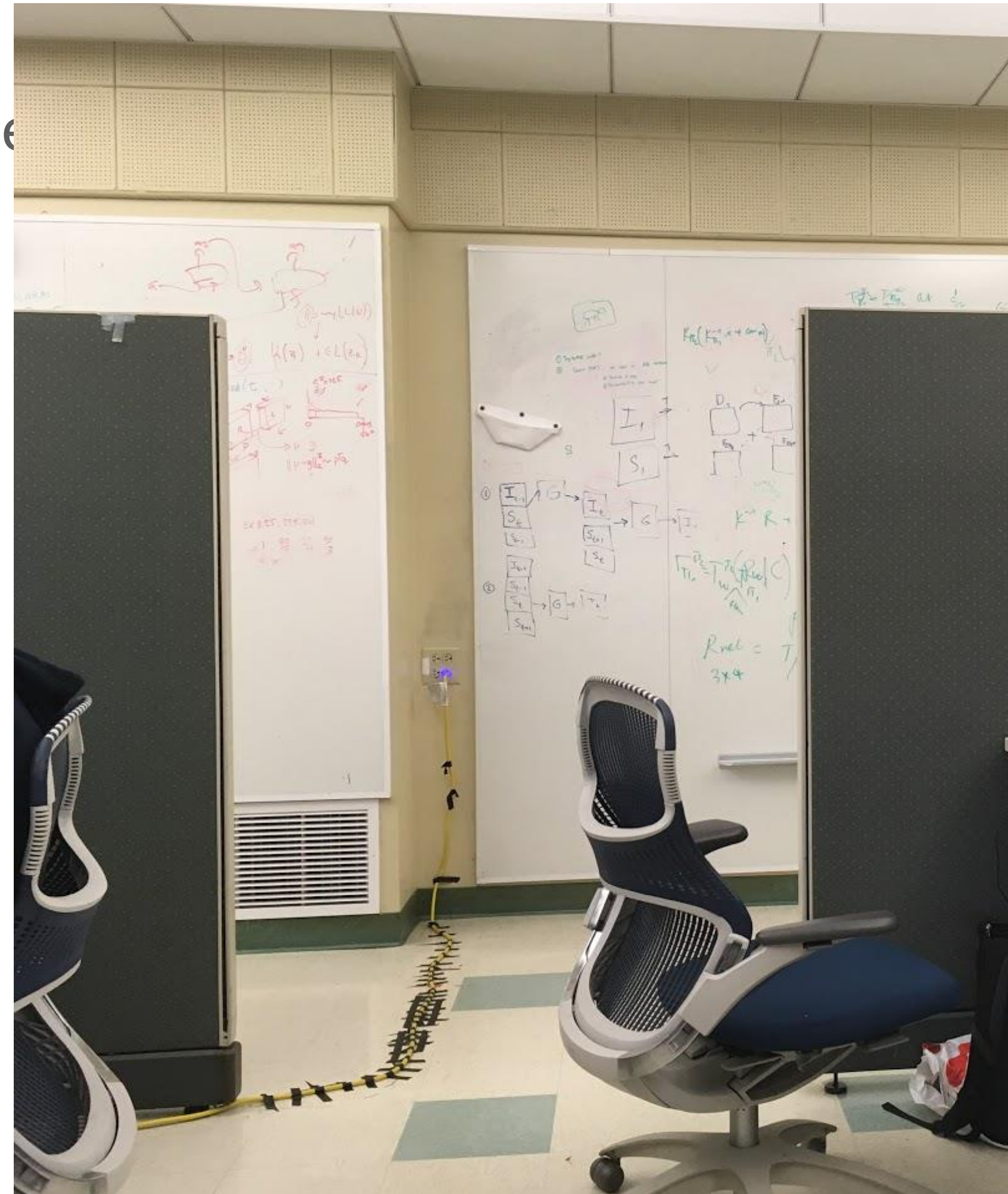
Do we need to tediously reconstruct everything on this table?

Video Credit: Mur-Artal and Tardos, TRobotics 2016. ORB-SLAM2: an Open-Source SLAM System for Monocular, Stereo and RGB-D Cameras.

Insufficient

Ge

World



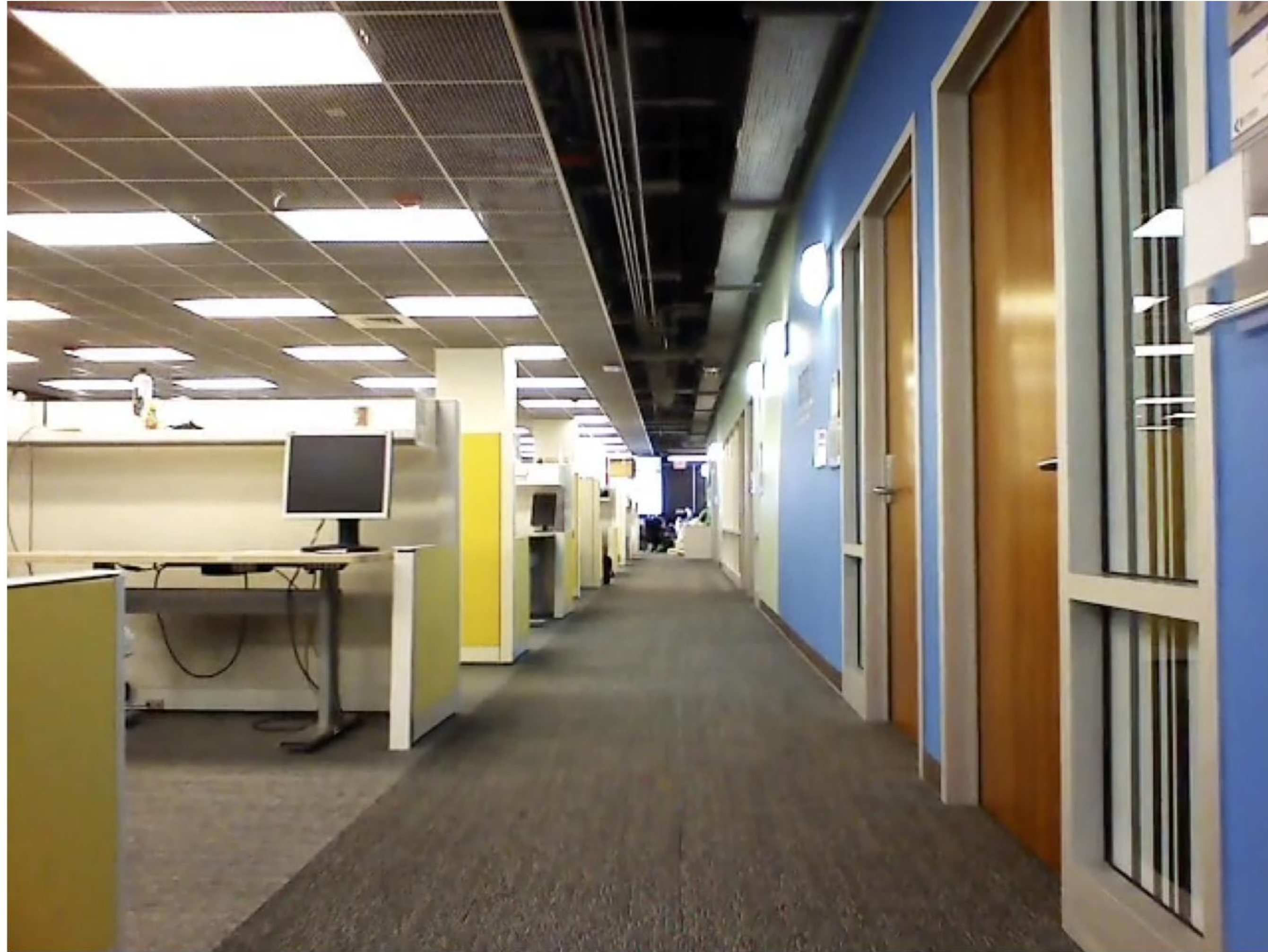
Can't speculate about space not directly observed.

Insufficient



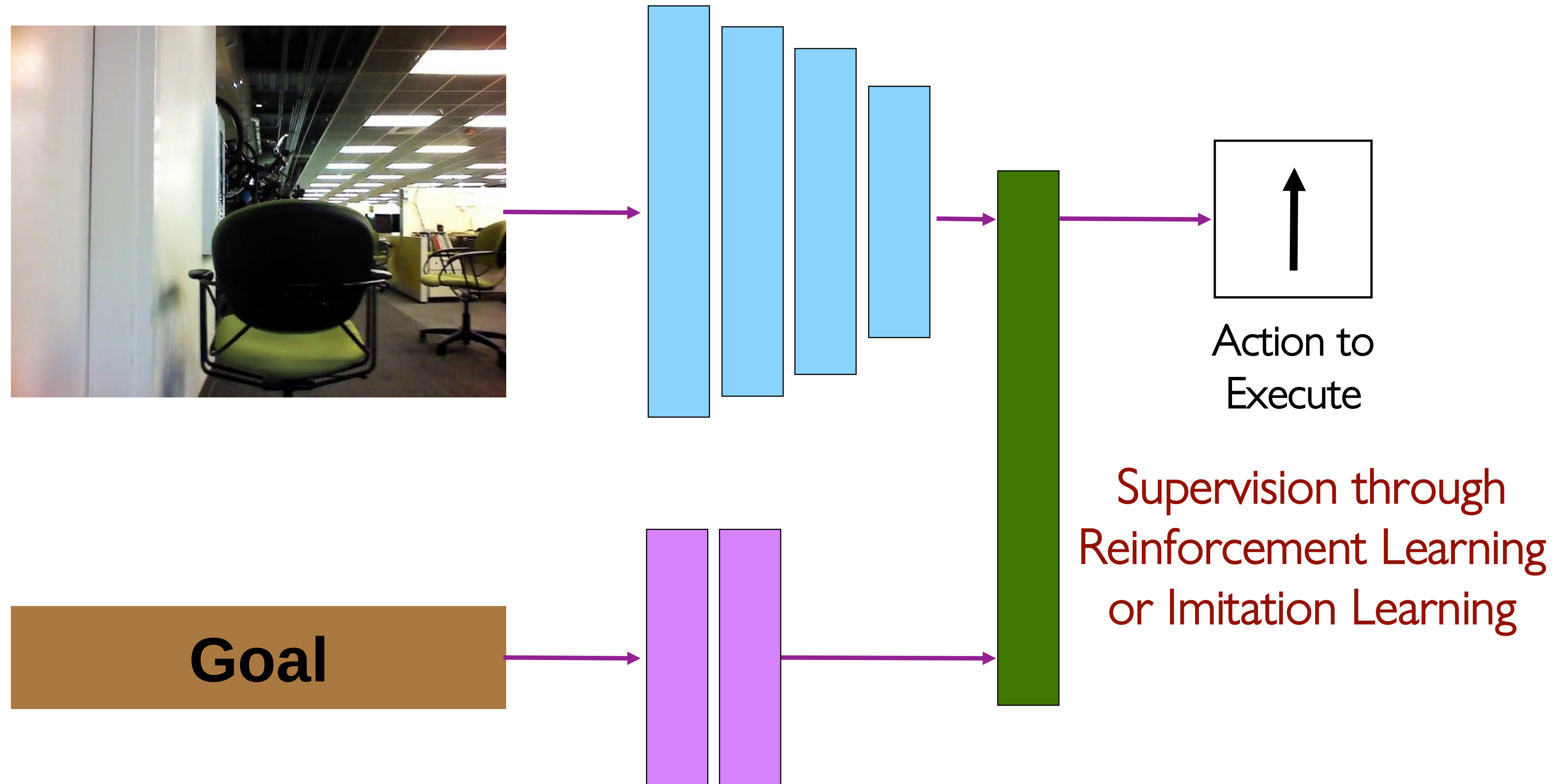
Can't exploit patterns in layout of indoor spaces.

Insufficient



Ignore navigational affordances.

Modern End-to-End Learned Navigation

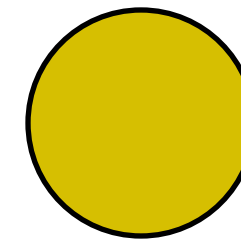
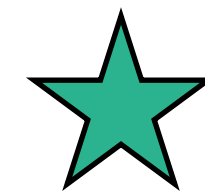
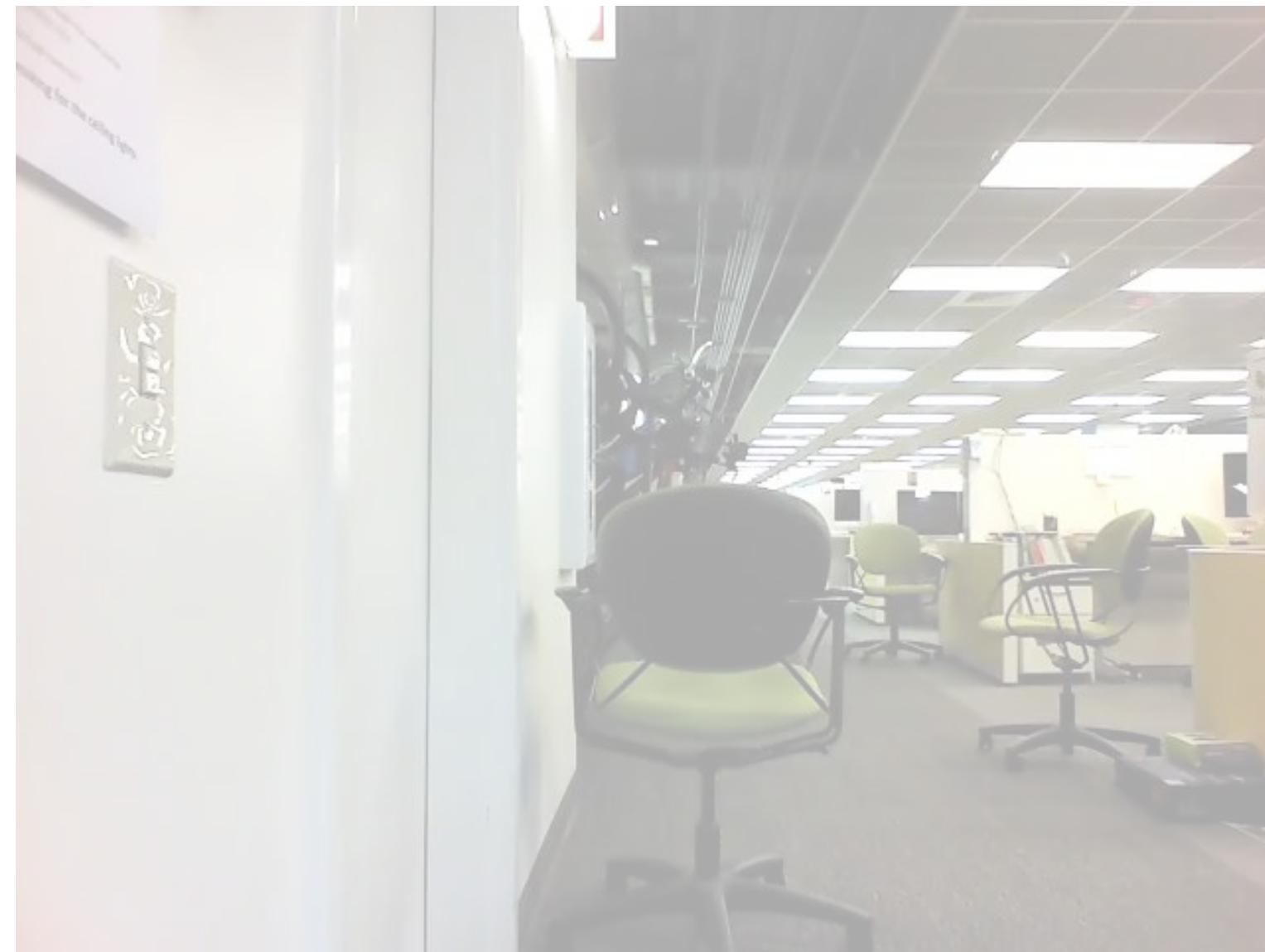


Mnih et al., Nature 2014. Human-level control through deep reinforcement learning.

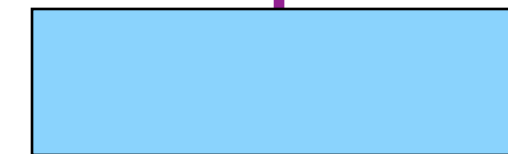
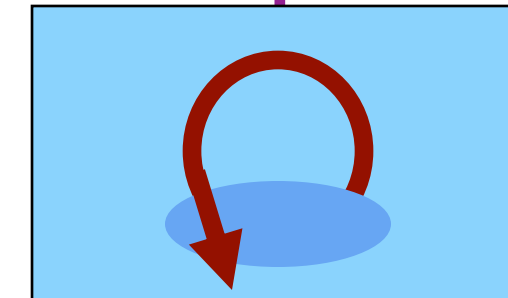
Levine et al., JMLR 2015. End-to-End Training of Deep Visuomotor Policies.

Zhu et al., ICRA 2017. Target-driven Visual Navigation in Indoor Scenes using Deep Reinforcement Learning.

Simple networks are not enough



Action to
Execute



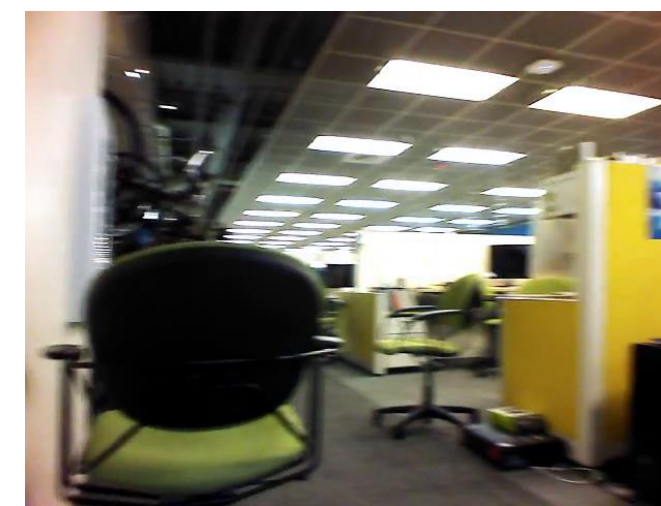
$t-4$

$t-3$

$t-2$

$t-1$

t



Mnih et al., Nature 2014. Human-level control through deep reinforcement learning.

Levine et al., JMLR 2015. End-to-End Training of Deep Visuomotor Policies.

Zhu et al., ICRA 2017. Target-driven Visual Navigation in Indoor Scenes using Deep Reinforcement Learning.

Navigating to Objects in the Real World



Theophile Gervet ¹



Soumith Chintala ⁴



Dhruv Batra ^{3,4}



Jitendra Malik ^{2,4}



Devendra Chaplot ⁴

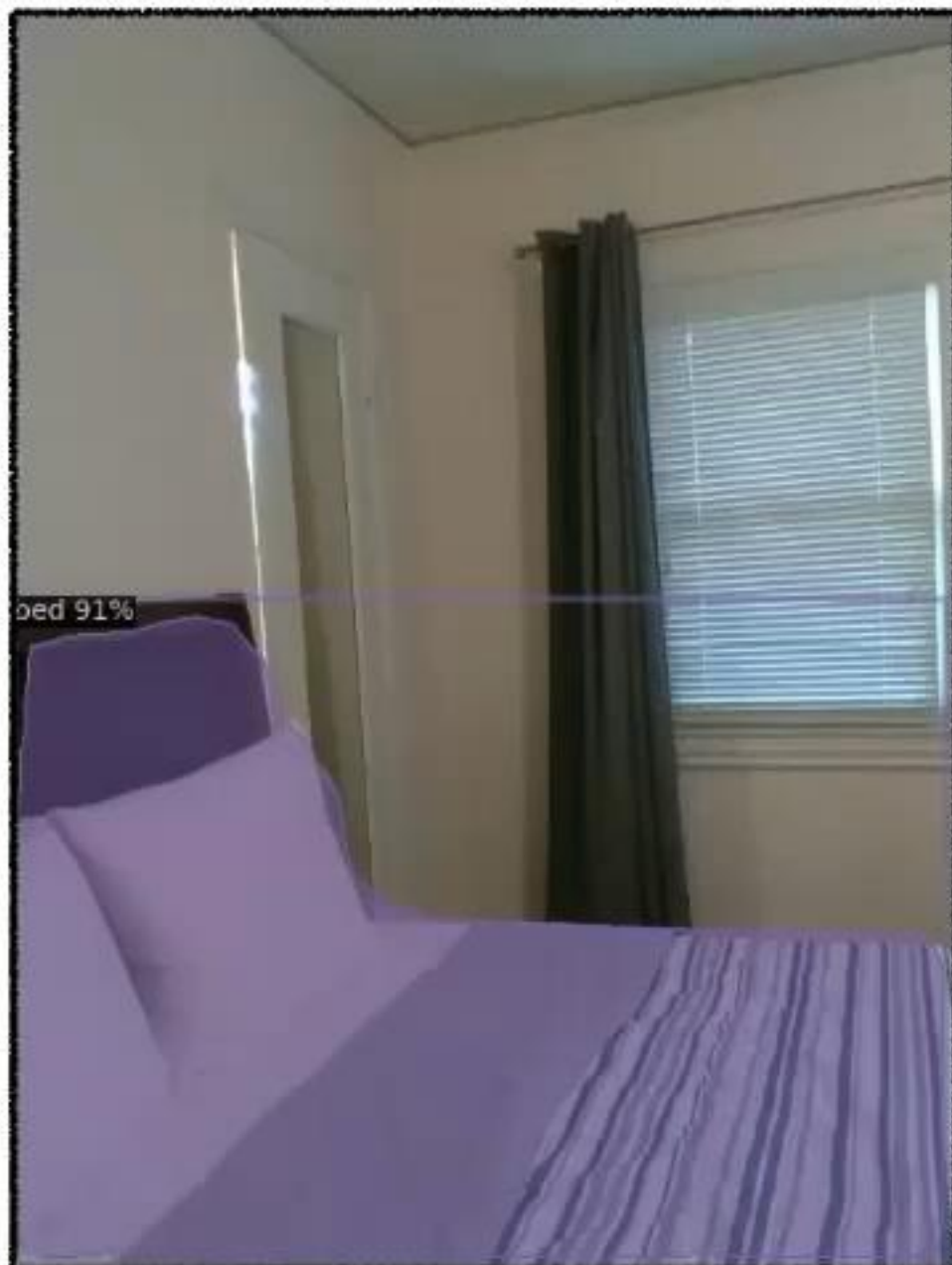
¹ **Carnegie
Mellon
University**

²  **Berkeley**
UNIVERSITY OF CALIFORNIA

³ **Georgia
Tech** 

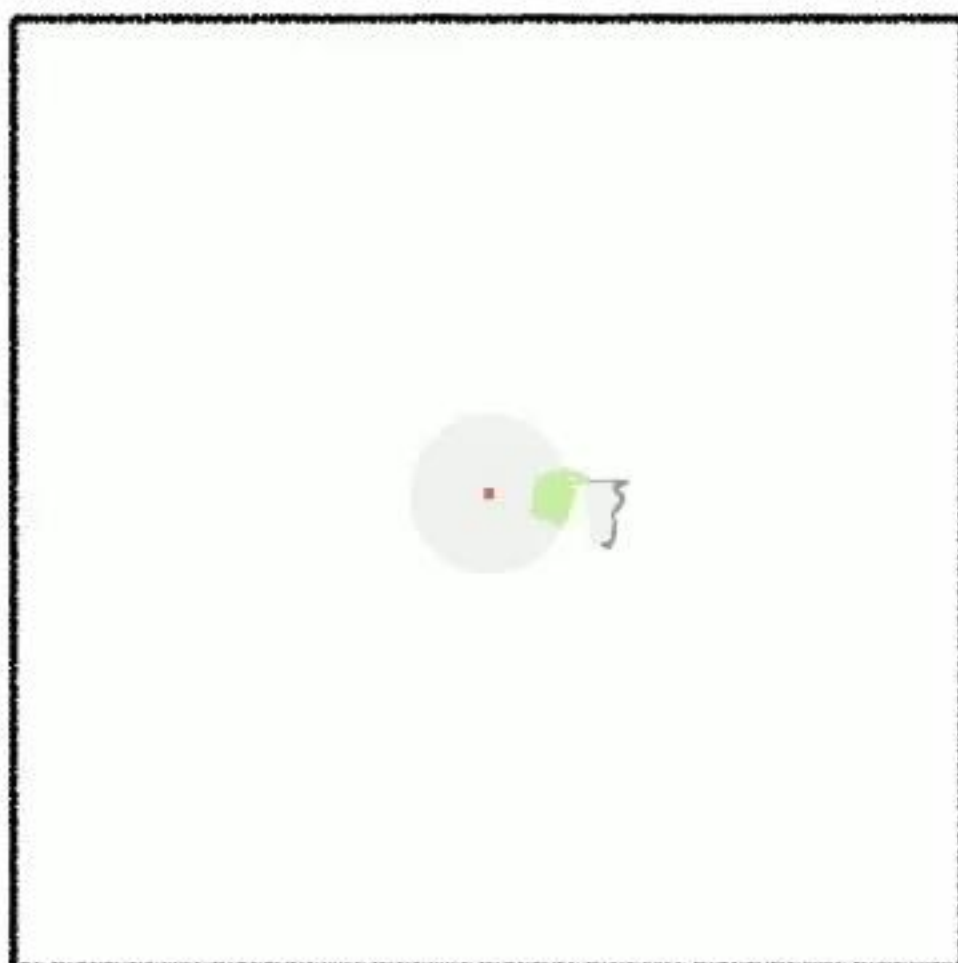
⁴  **Meta AI**

Observation



Goal: *plant*

Predicted Semantic Map



Third-person view



0: Navigable Area

1: chair

2: couch

3: potted plant

4: bed

5: toilet

6: tv

7: dining-table

8: oven

9: sink

10: refrigerator

11: book

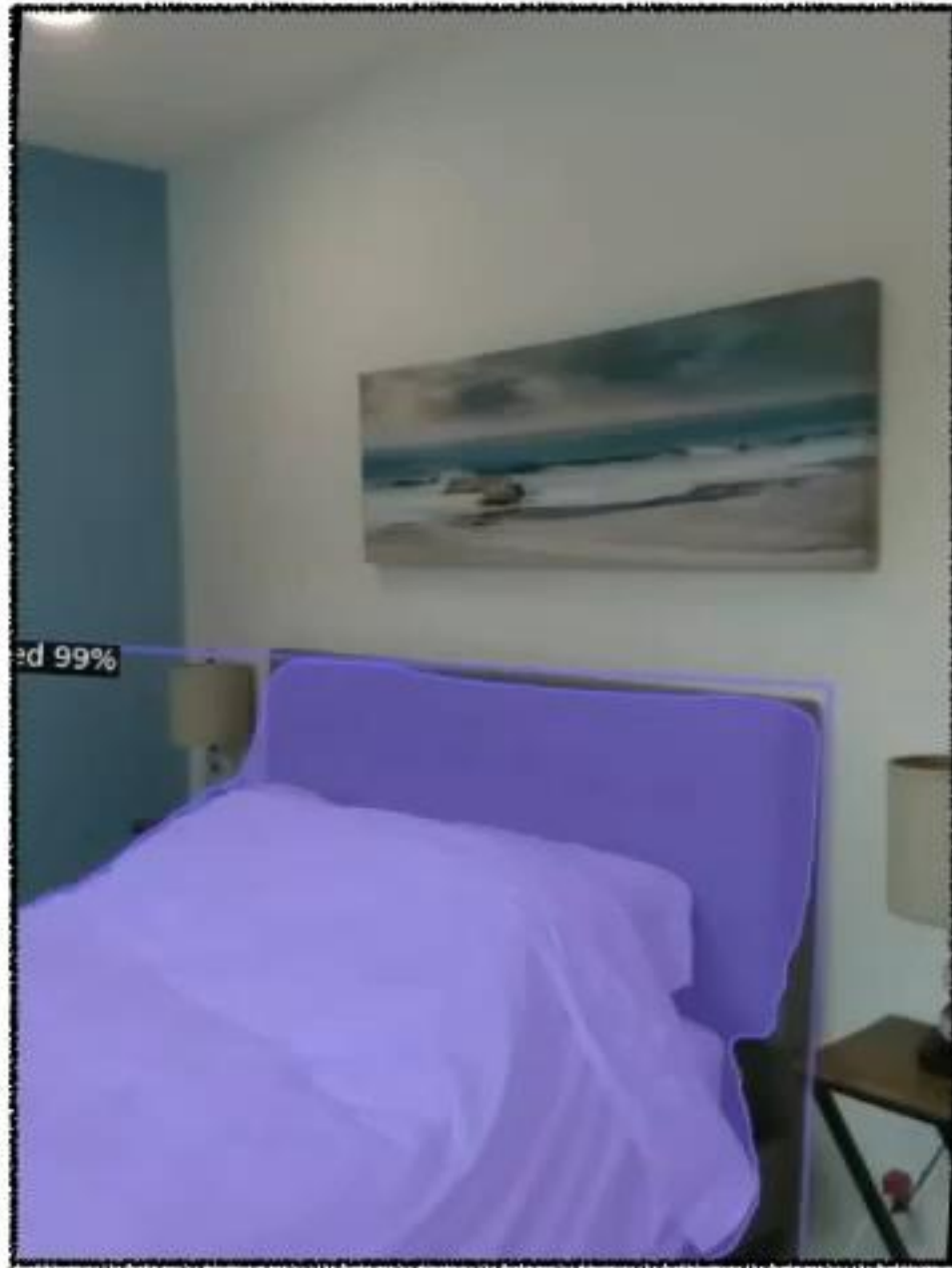
12: clock

13: vase

14: cup

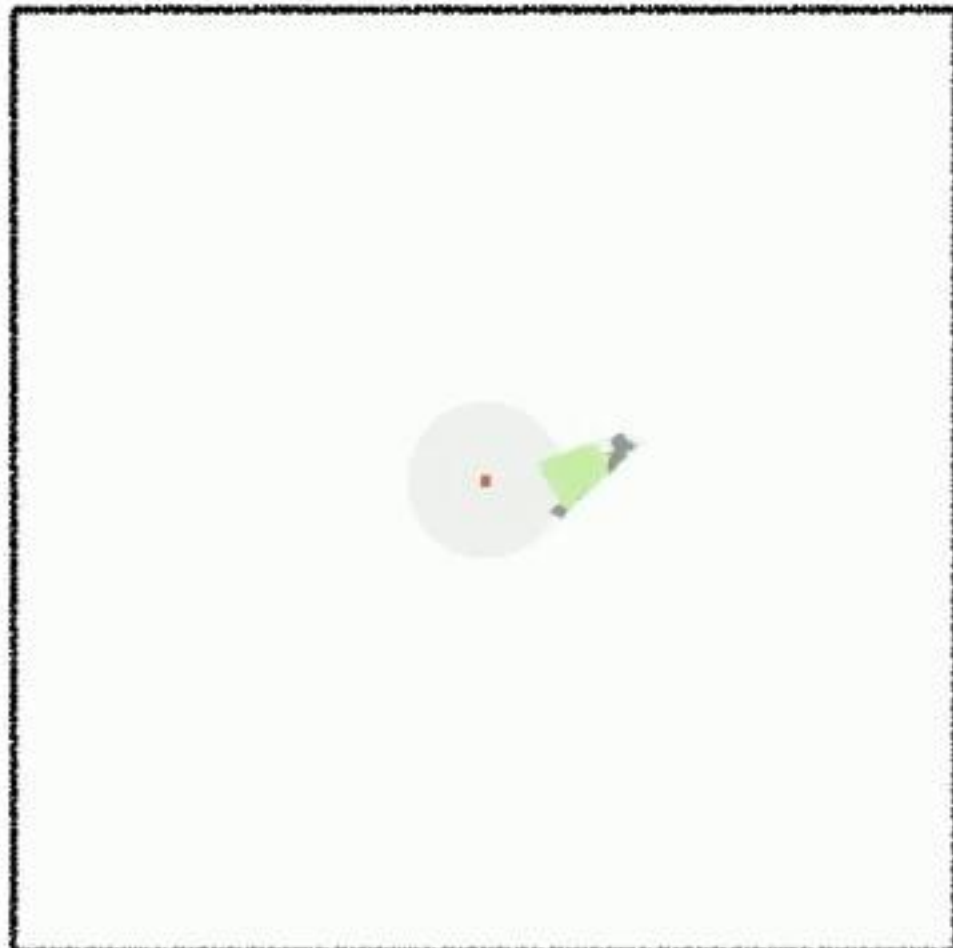
15: bottle

Observation



Goal: *chair*

Predicted Semantic Map



Third-person view



0: Navigable Area

1: chair

2: couch

3: potted plant

4: bed

5: toilet

6: tv

7: dining-table

8: oven

9: sink

10: refrigerator

11: book

12: clock

13: vase

14: cup

15: bottle

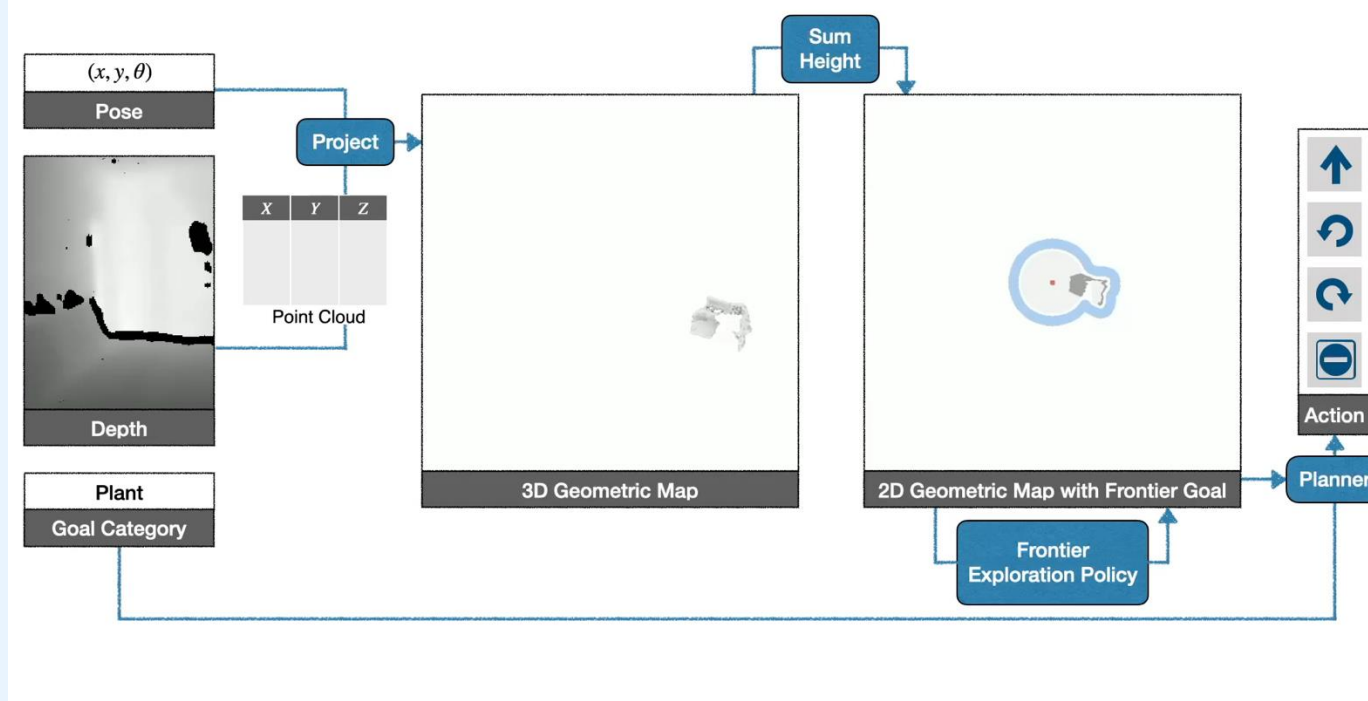


Empirical Evaluation
3 Approaches
6 Unseen Homes
6 Goal Object Categories



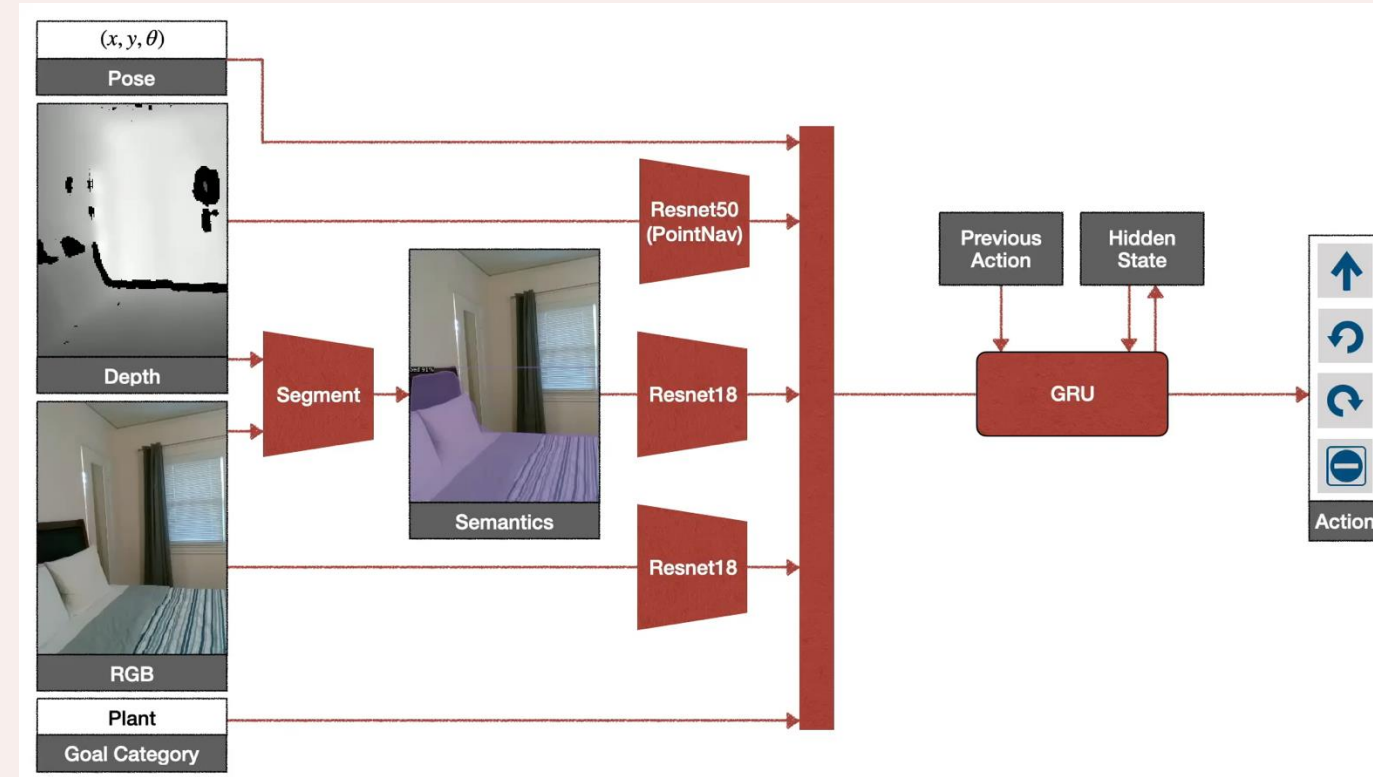
Methods

Classical



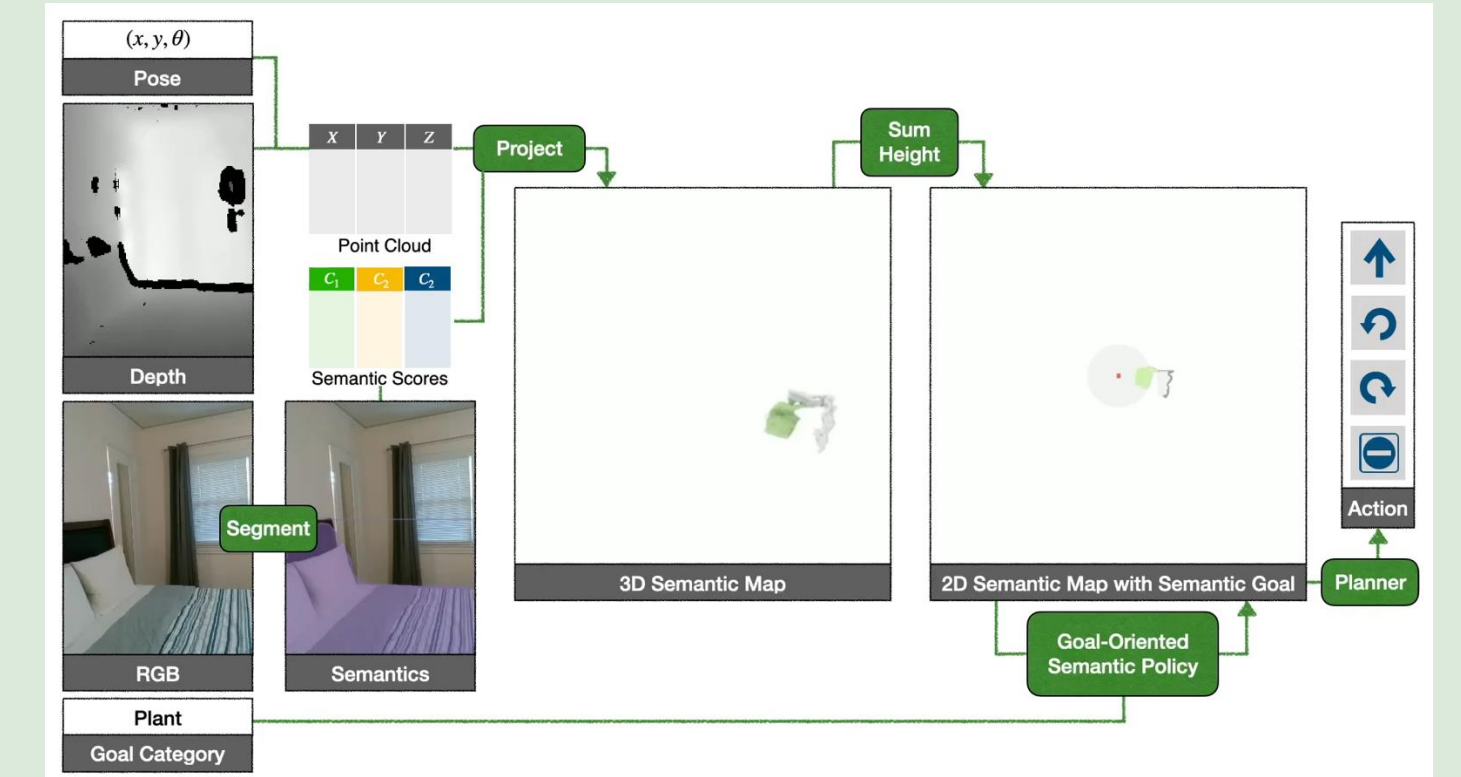
- Geometric Map
- Heuristic Exploration
 - No Training

End-to-end Learning



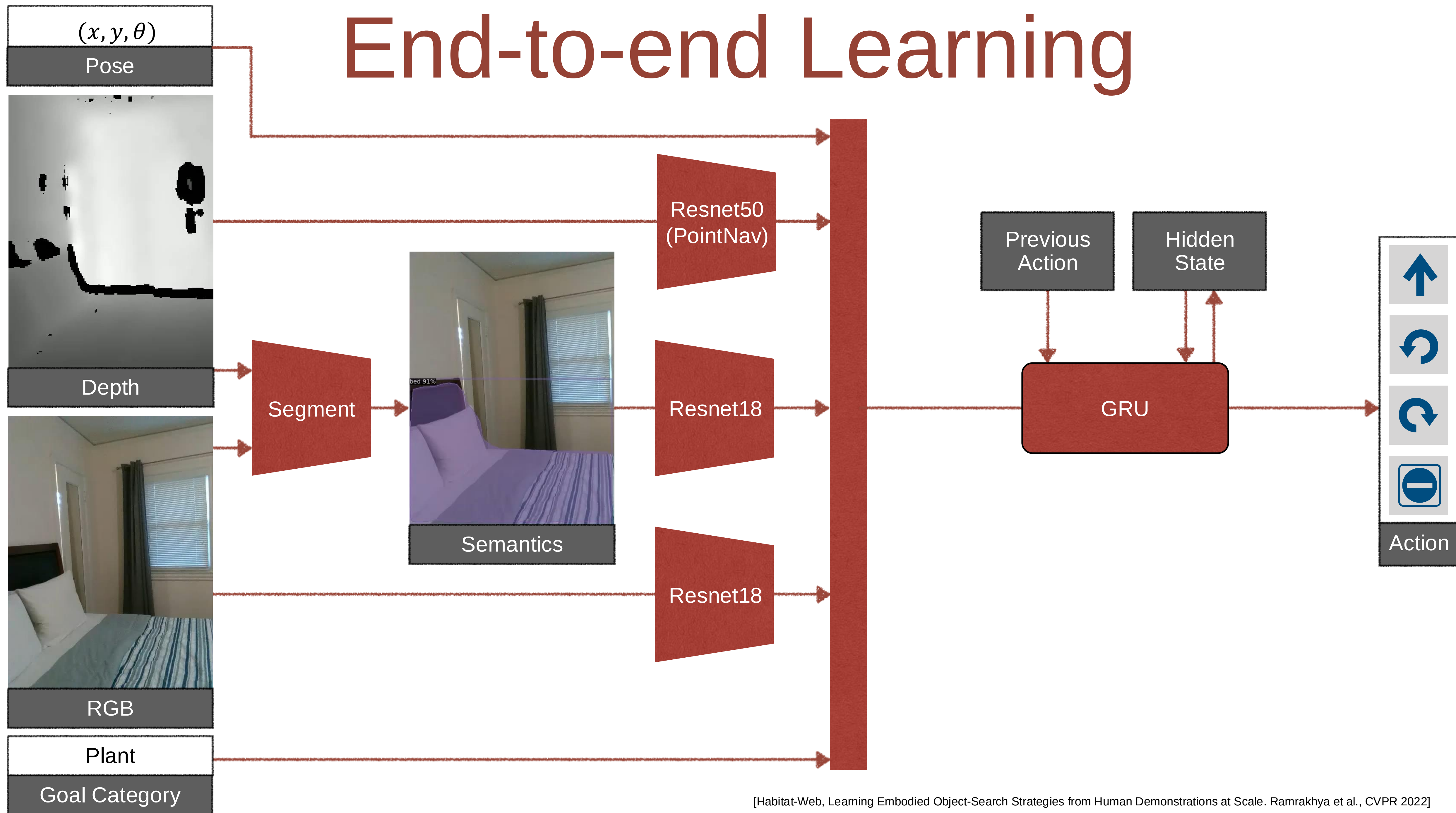
- End-to-end
- Large-scale IL + RL fine-tuning
 - 77,000 human trajectories
 - 200M frames of RL

Modular Learning

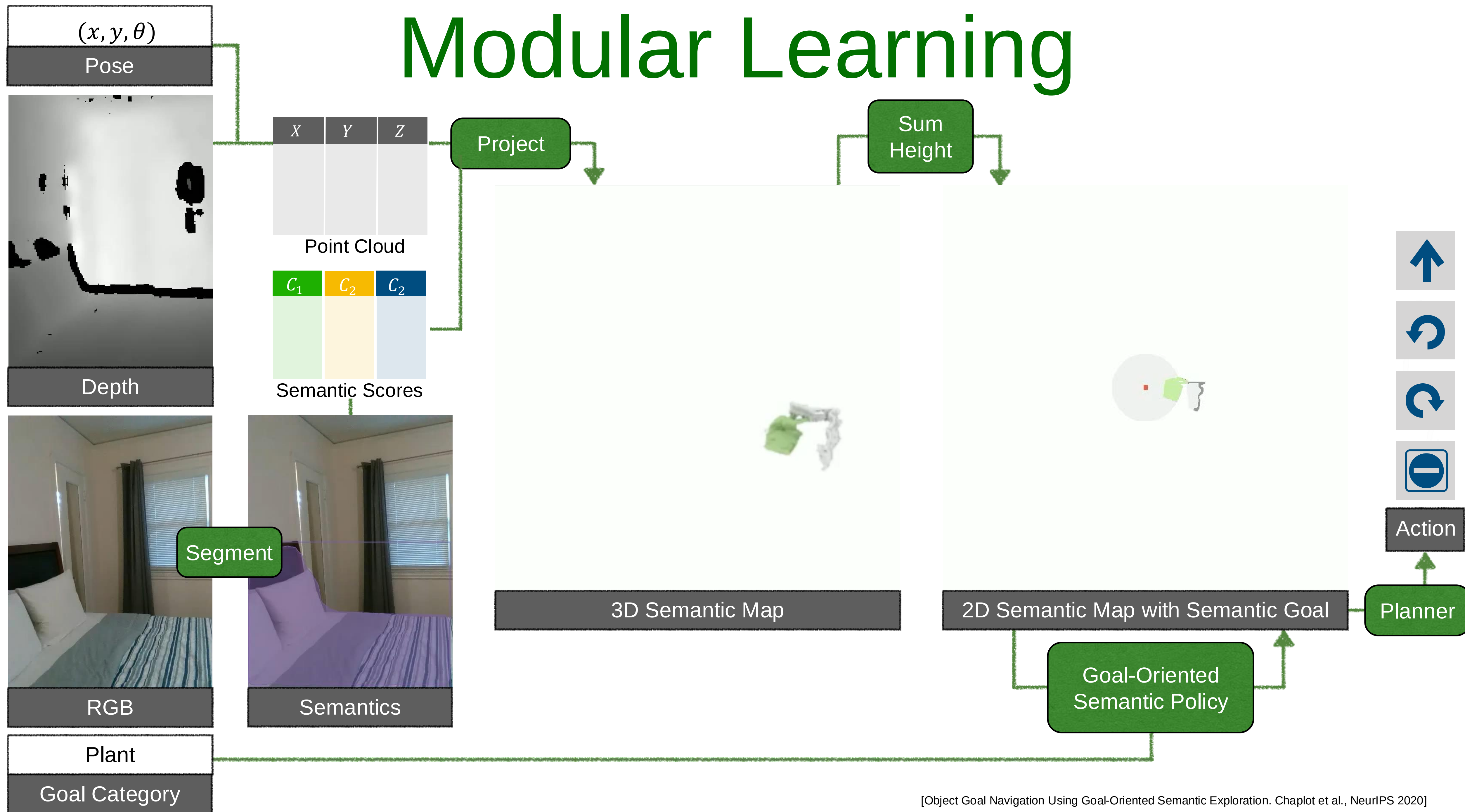


- Semantic Map
- Goal-Oriented Exploration
 - 10M frames of RL

End-to-end Learning



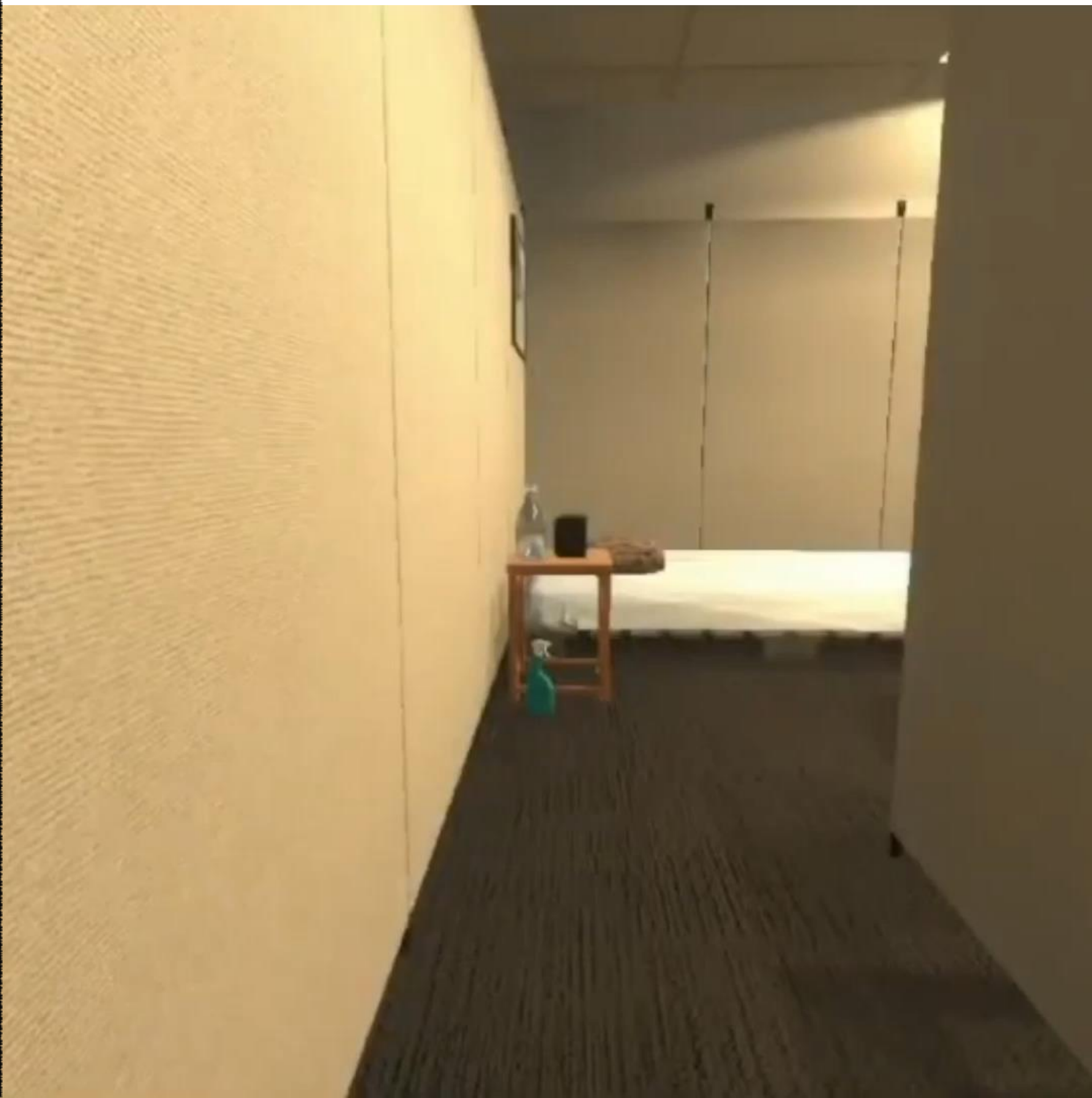
Modular Learning



Habitat



AI2-Thor



Results

Success Rate

SPL

■ Sim ■ Real World

Modular Learning

Classical

End-to-end Learning

0.00

0.25

0.50

0.75

1.00

Goal: *couch*

SPL: 0.74, 78 steps

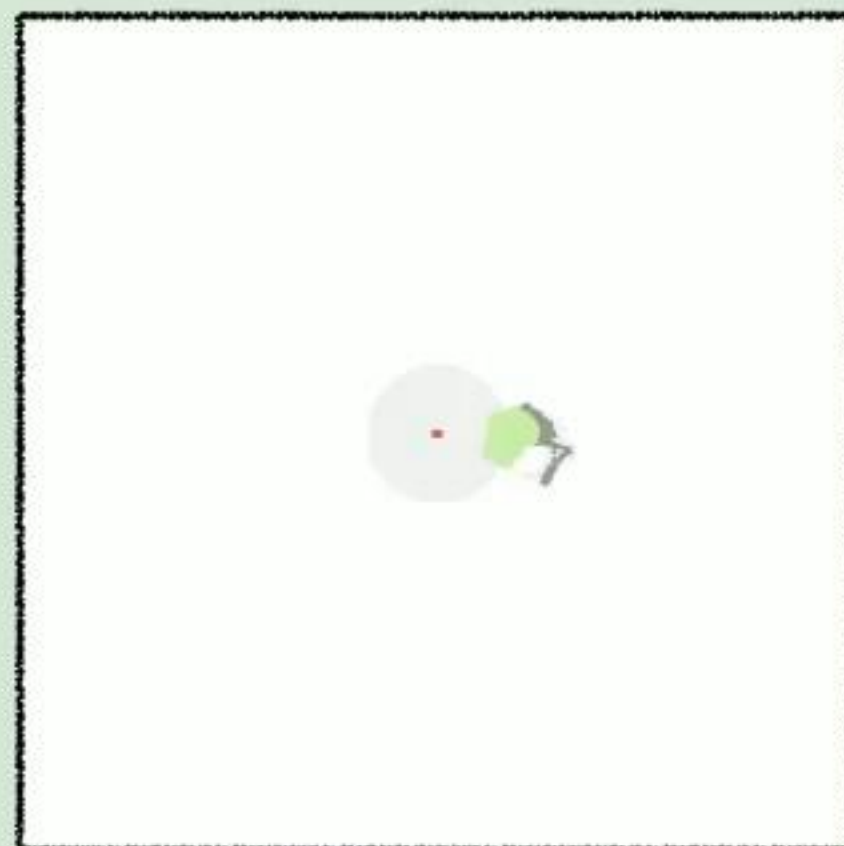
Modular

Third-person view

✓ Success



Observation



Predicted
Semantic Map

SPL: 0.0, 121 steps

End-to-End

Third-person view

✗ Failure



SPL: 0.33, 181 steps

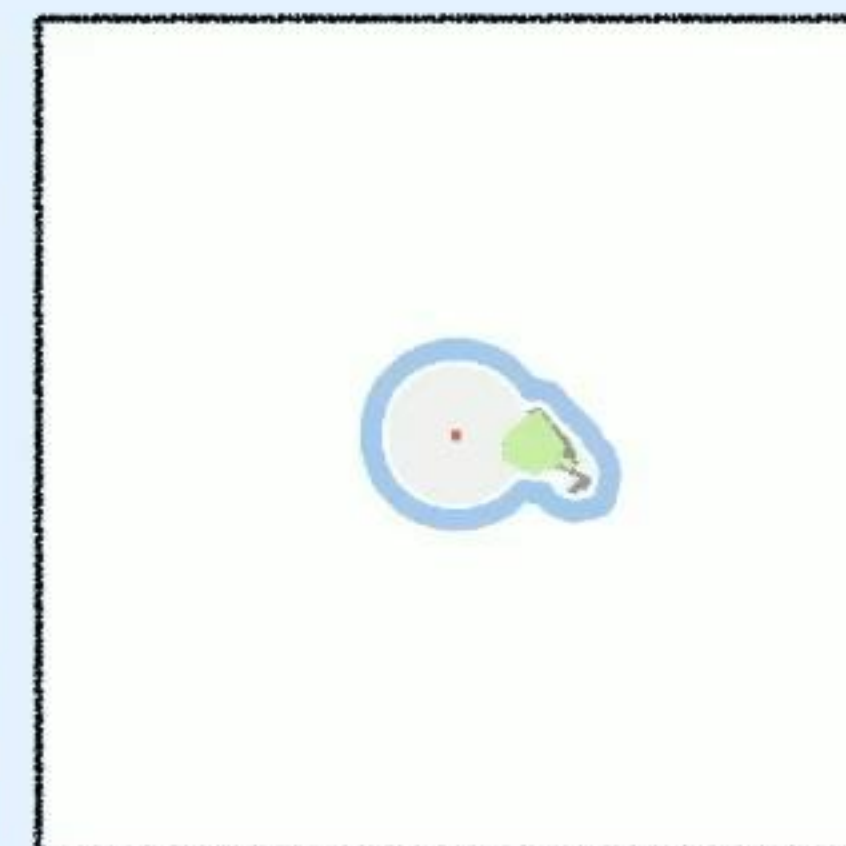
Classical

Third-person view

✓ Success

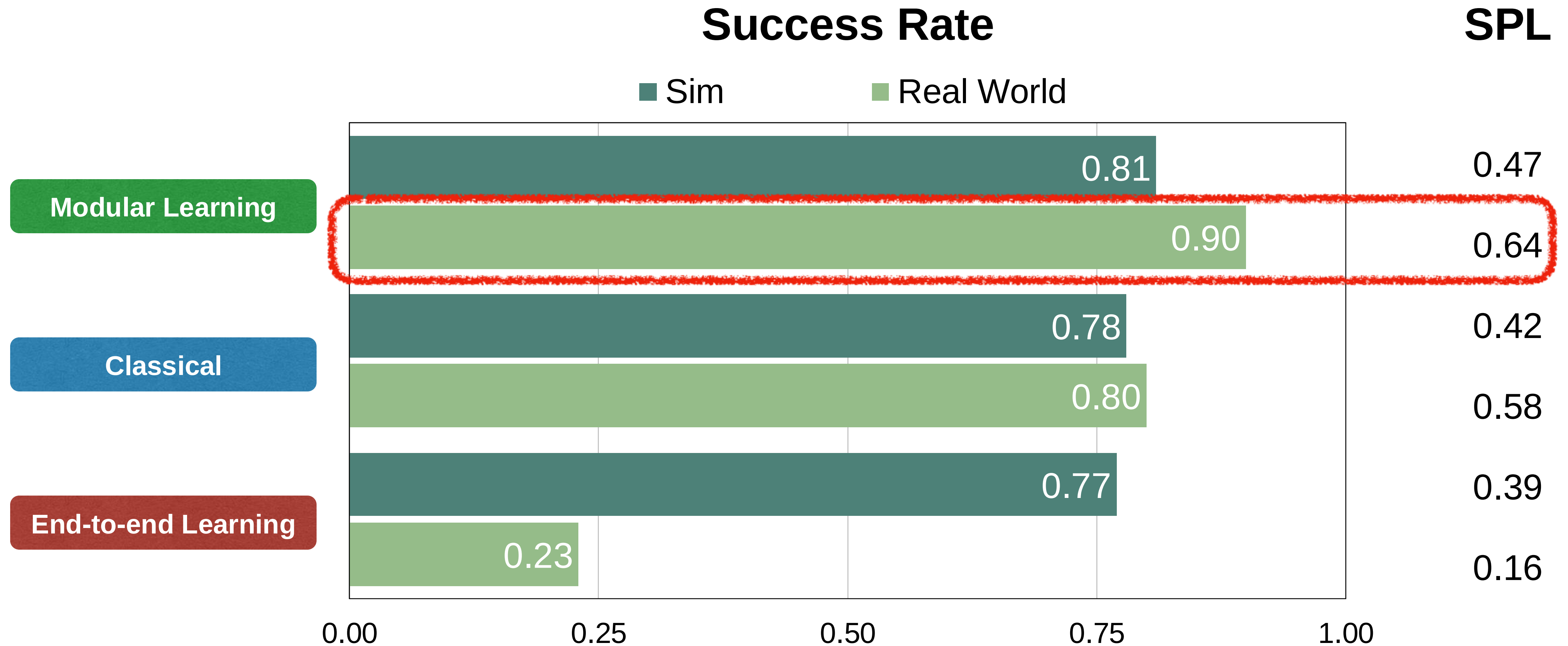


Observation



Predicted
Semantic Map

Modular Learning is Reliable



Observation



Goal: *toilet*

Predicted Semantic Map



Third-person view



— Navigable Area

— 0: chair

— 1: couch

— 2: potted plant

— 3: bed

— 4: toilet

— 5: tv

— 6: dining-table

— 7: oven

— 8: sink

— 9: refrigerator

— 10: book

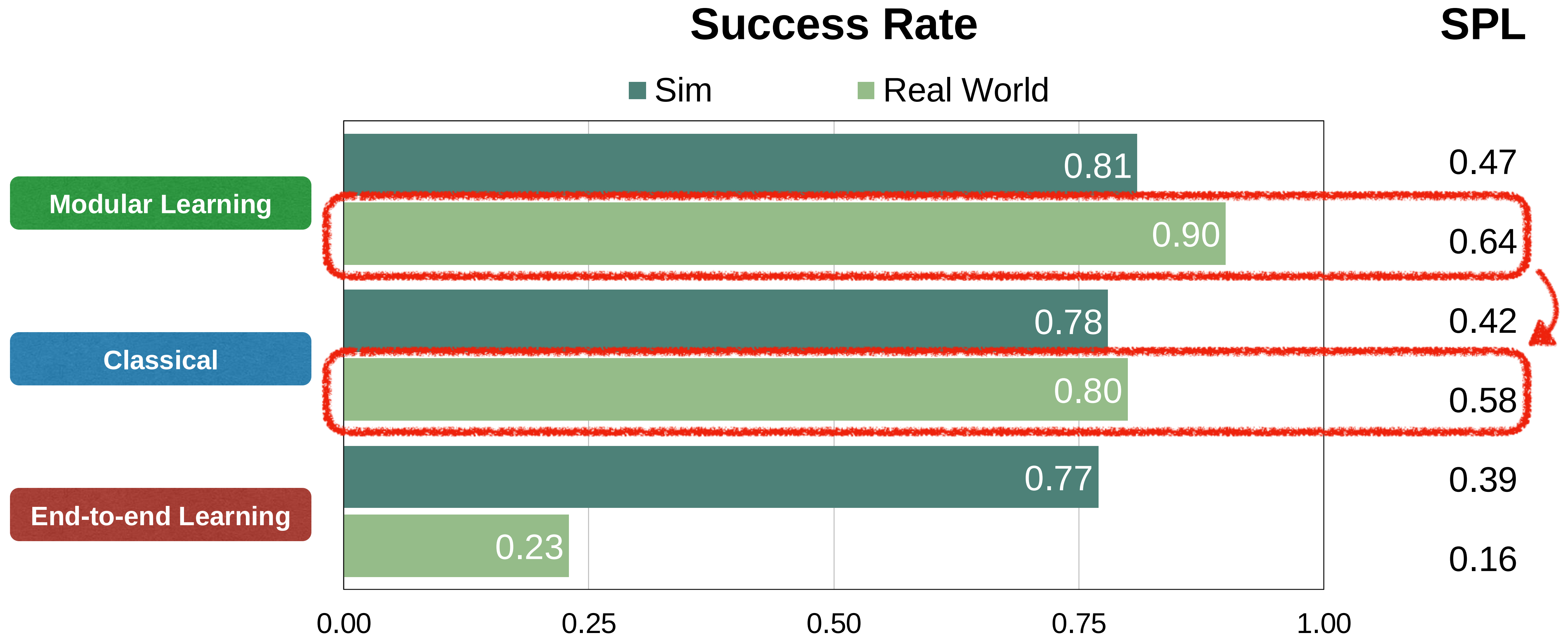
— 11: clock

— 12: vase

— 13: cup

— 14: bottle

Classical vs Modular Learning

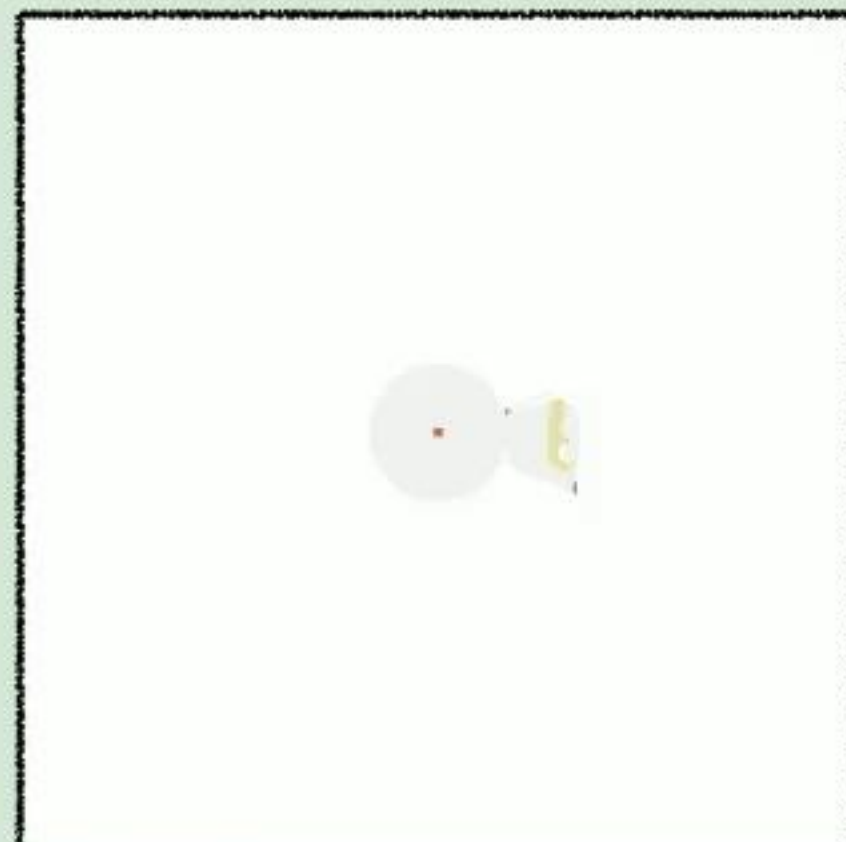


Classical vs Modular Learning

Goal: *bed*

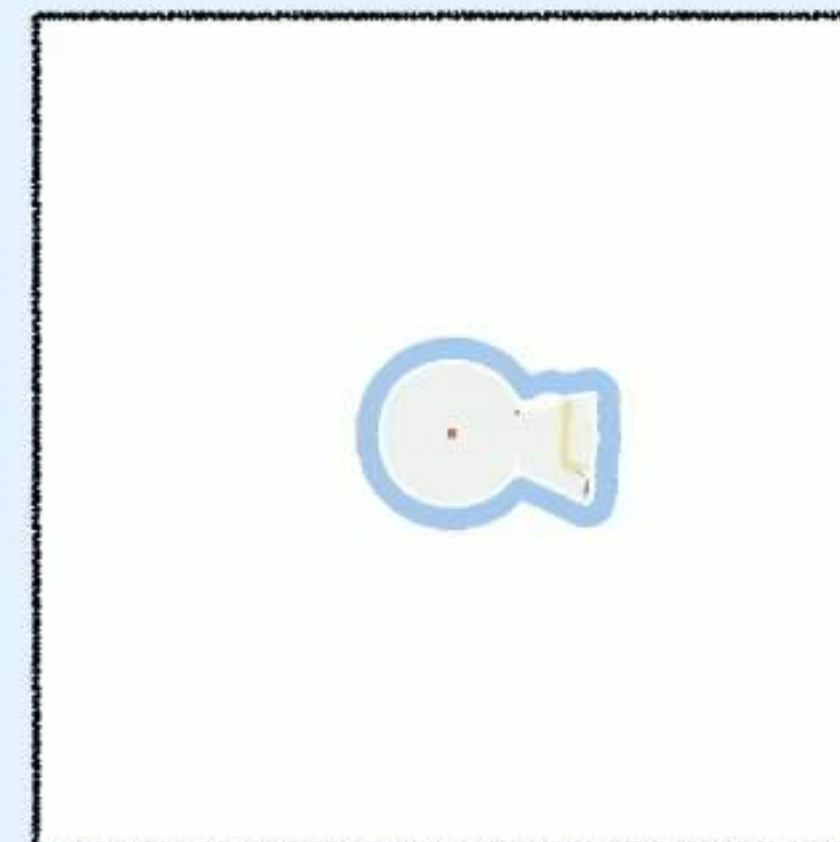
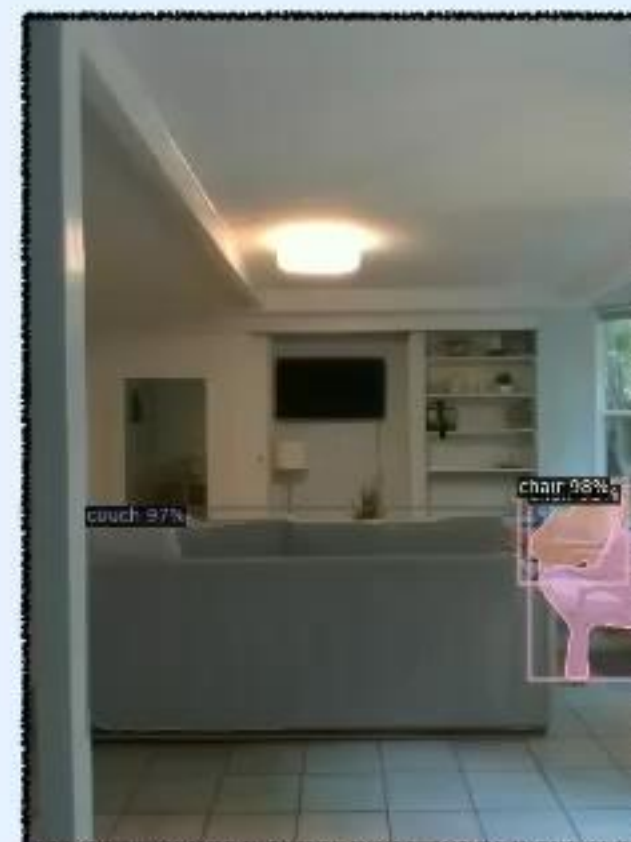
SPL: 0.90, 98 steps

Semantic Exploration

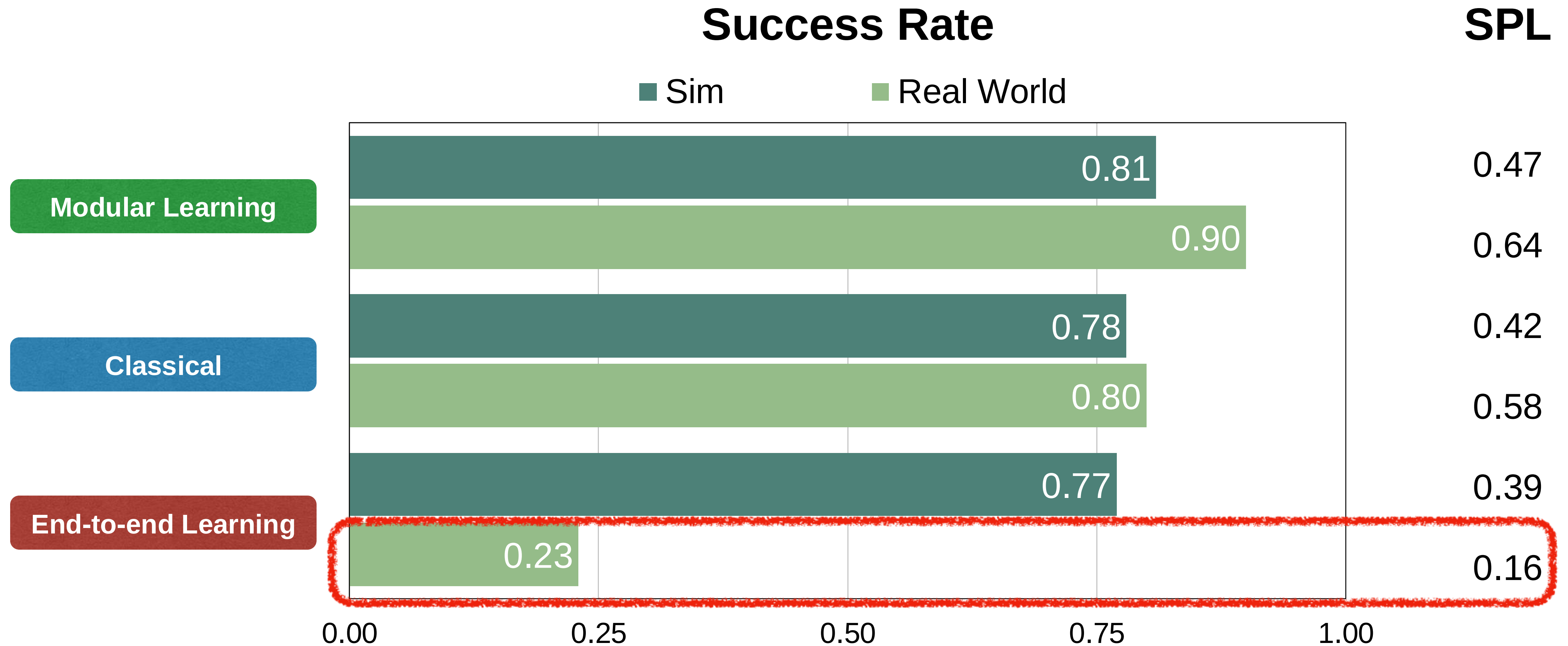


SPL: 0.52, 152 steps

Frontier Exploration



End-to-end fails to Transfer



End-to-end Failures

Goal: *TV*



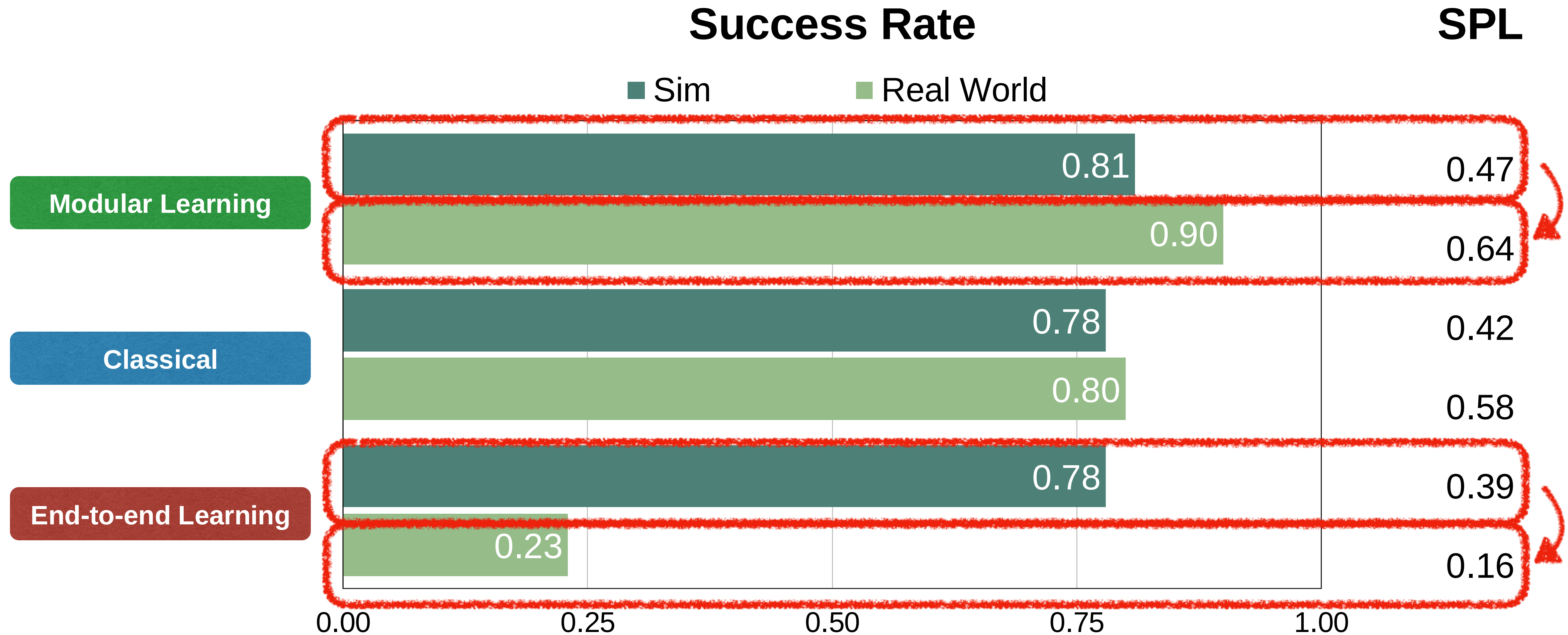
Goal: *Toilet*



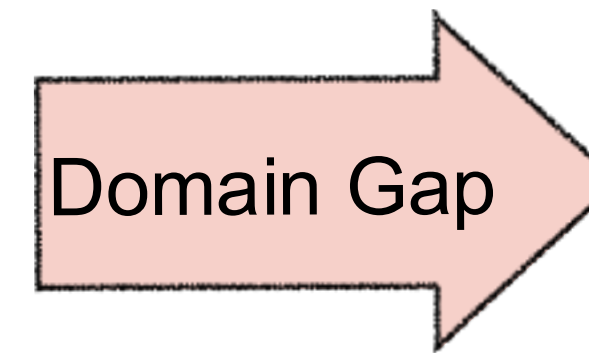
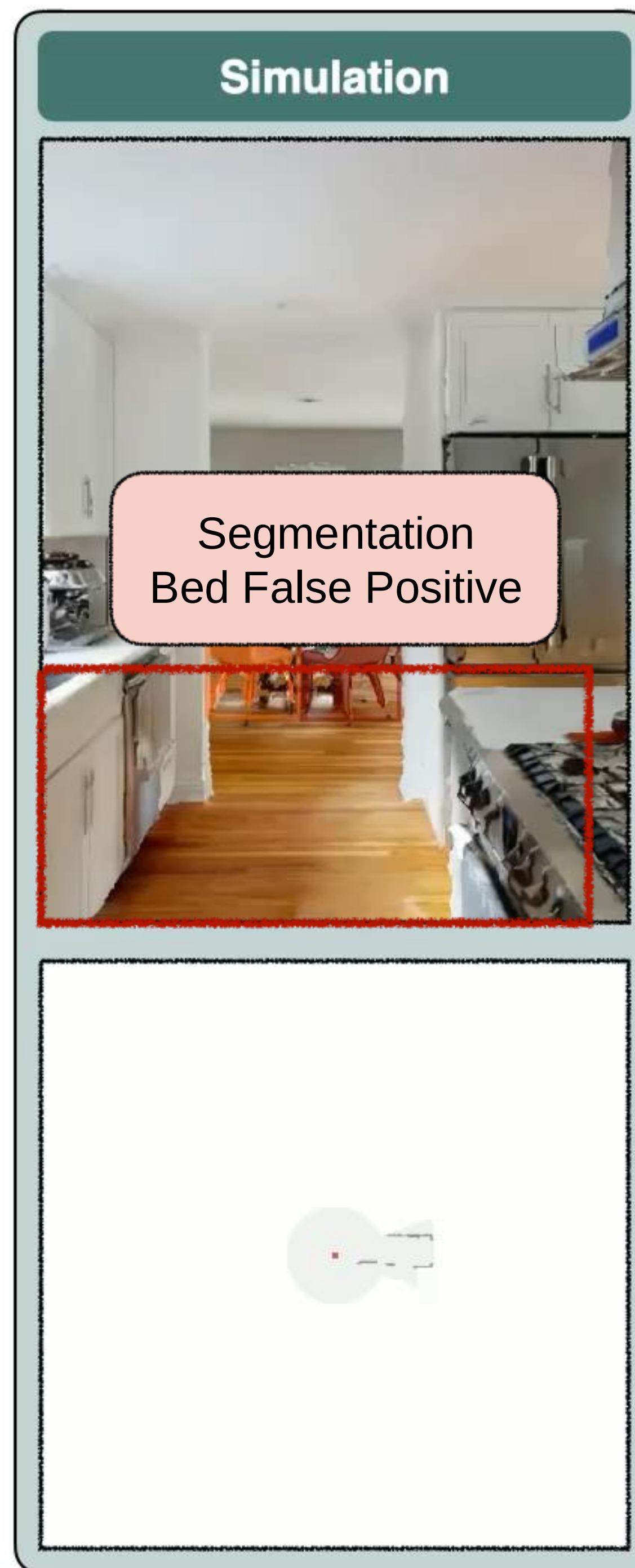
Goal: *Plant*



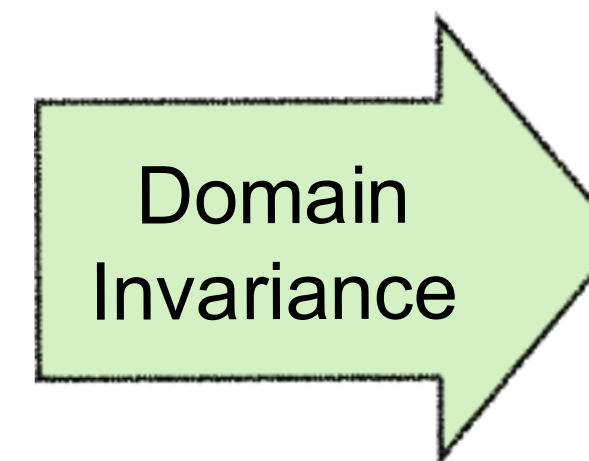
Modular vs End-to-end Transfer



Simulation vs Reality



*End-to-end
Learning
Input*

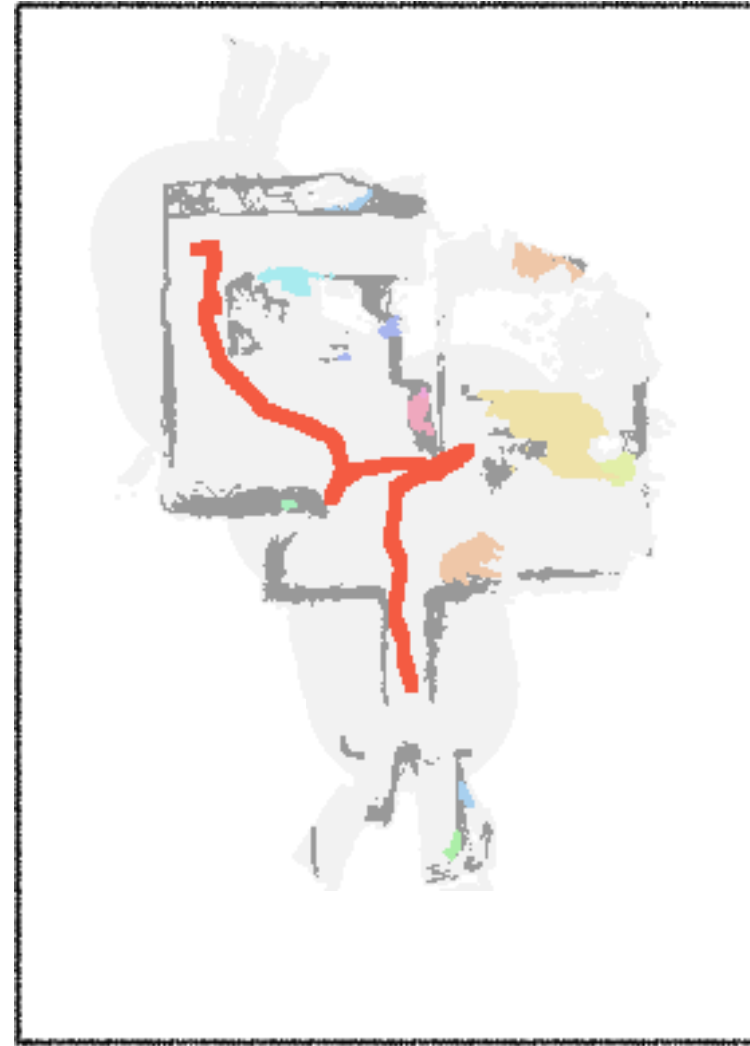


*Modular
Learning
Input*



Real World

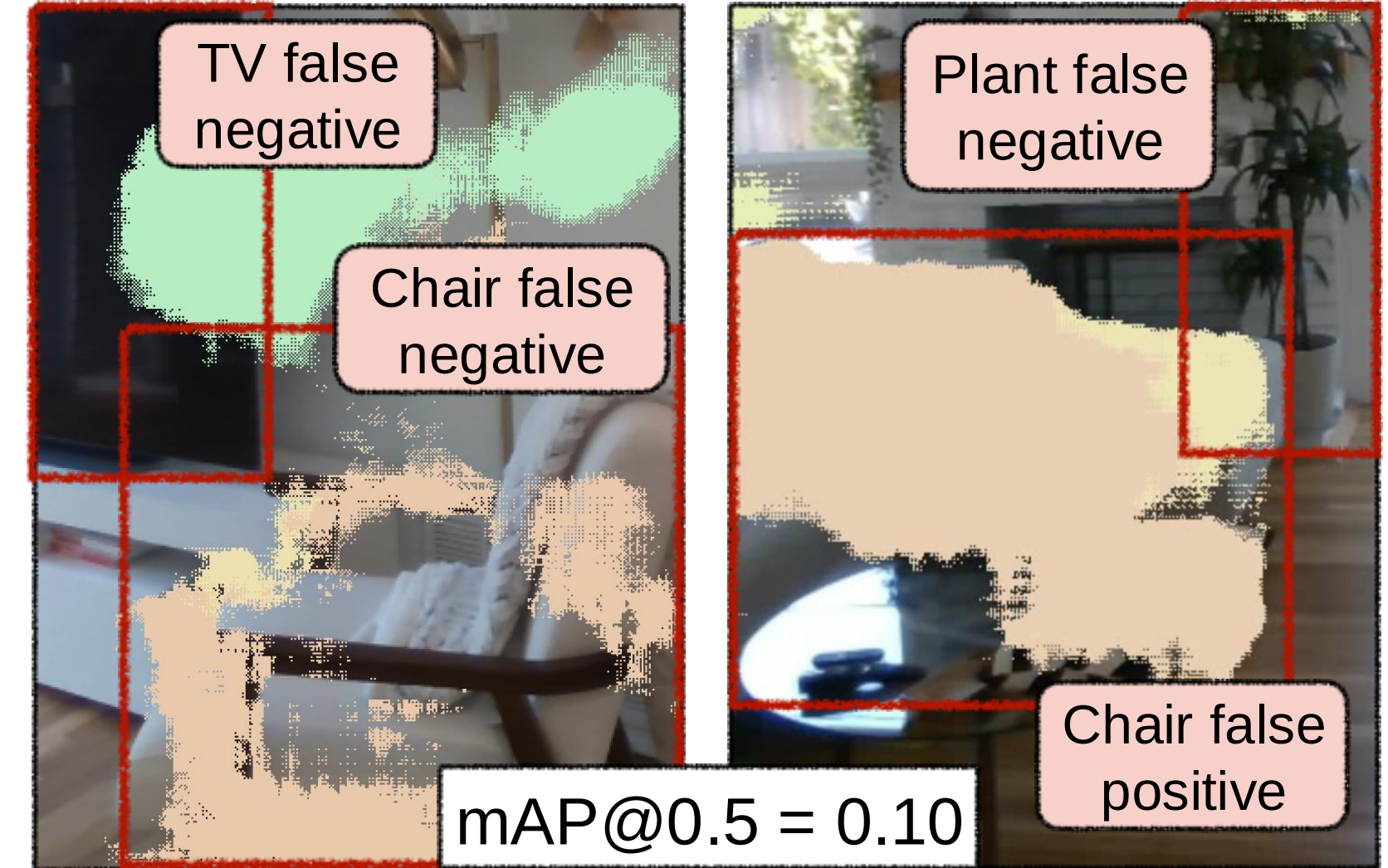
Predicted Semantic Map



Segmentation Model Trained in Real World



Segmentation Model Trained in Simulation



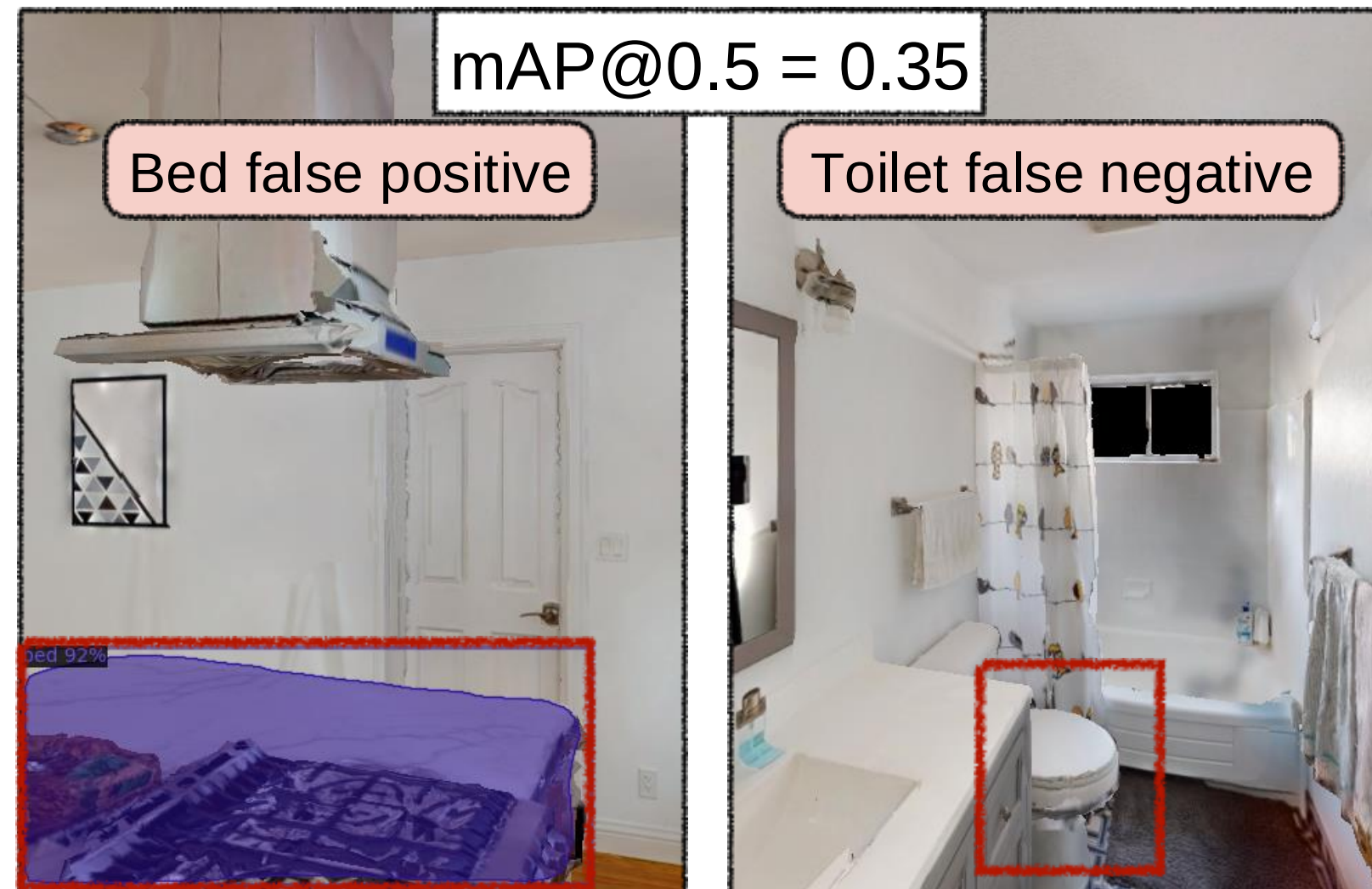
Domain Invariance

Domain Gap

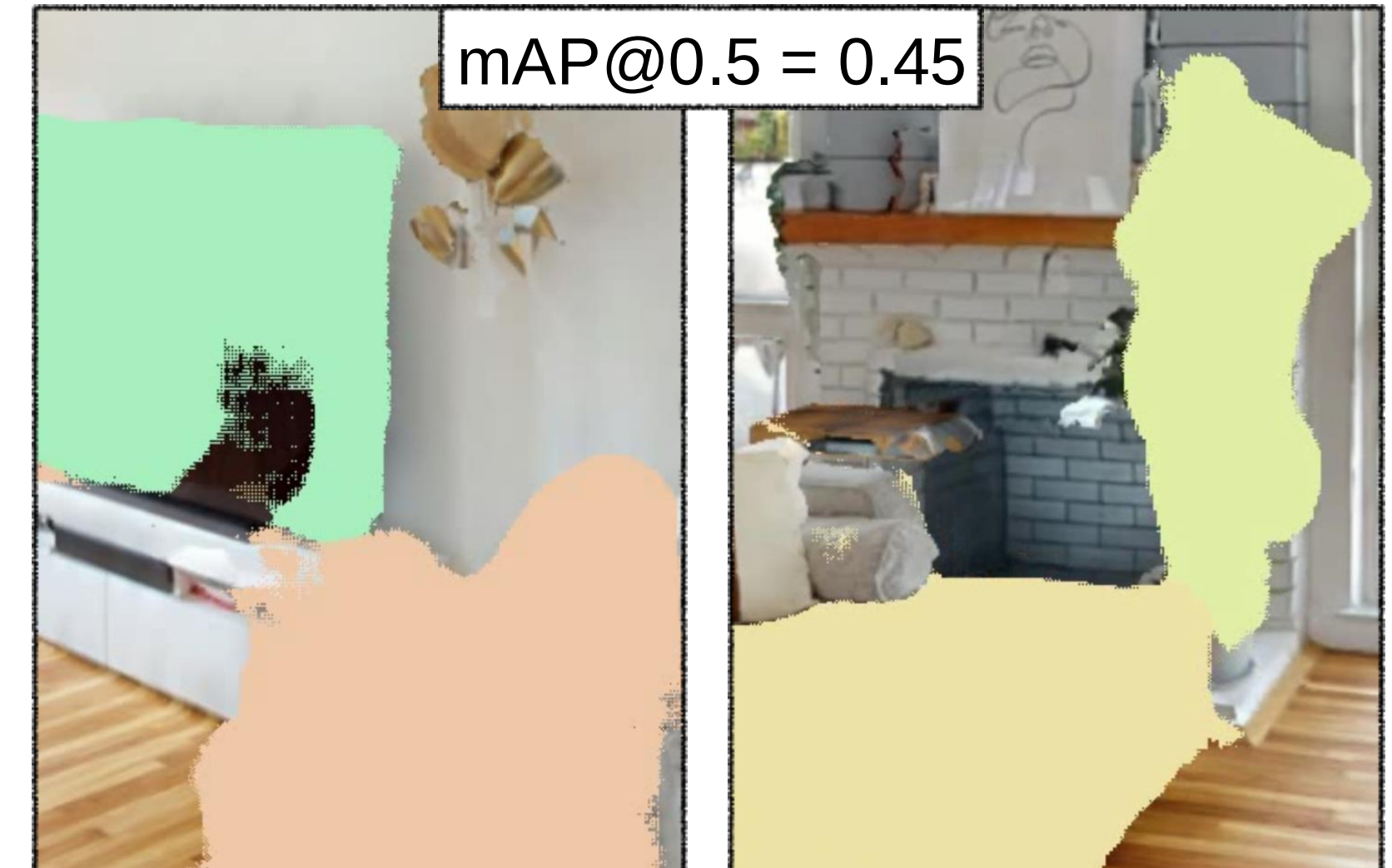
Domain Gap

- 0: chair
- 1: couch
- 2: potted plant
- 3: bed
- 4: toilet
- 5: tv

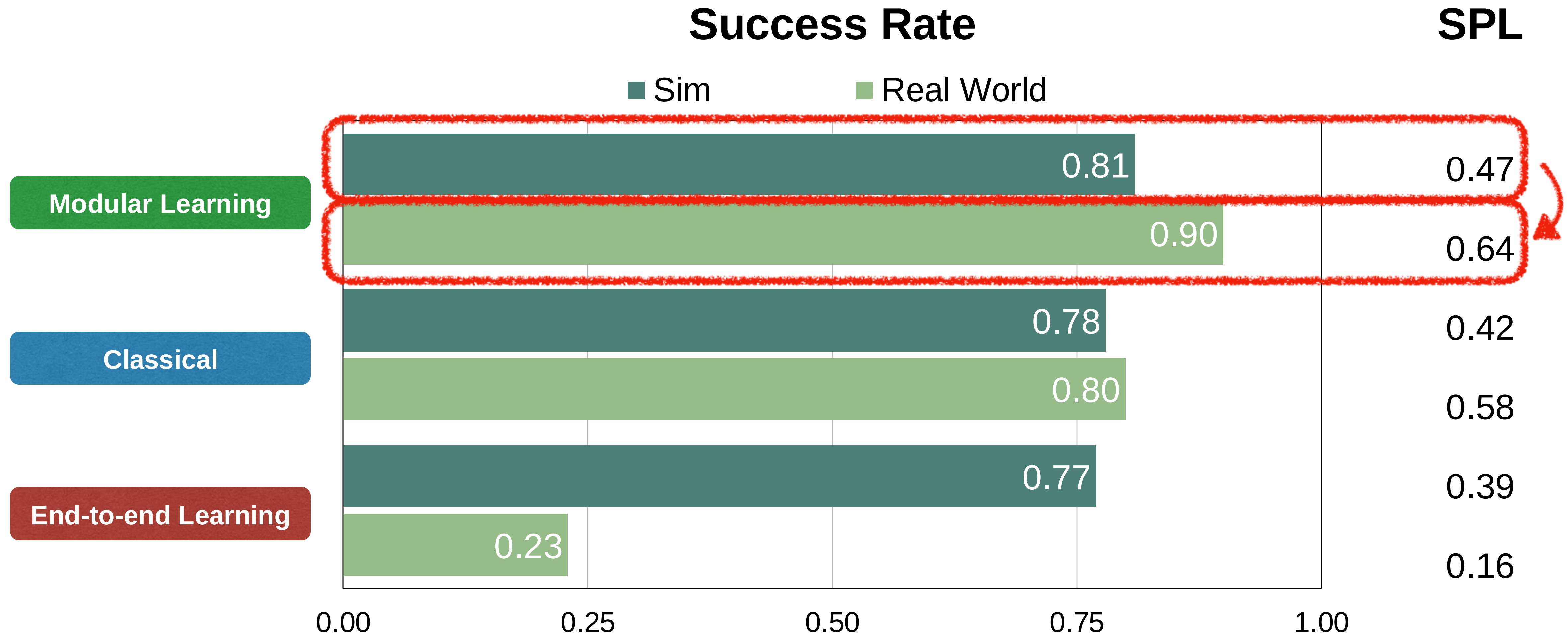
Simulation



mAP@0.5 = 0.45



Modular Learning Sim vs Real



Takeaways

For practitioners:

- Modular learning can reliably navigate to objects with 90% success

For researchers:

- Models relying on RGB images are hard to transfer from sim to real → *leverage modularity and abstraction in policies*
- Disconnect between sim and real error modes → *evaluate semantic navigation on real robots*



GOAT: GO to Any Thing

Matthew Chang*, Theophile Gervet*, Mukul Khanna*, Sriram Yenamandra*, Dhruv Shah, Tiffany Min, Kavit Shah, Chris Paxton, Saurabh Gupta, Dhruv Batra, Roozbeh Mottaghi, Jitendra Malik*, Devendra Singh Chaplot*

Third-person view



1. GOAT Problem

GOAT System Architecture

Results

Applications

Pick & Place

Social Navigation

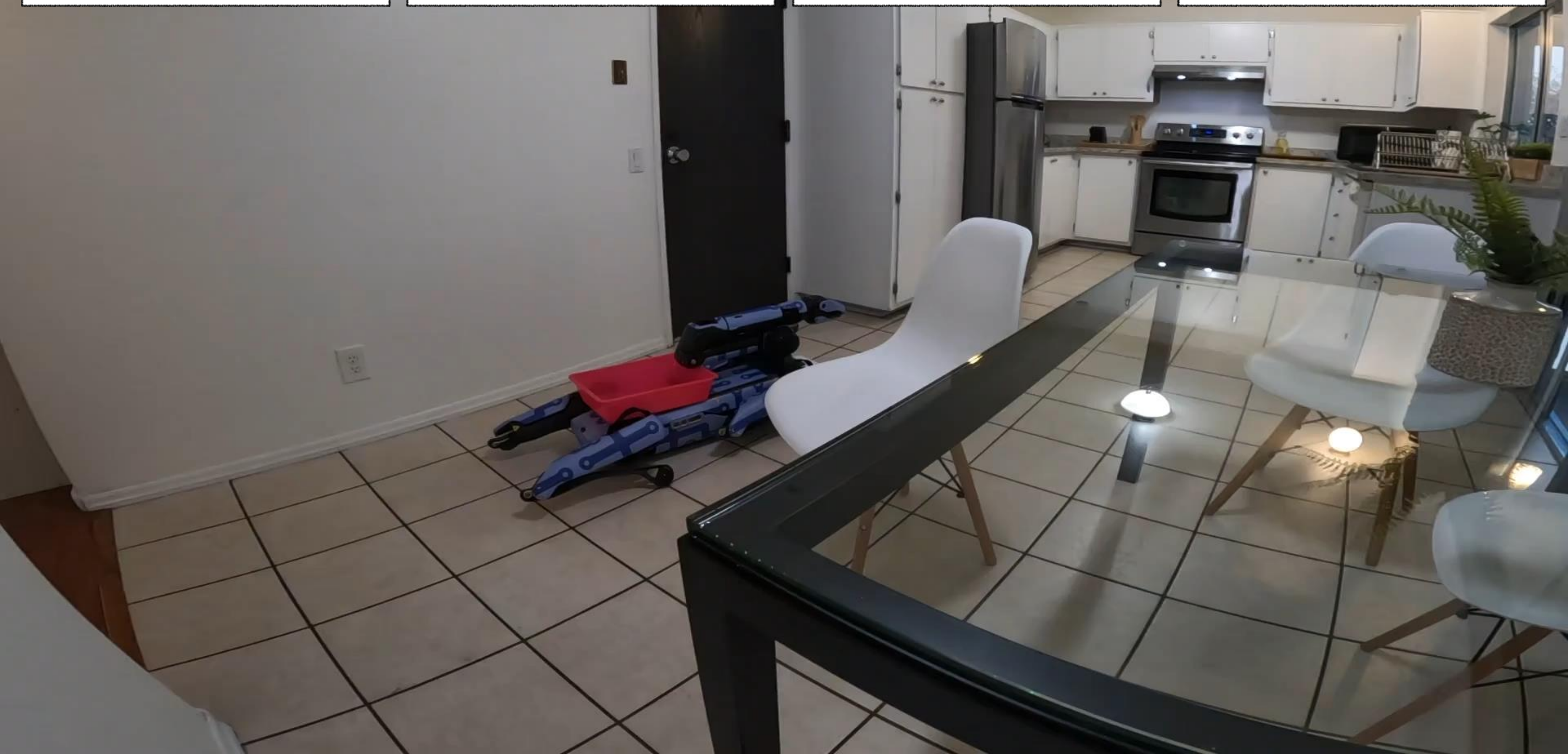
Platform Agnostic

Unknown Environment
Explore

Perception
*Detect and Localize
Objects*

Lifelong Memory
*Remember Object
Locations*

Control
*Navigate to / Pick & Place
Objects*



Multimodal:

Reach Any Object Specified in Any Way

Image



Language

Find **the fruit basket** on the **kitchen counter**

Category

Bring me **a CUP**

Lifelong:

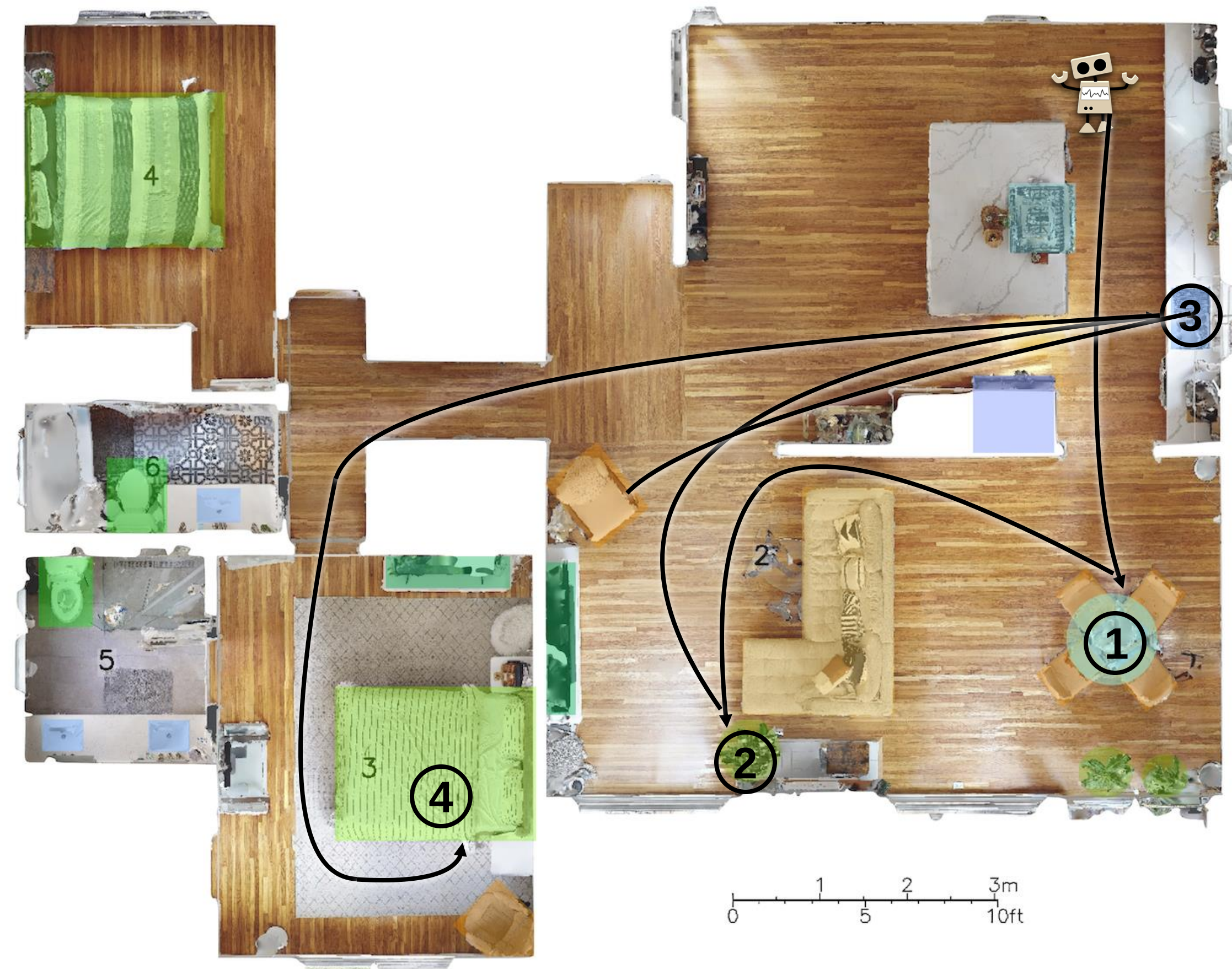
Remember Object Locations



② Go to the potted plant next to the couch

③ Go to a SINK

④ Go to the black and white striped bed



1. GOAT Problem

GOAT System Architecture

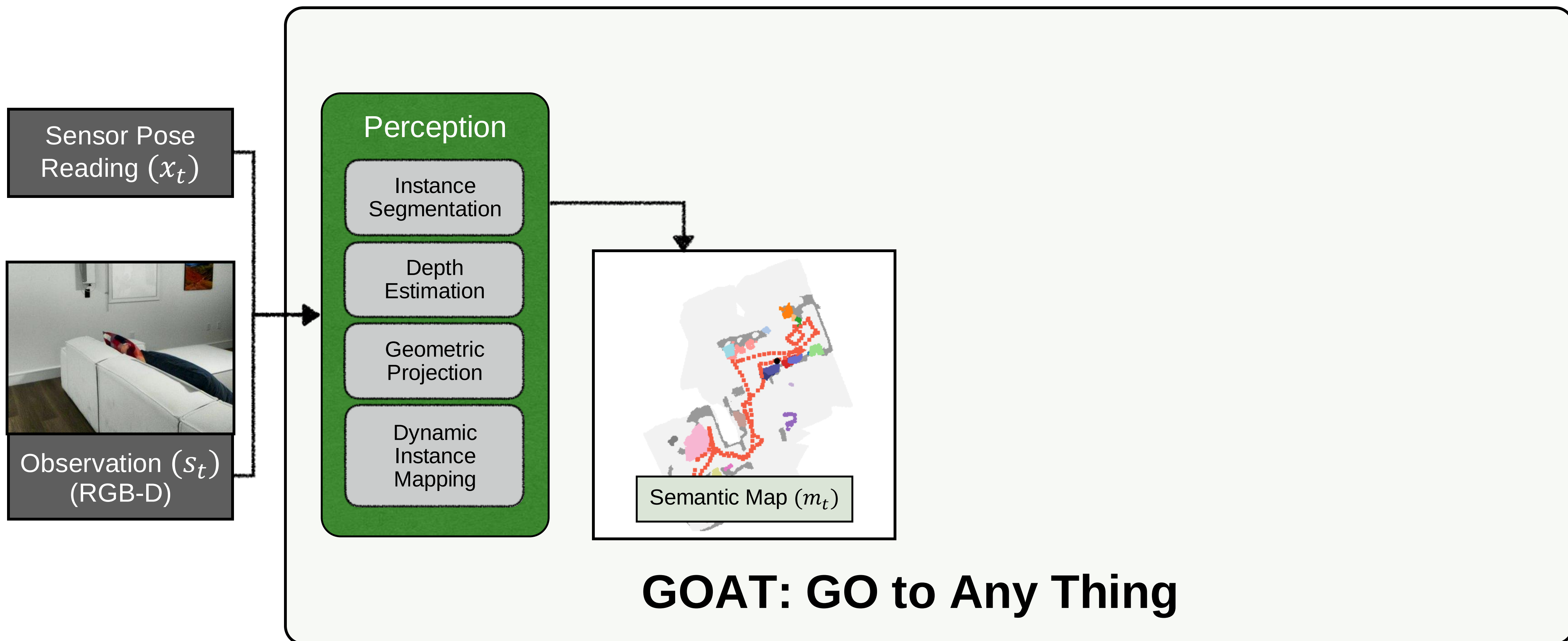
Results

Applications

Pick & Place

Social Navigation

Platform Agnostic



Language Goal

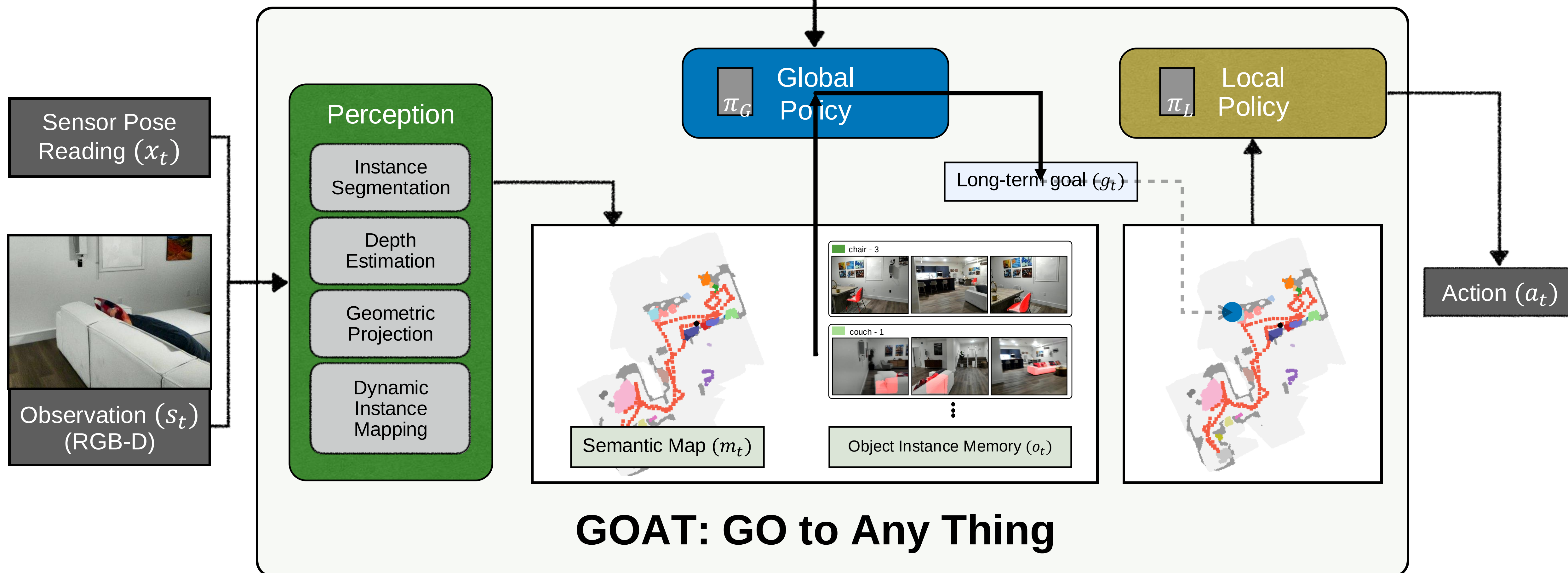
Brown office chair
Bed with white sheets
Cup on the kitchen counter

Image Goal

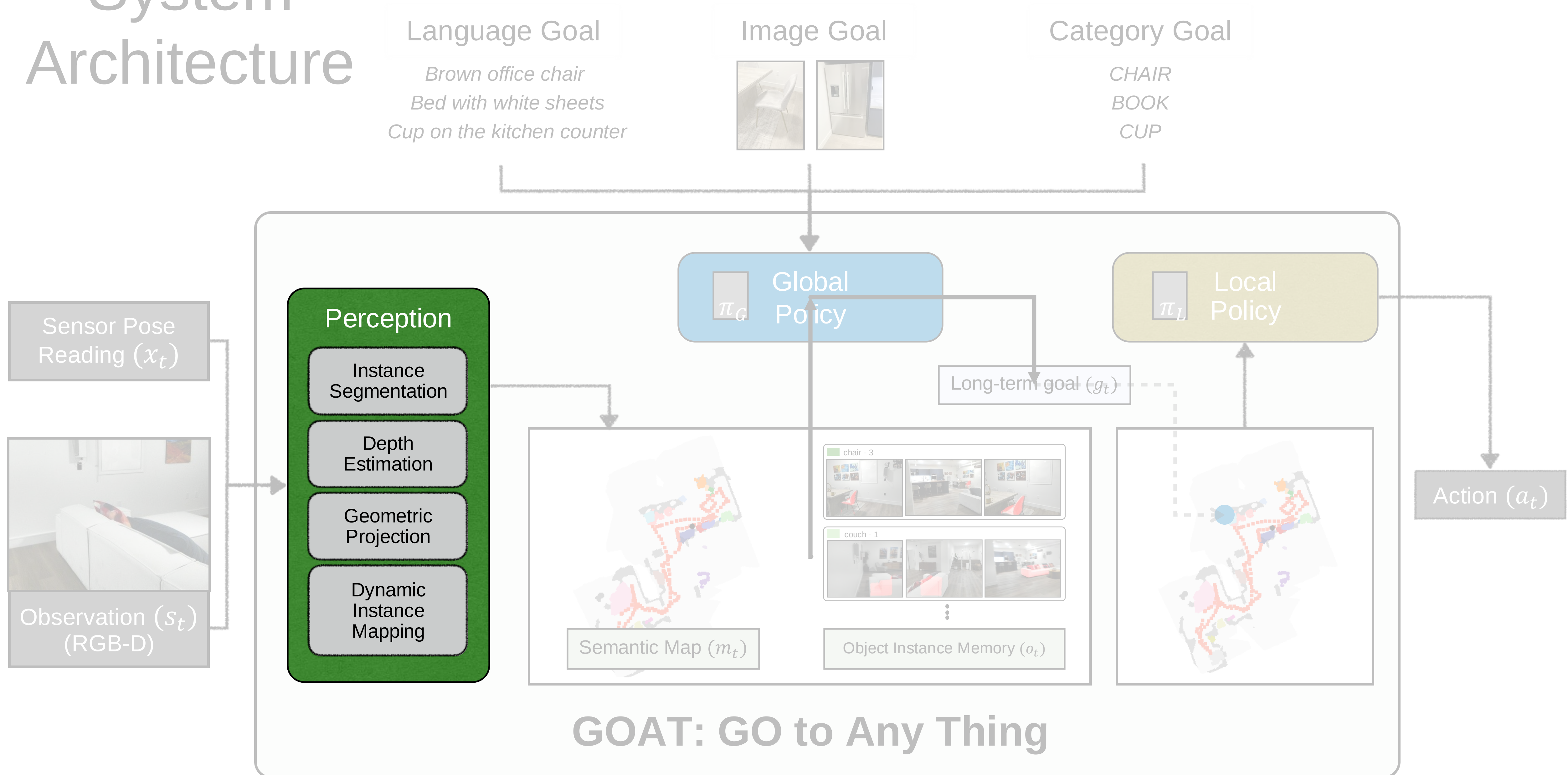


Category Goal

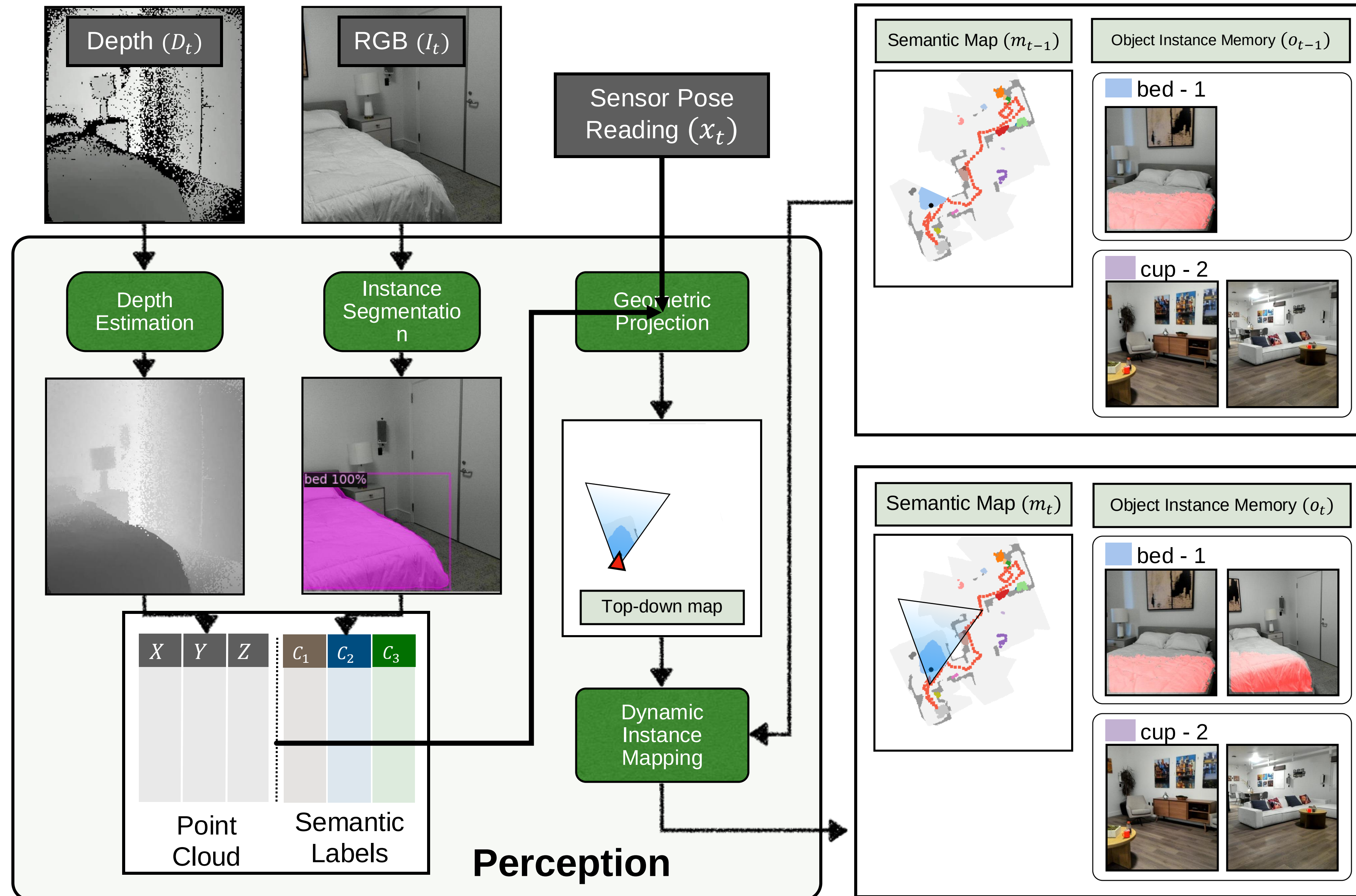
CHAIR
BOOK
CUP



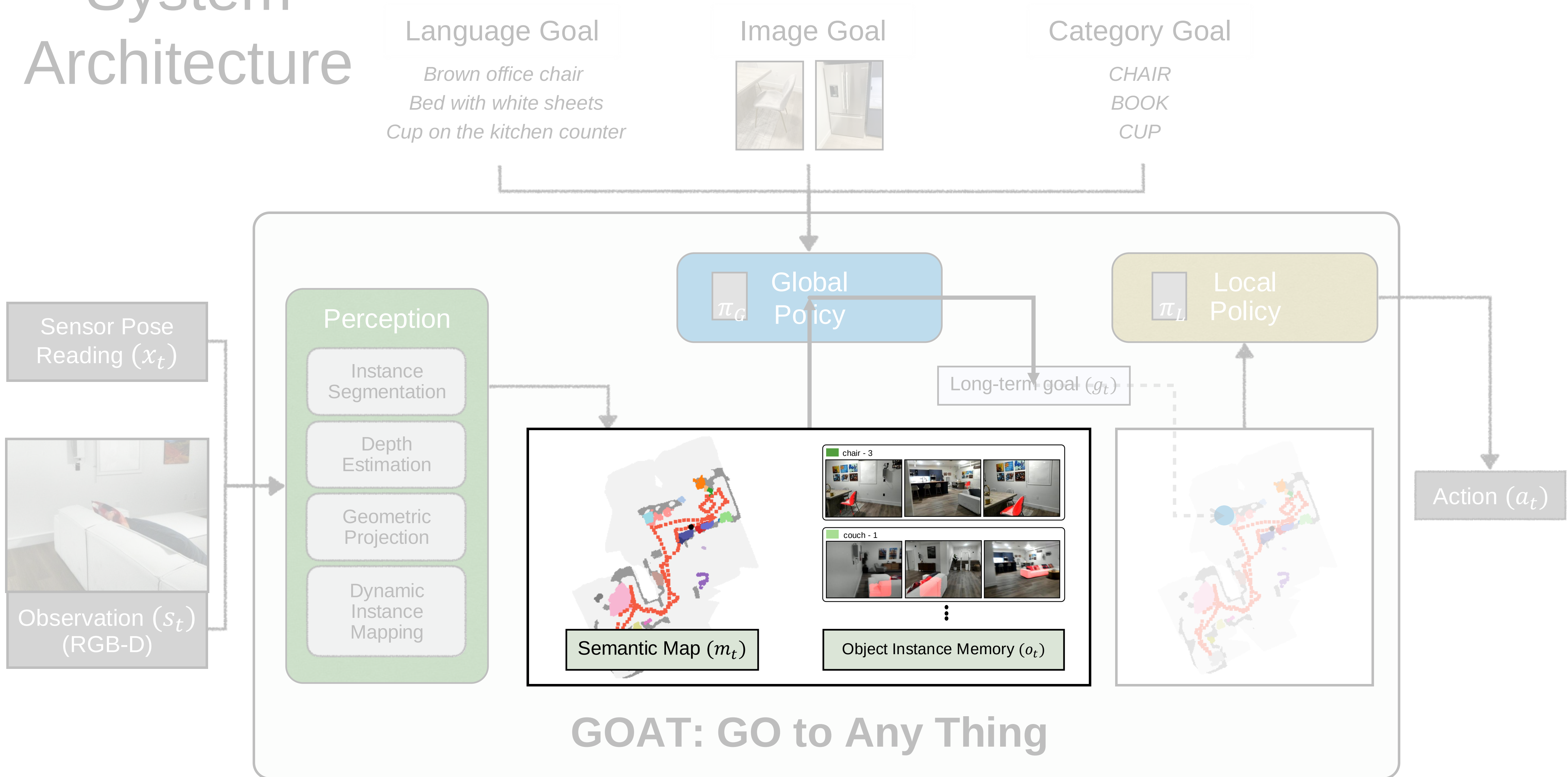
System Architecture



Perception System

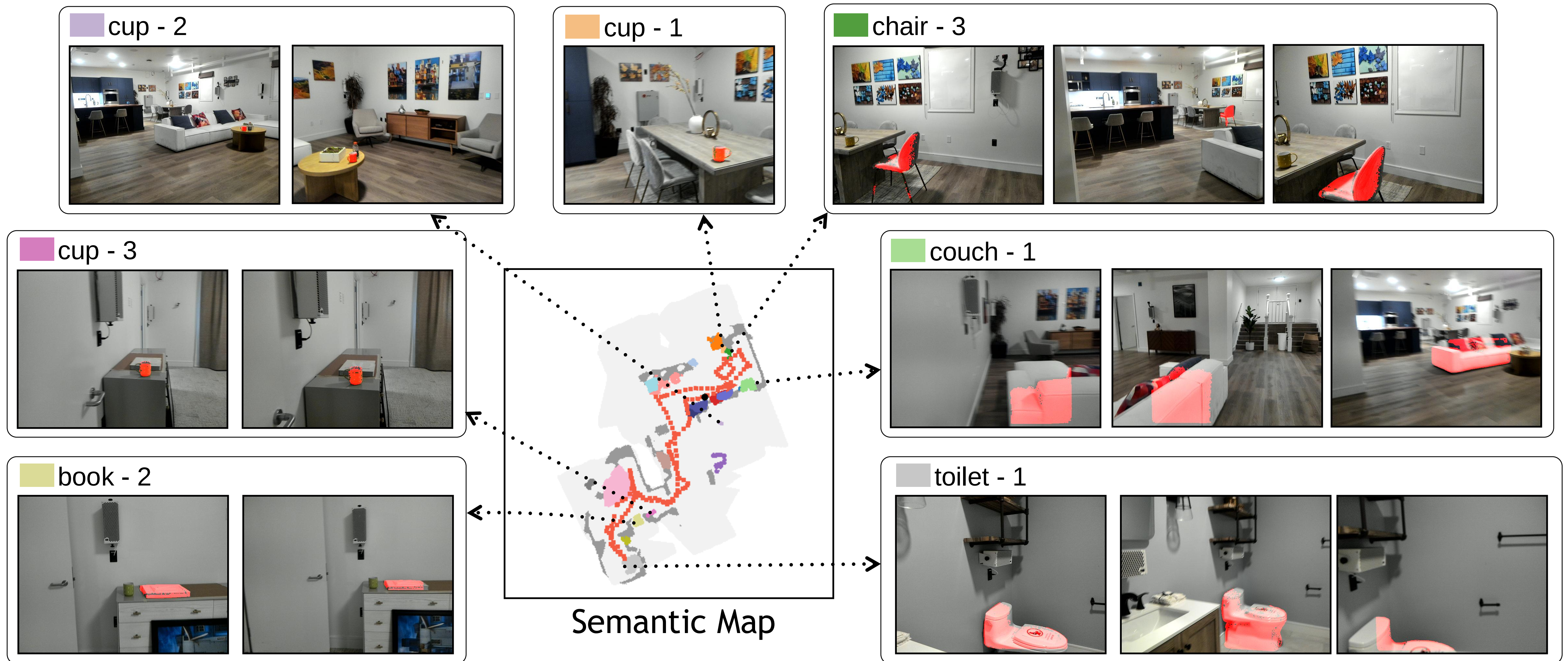


System Architecture



GOAT Memory Representation

Semantic map with associated Object Instance memory

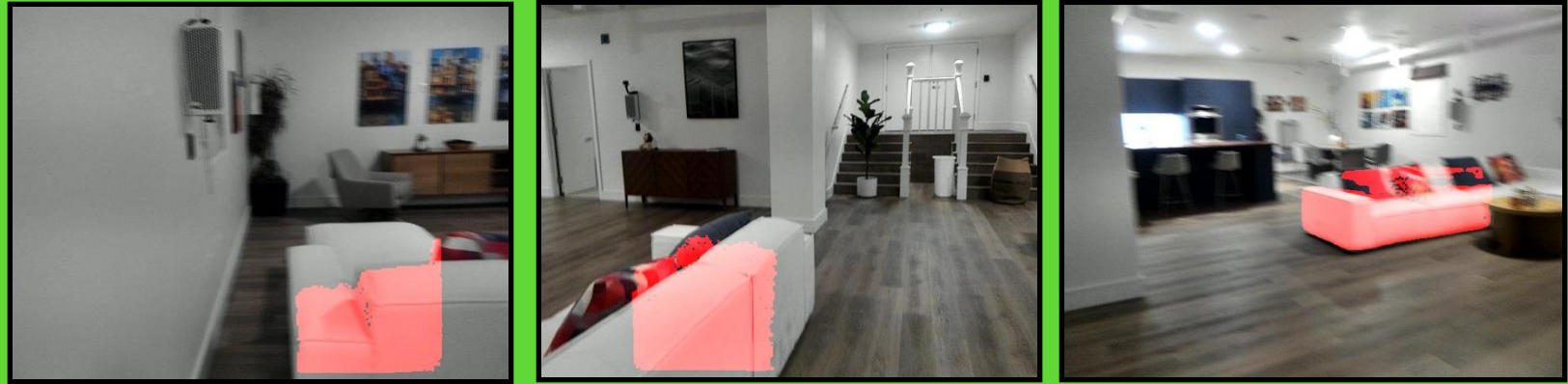


Object Instance Memory

chair - 3



couch - 1



toilet - 1

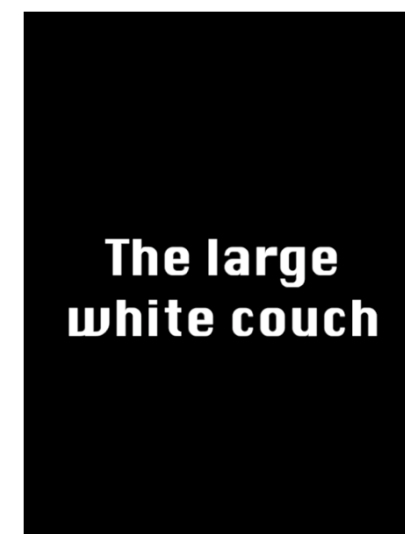


book - 2



⋮

Goals



Goal Localization

Goal Localization

Goal Localization

Localized

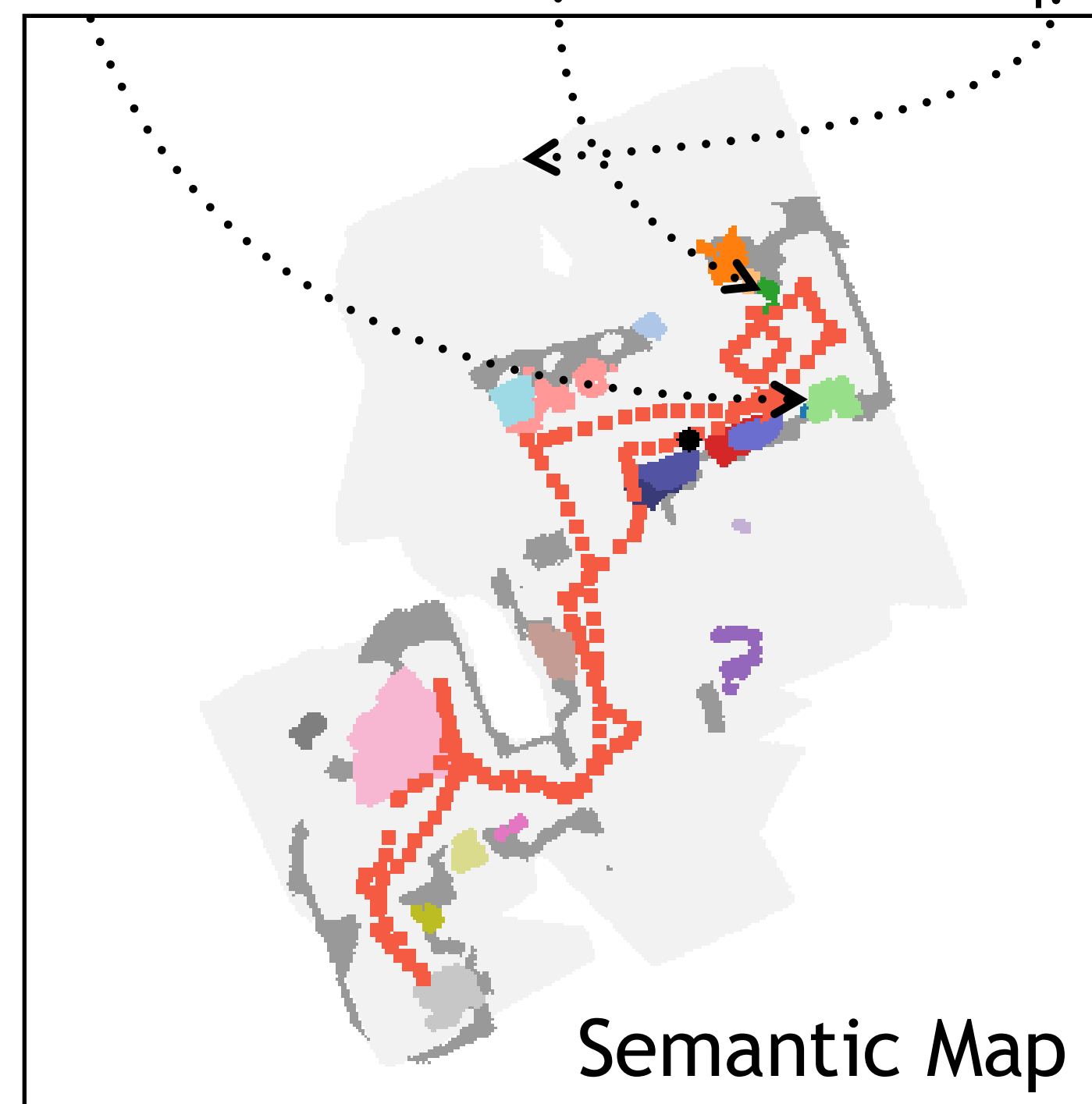
Localized

Not localized

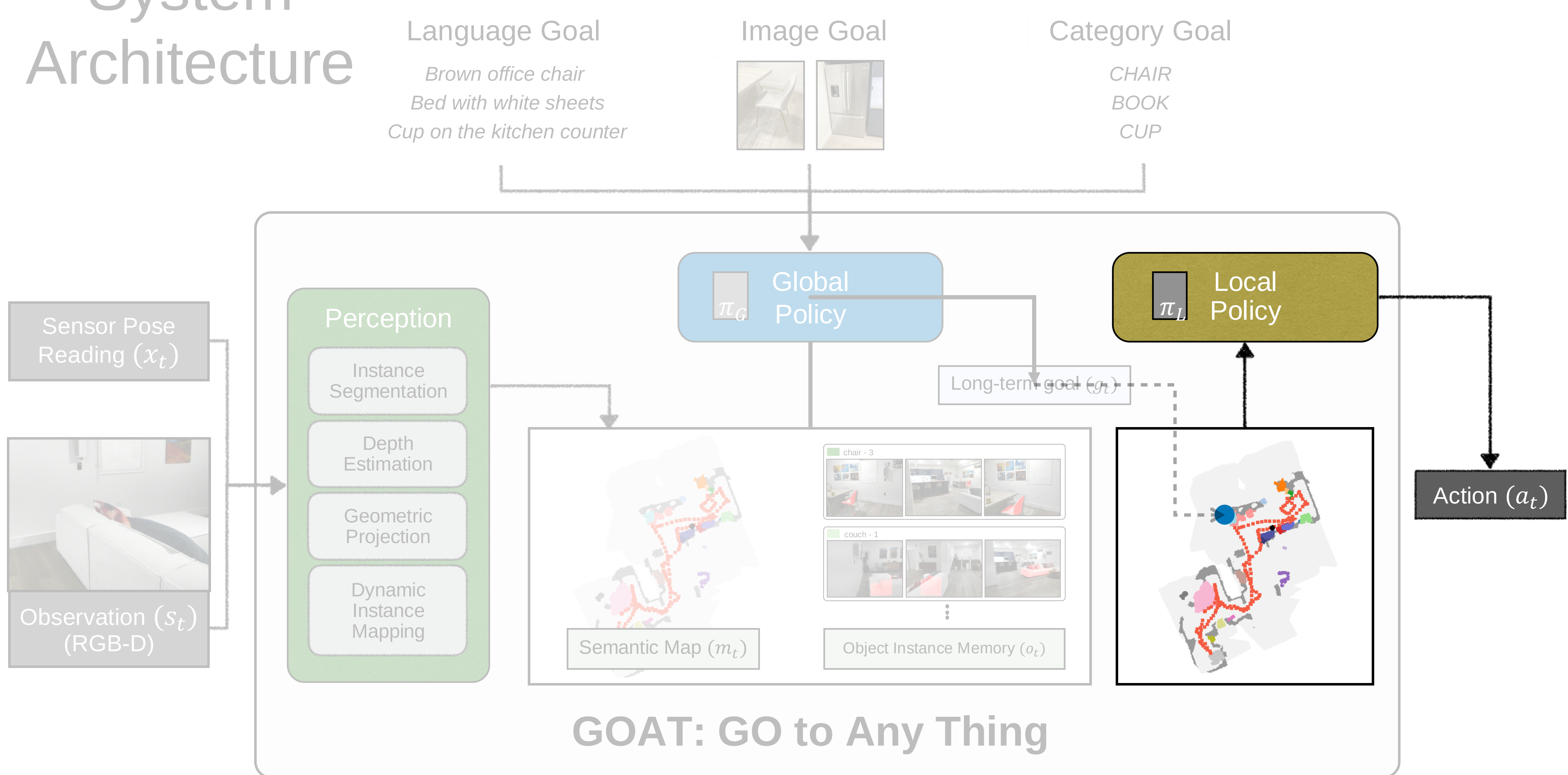
couch - 1

chair - 3

Explore



System Architecture



1. GOAT Problem

GOAT System Architecture

Results

Applications

Pick & Place

Social Navigation

Platform Agnostic



“In the Wild”
Empirical Evaluation
9 Unseen Homes
4 Methods
10 Trajectories per Home
5-10 Goals per Trajectory
~90h of Experiments



The bed with the white blanket pulled back halfway and grey sheets.



The bunk bed with stars on the blanket.



The bed with blue and white sheets.



The with the blue and yellow painting above it and plant on the right.



The bed with the blue blanket and blue pillows.



The bed with grey sheets and light green trim.



The bed with the white blanket and red pillows



The large grey living room couch with many pillows.



The green couch.



The white rectangular couch with no pillows.



The large grey living room couch with many pillows.



The light brown couch with blue pillows.



The large grey couch in front of the yellow wall.



The couch covered in a white blanket.



The green cup on the kitchen counter



The red cup on the kitchen counter



The green mug on the plastic chair.



The grey cup on the nightstand.



The red cup on top of the glass table.



The red cup on the kitchen counter.



The light blue cup.



The plant in front of the window.



The potted plant next to the stairs.



The large potted plant in the bedroom.



The group of plants in front of the curtain.



The large potted plant next to the foosball table.



The large potted plant in front of the mirror.



The small potted plant on the hallway table



The grey dining table chair.



The green lawn chair.



The black office chair.



The small wooden chair at the desk.



The black leather chairs in the kitchen.



The bar-height chair at the kitchen island.



a wooden seat.



wood desk.



the kitchen counter



The light blue bowl on the living room coffee table.



The lavender bowl on the kitchen counter.



The light blue bowl on the couch.



The green cover book on the coffee table.



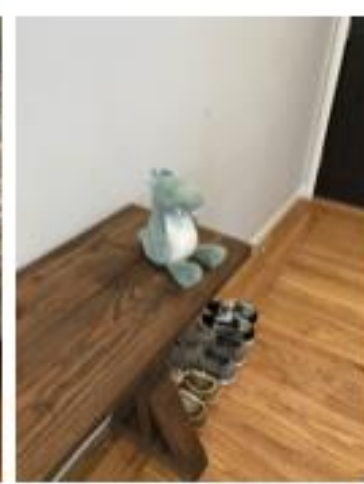
The book on the desk. It has a car on the cover.



The beige teddy bear.



The stuffed lion toy.



The green dinosaur stuffed toy.



The stuffed lion.



The beige teddy bear.



The stuffed lion toy.



The green dinosaur stuffed toy.



The refrigerator.



The refrigerator.



The refrigerator.



The refrigerator.



The toilet next to the shower curtain with blue fish.



The toilet.



The toilet next to the glass shower door.



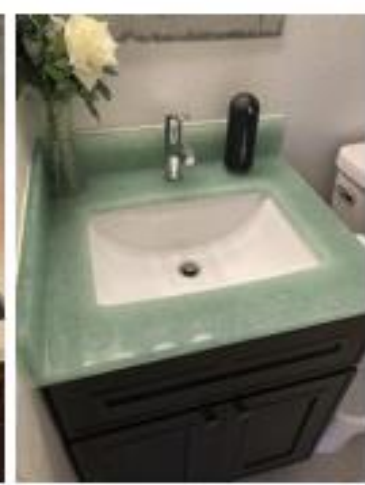
The kitchen sink.



The bathroom sink with marble top.



The kitchen sink.



The bathroom sink with green counter.



The bathroom sink.



The kitchen sink.



The bathroom sink with black and white counter.



The television.



The television mounted on a yellow wall.



The television mounted on a white wall.



The oven.



The oven.



The oven.



The oven.

“In the Wild” Empirical Evaluation 200+ *Object Instances*

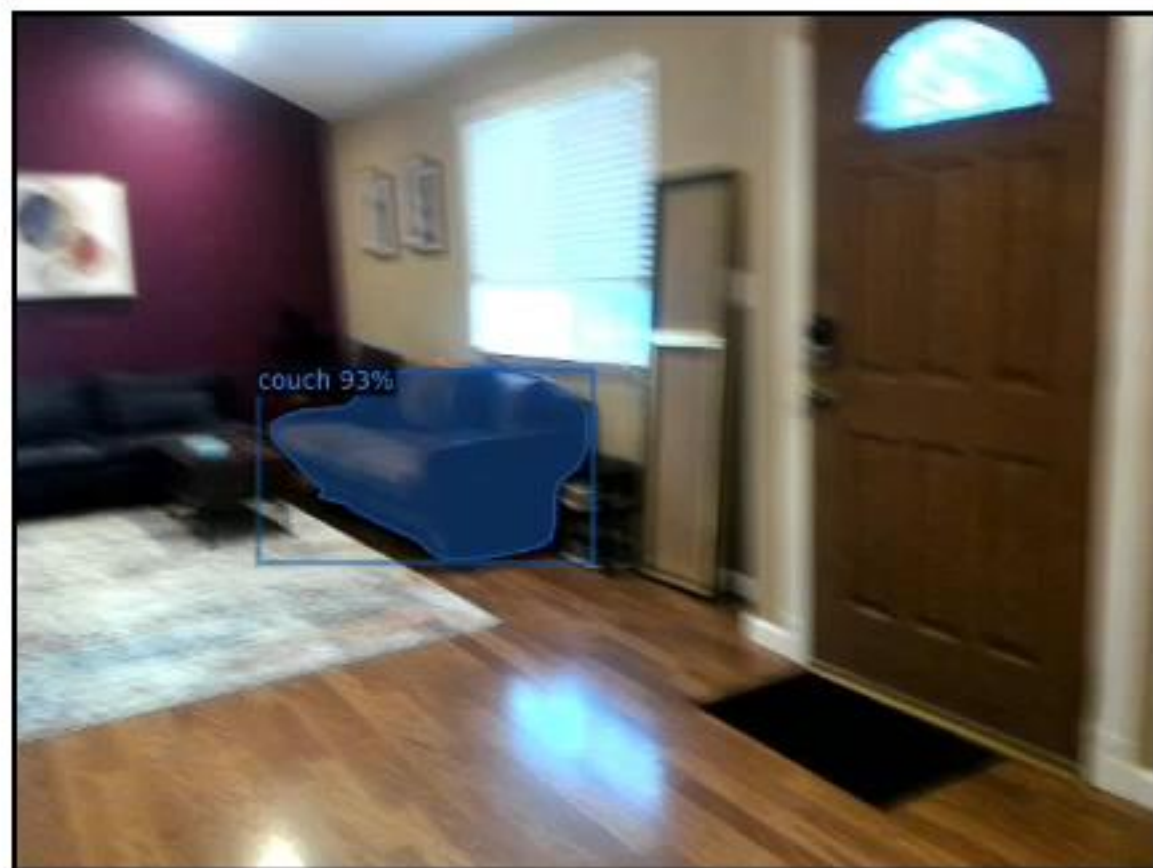
Observation



Third-person view



Observation



Third-person view



Semantic Map



couch - 1	chair - 1	chair - 2	couch - 2	chair - 3	chair - 4	chair - 5
refrigerator - 1	chair - 6	dining table - 1	potted plant - 1	potted plant - 2	sink - 1	oven - 1
bottle - 1	bottle - 2	chair - 7	tv - 1	chair - 8	teddy bear - 1	chair - 9
cup - 1						

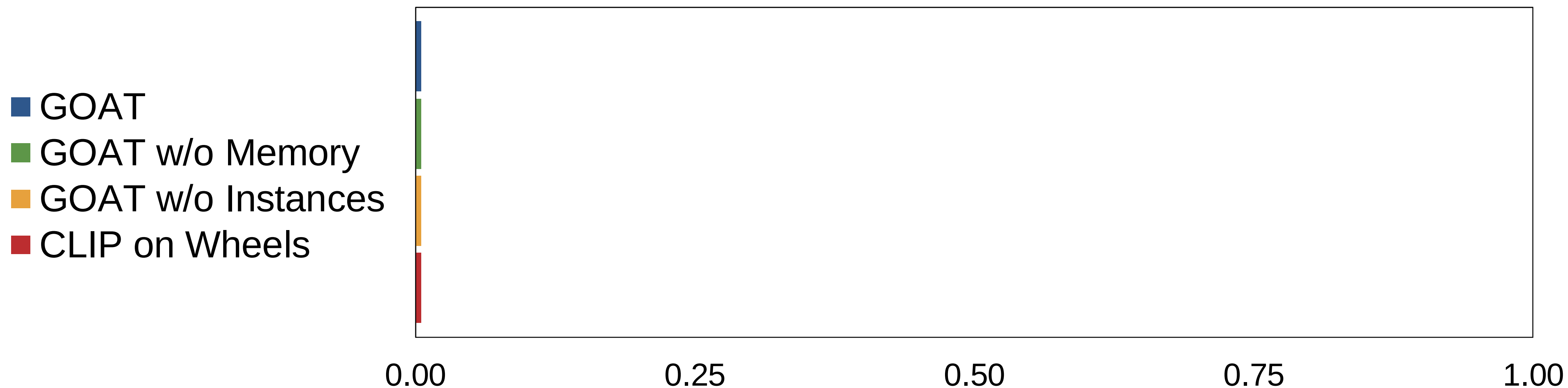
Third-person view



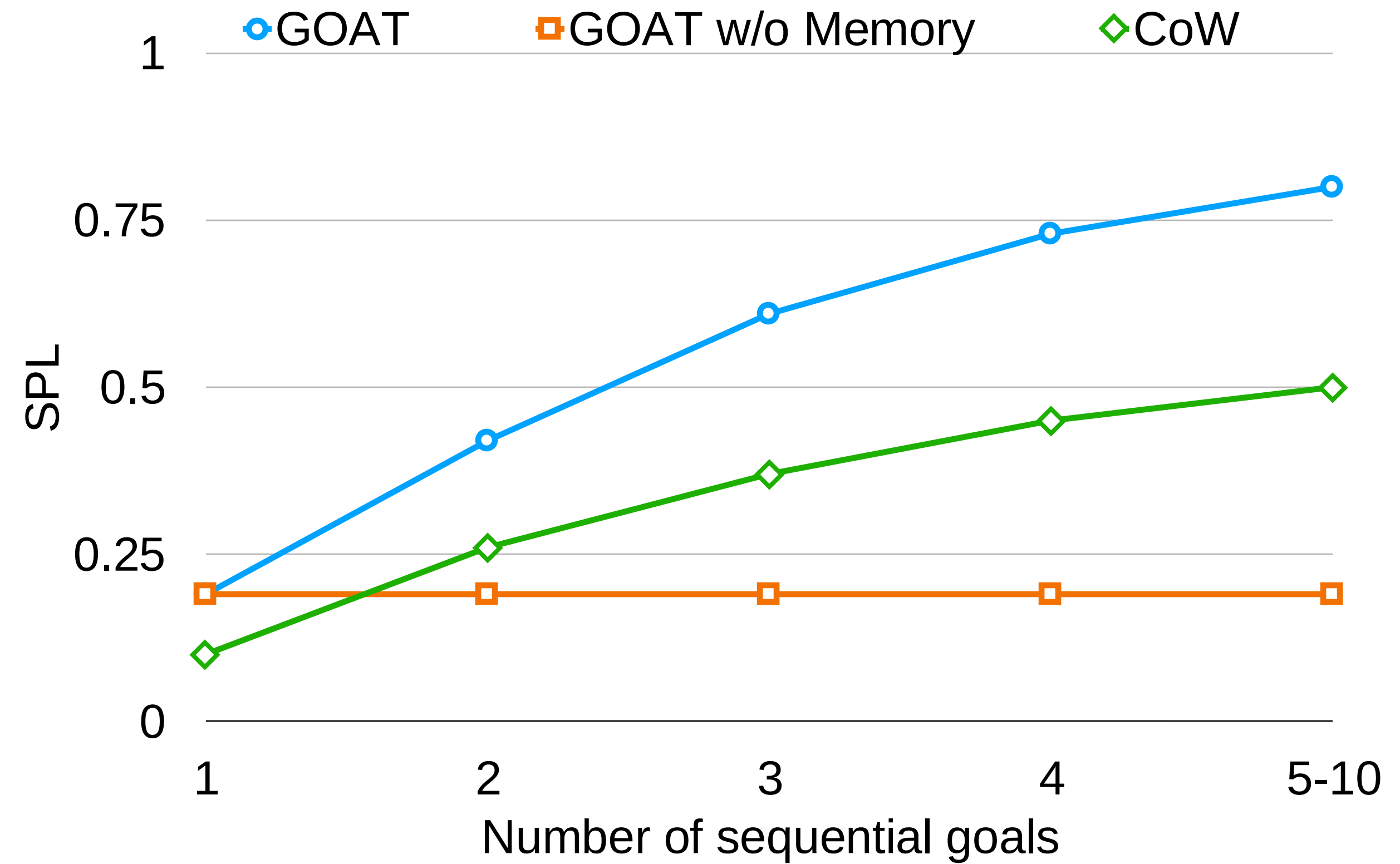
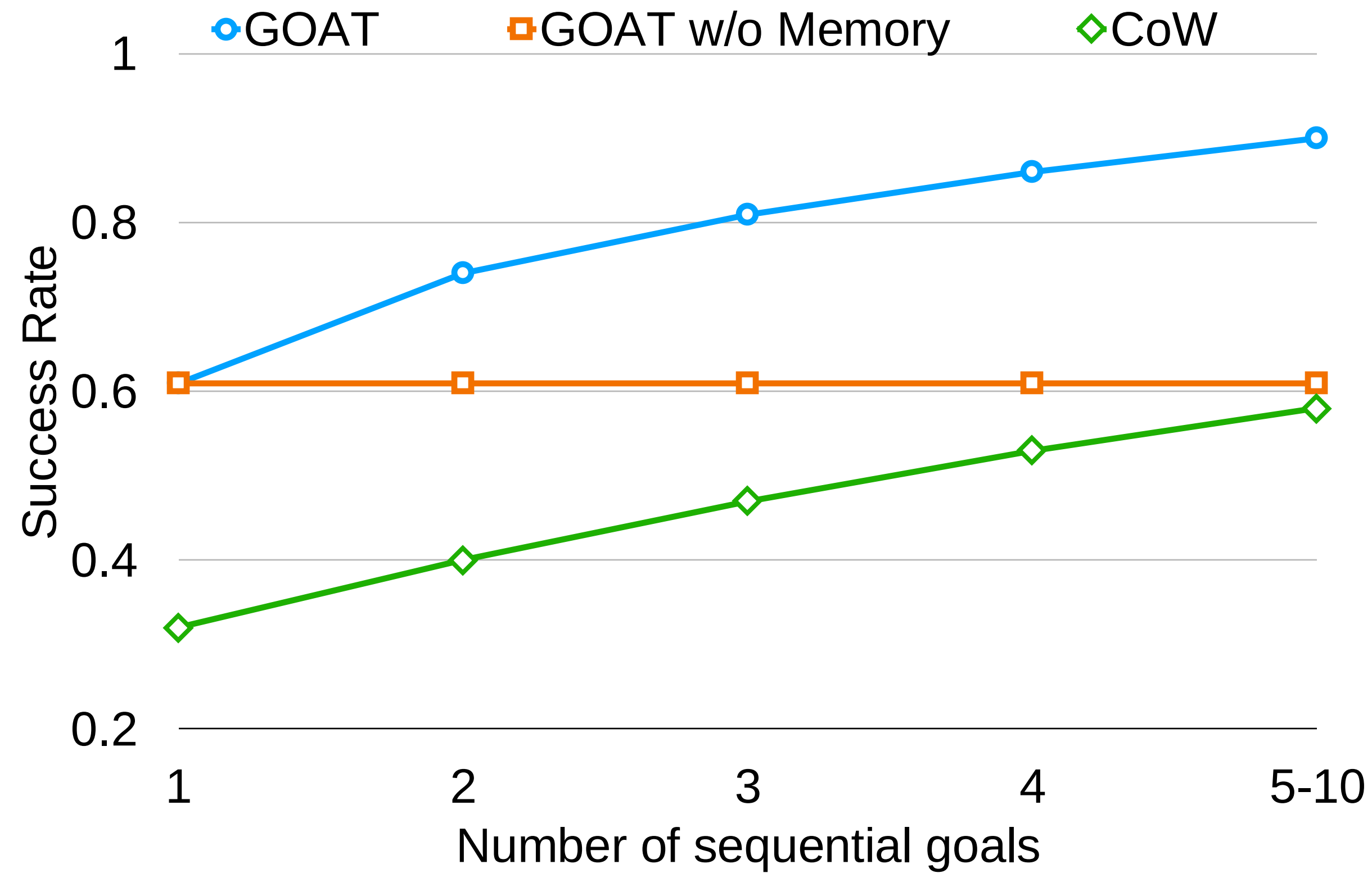
Results

Success Rate

SPL



Performance Across Episode



Goal:

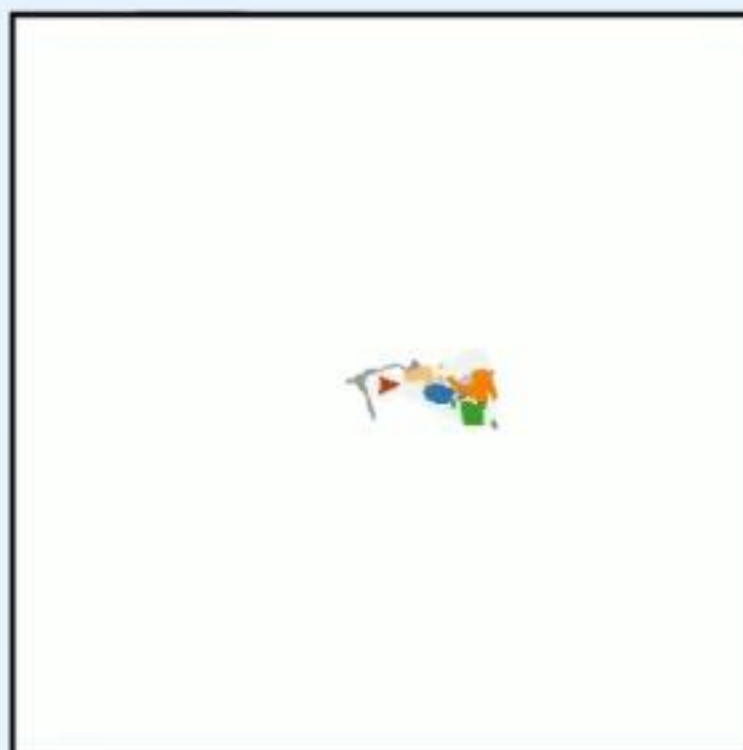


Baselines

Ours



Observation



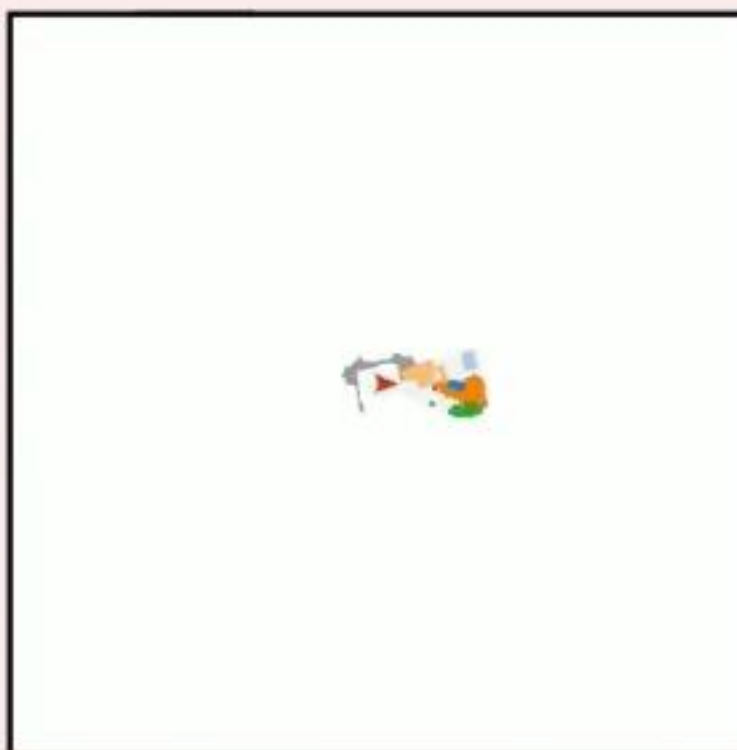
Instance Map

Success: 6/6 SPL: 0.78

No Memory



Observation



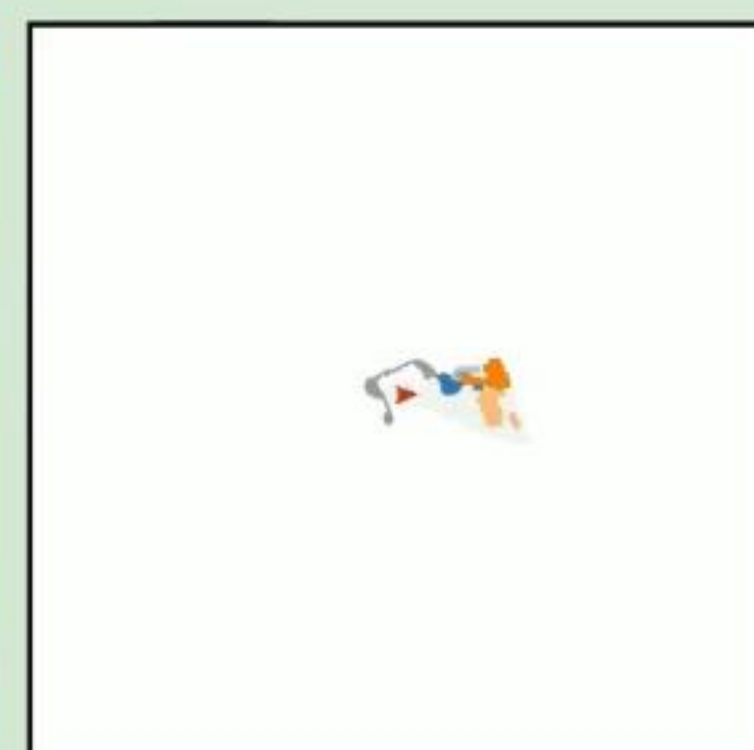
Instance Map

Success: 4/6 SPL: 0.40

CoW



Observation



Instance Map

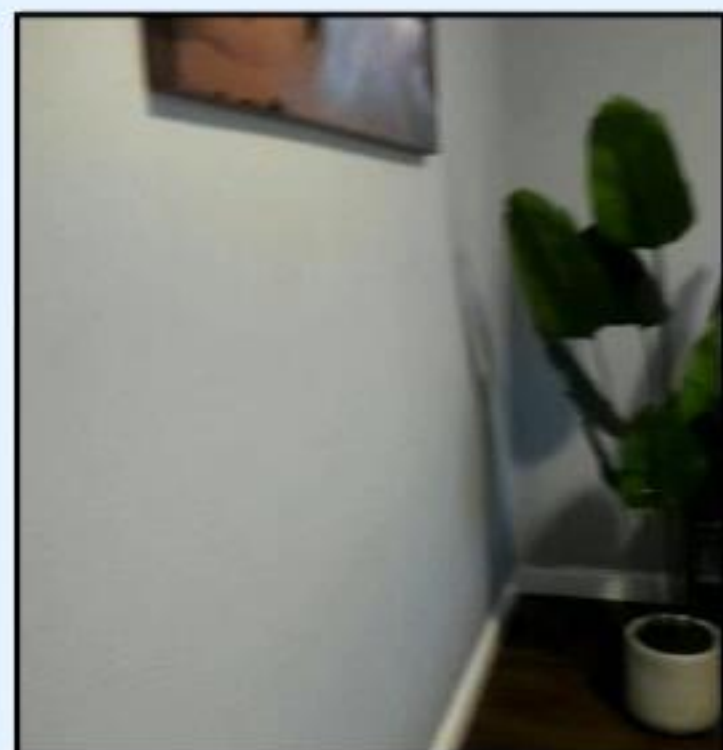
Success: 1/6 SPL: 0.16

Goal:



Baselines

Ours



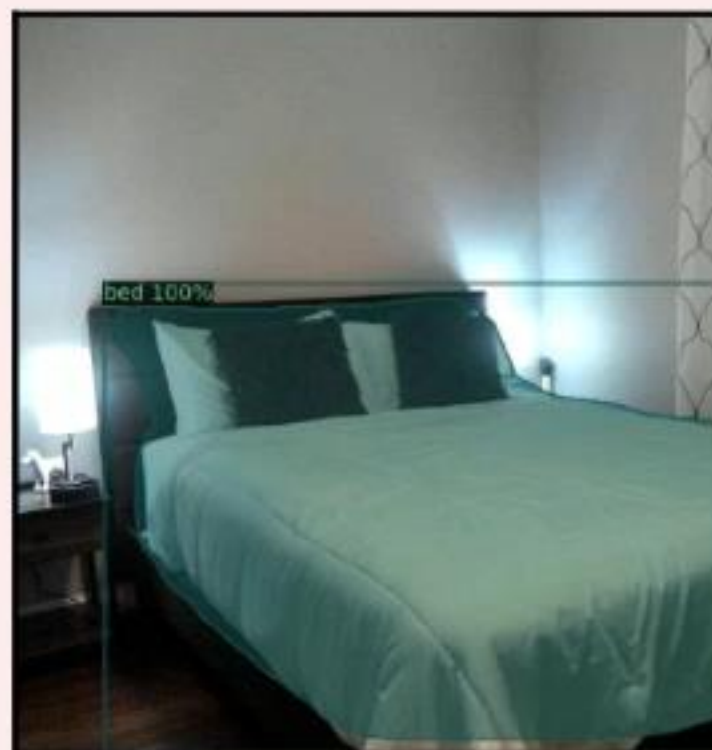
Observation



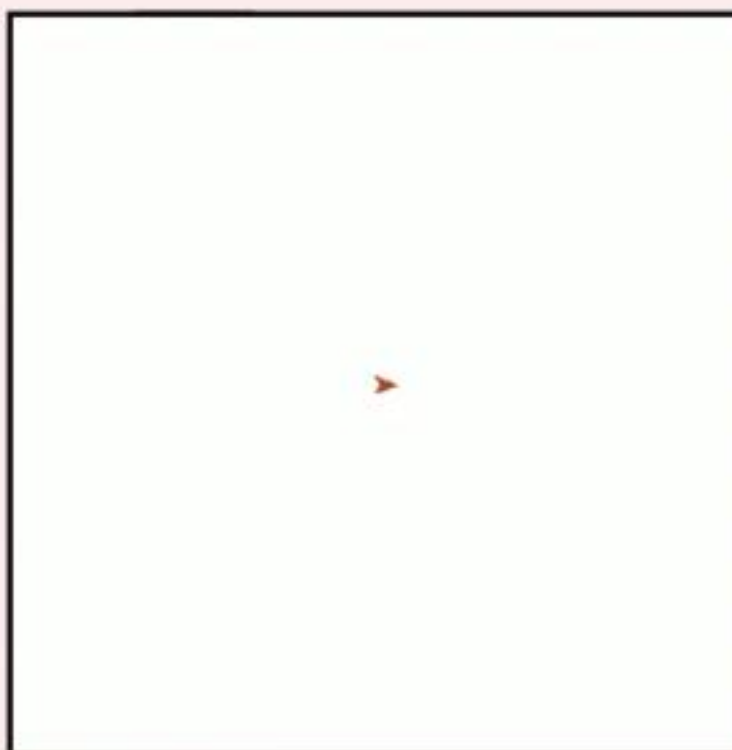
Instance Map

Success: 6/6 SPL: 0.78

No Memory



Observation



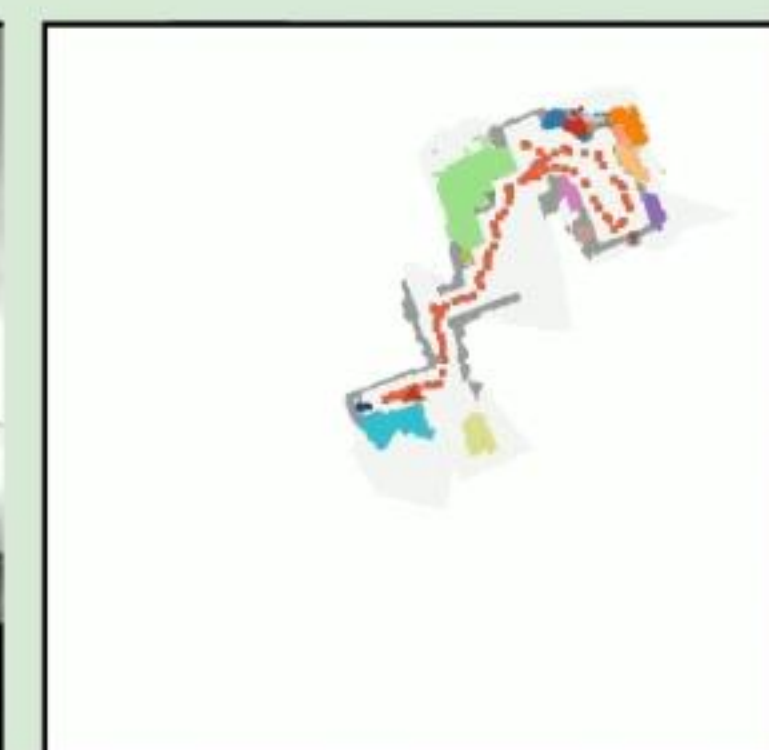
Instance Map

Success: 4/6 SPL: 0.40

CoW



Observation



Instance Map

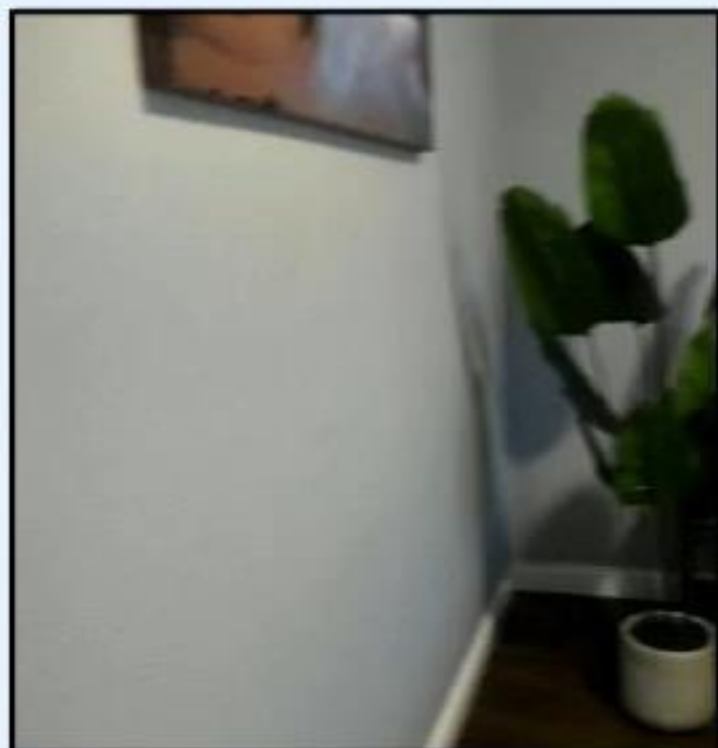
Success: 1/6 SPL: 0.16

Goal:

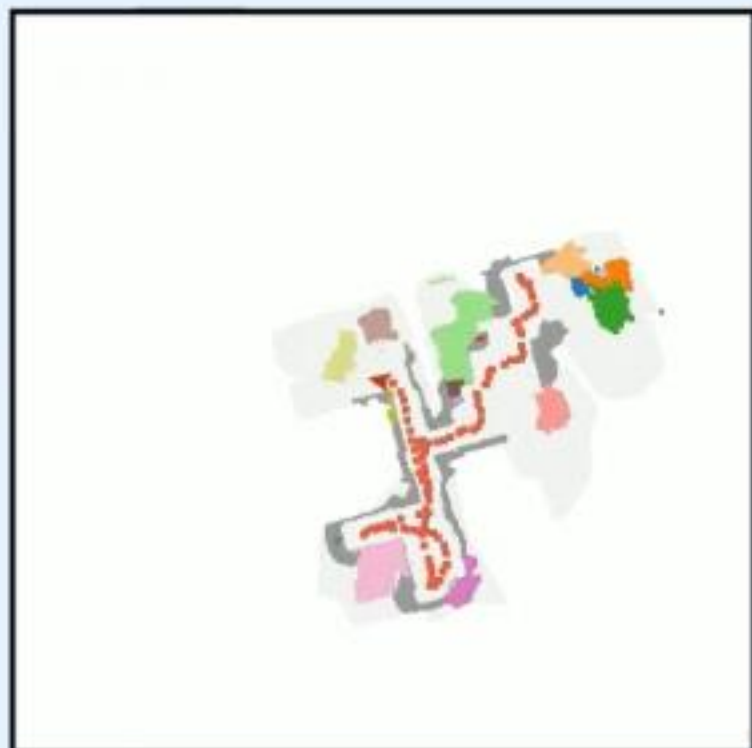


Baselines

Ours



Observation



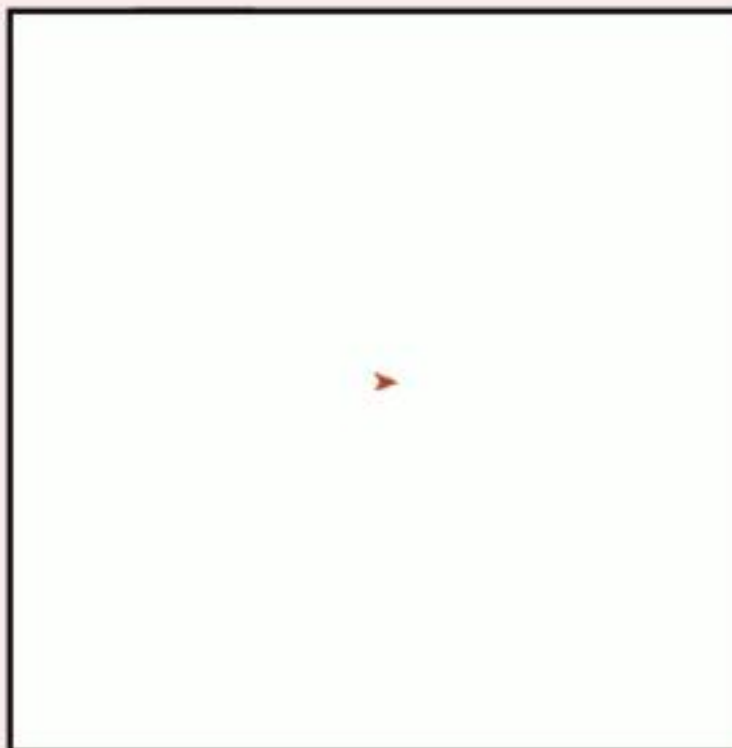
Instance Map

Success: 6/6 SPL: 0.78

No Memory



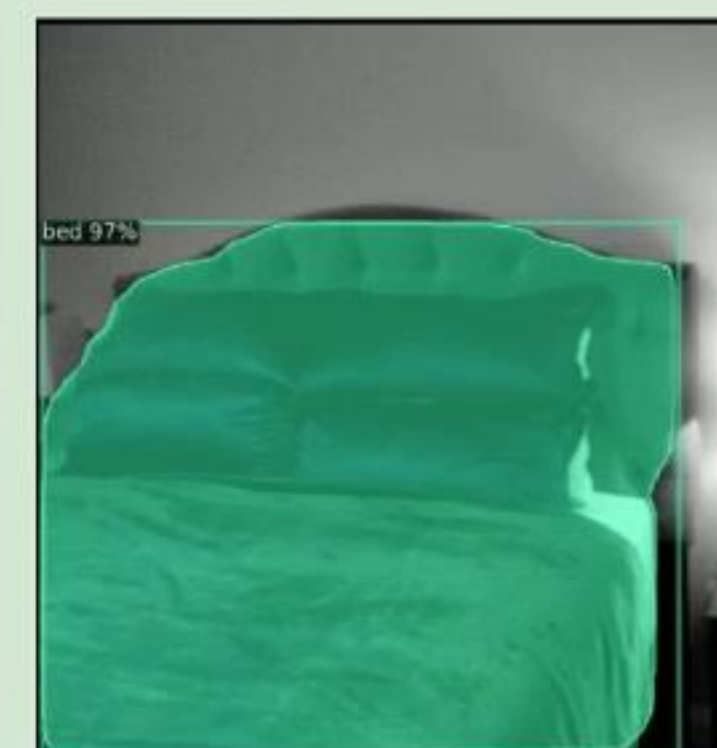
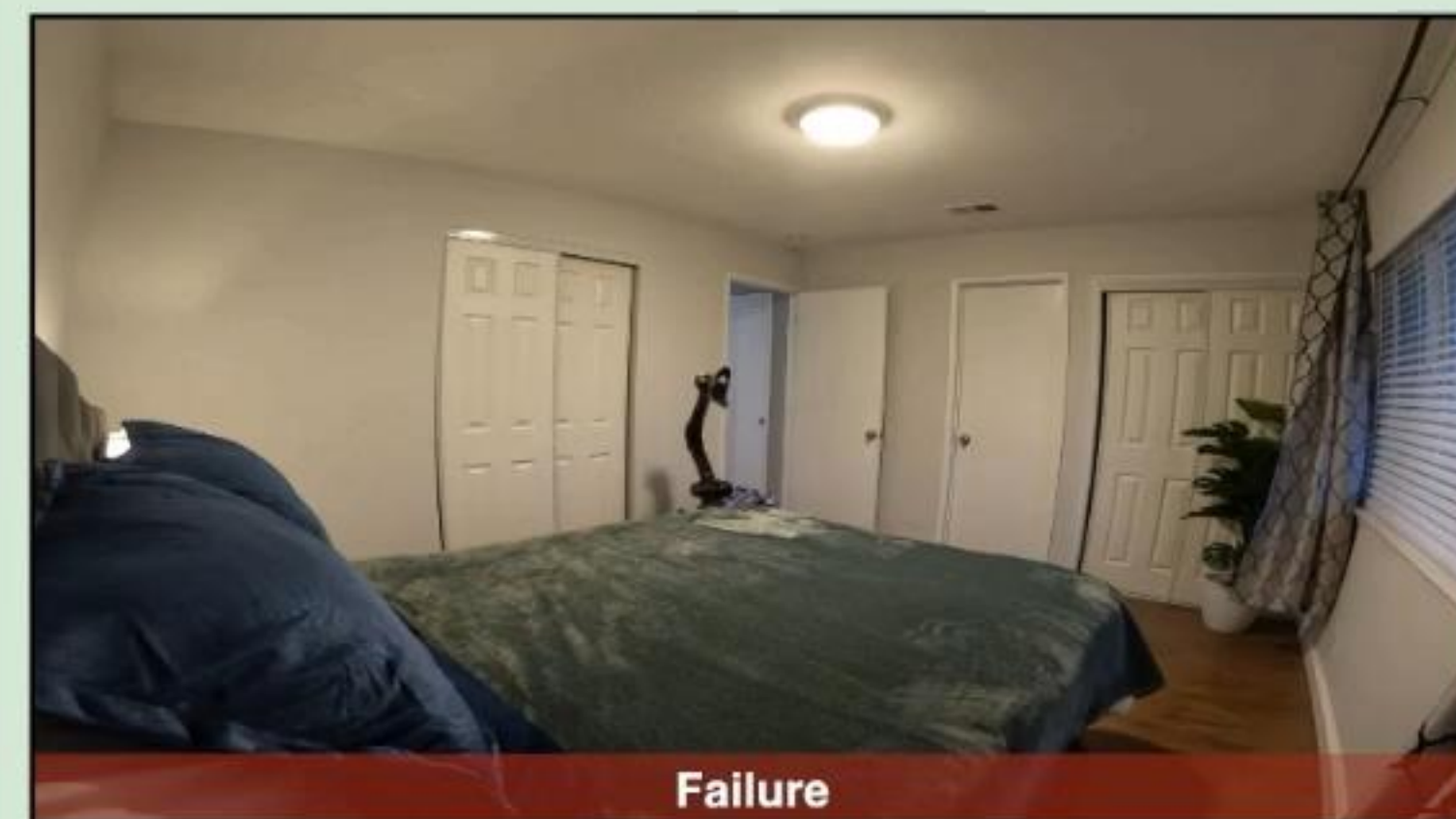
Observation



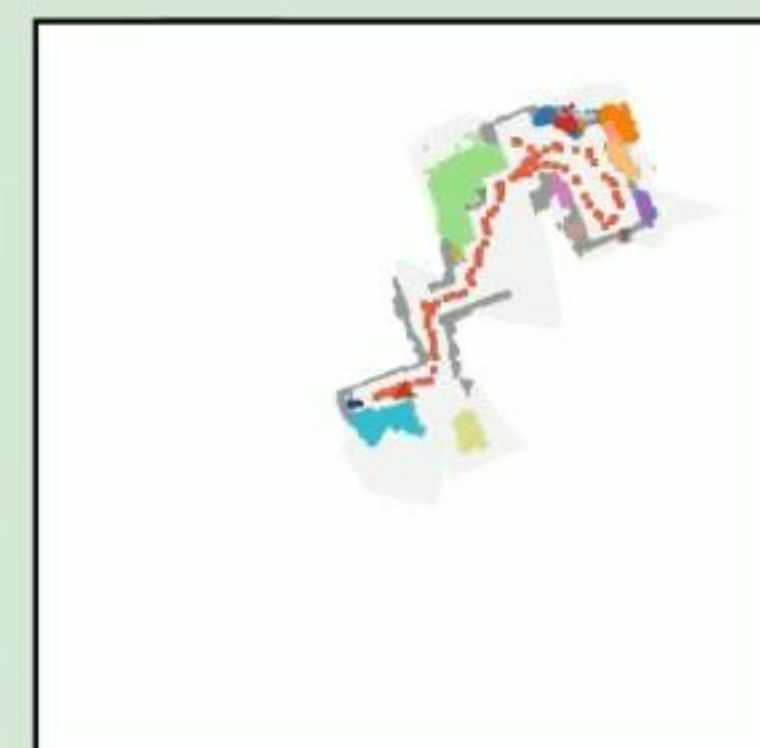
Instance Map

Success: 4/6 SPL: 0.40

CoW



Observation



Instance Map

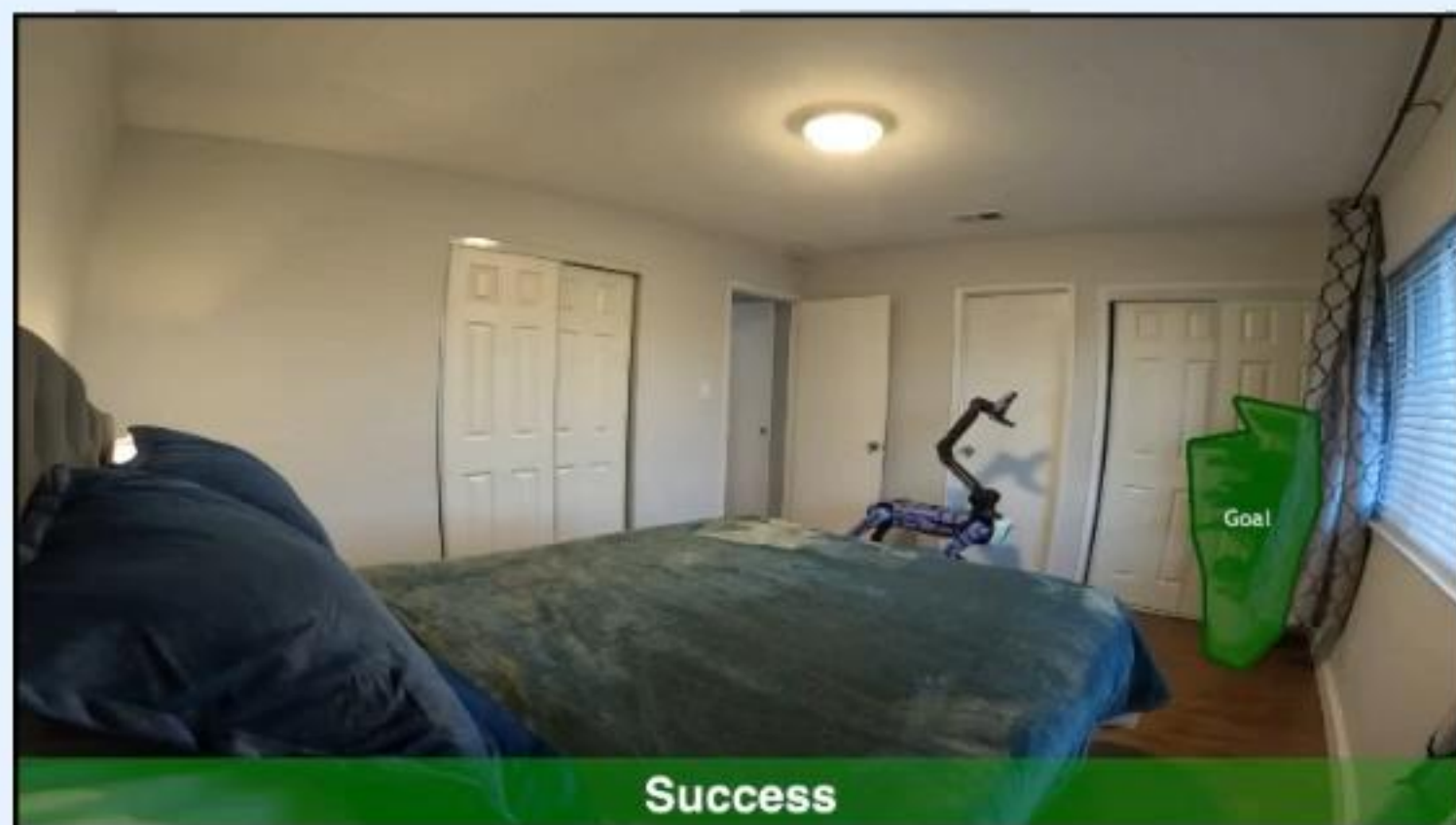
Success: 1/6 SPL: 0.16

Goal:

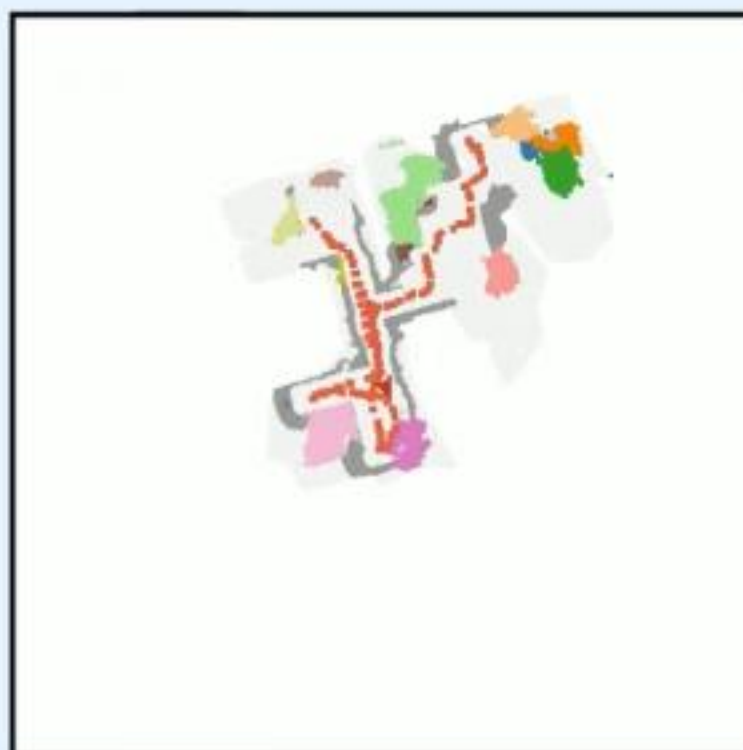


Baselines

Ours



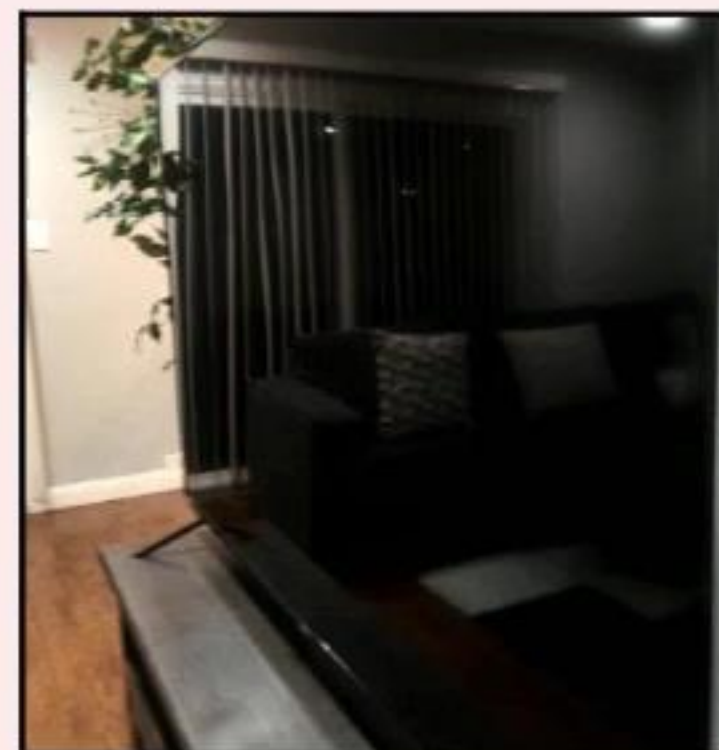
Observation



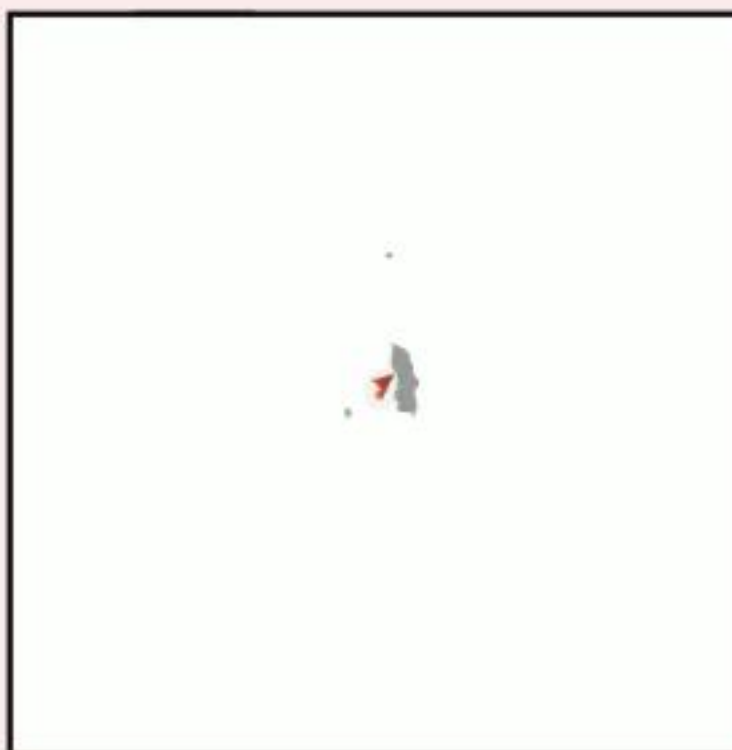
Instance Map

Success: 6/6 SPL: 0.78

No Memory



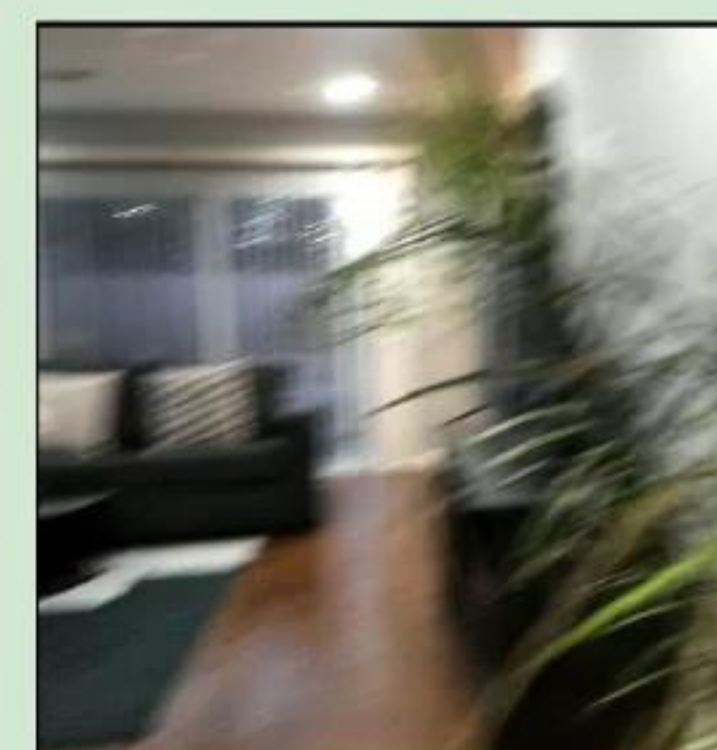
Observation



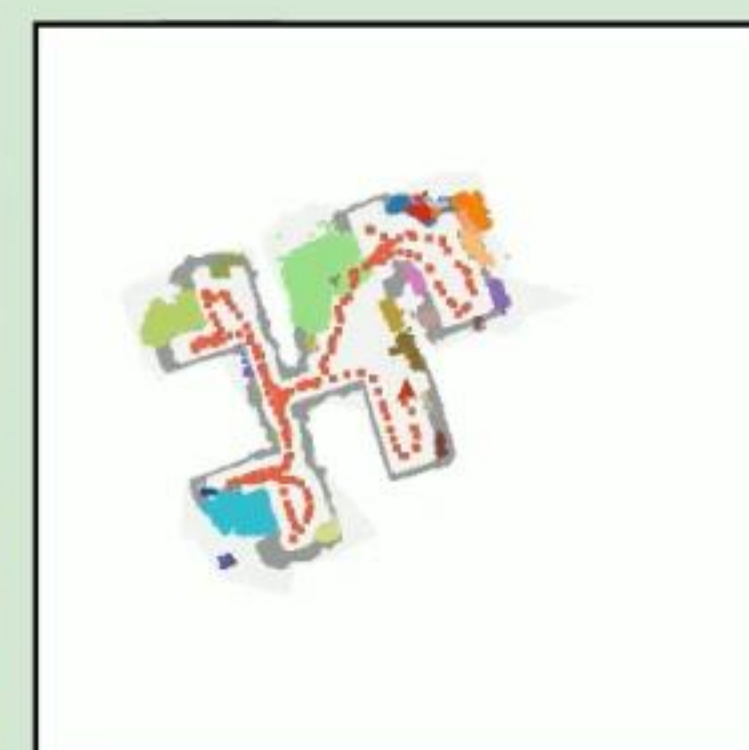
Instance Map

Success: 4/6 SPL: 0.40

CoW



Observation



Instance Map

Success: 1/6 SPL: 0.16

1. GOAT Problem

GOAT System Architecture

Results

Applications

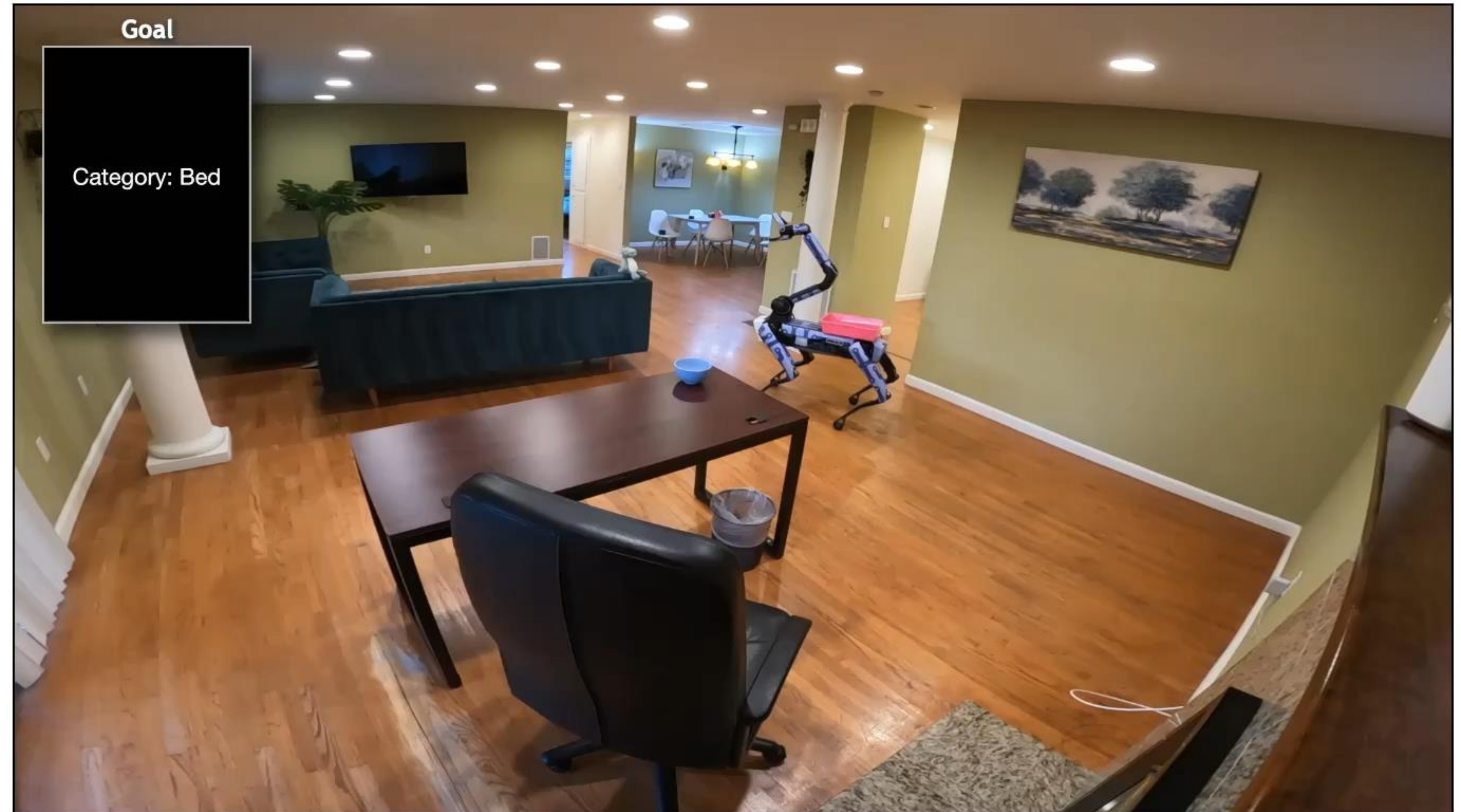
Pick & Place

Social Navigation

Platform Agnostic

Pick & Place

Third-person view



Social Navigation

Observation



Third-person view



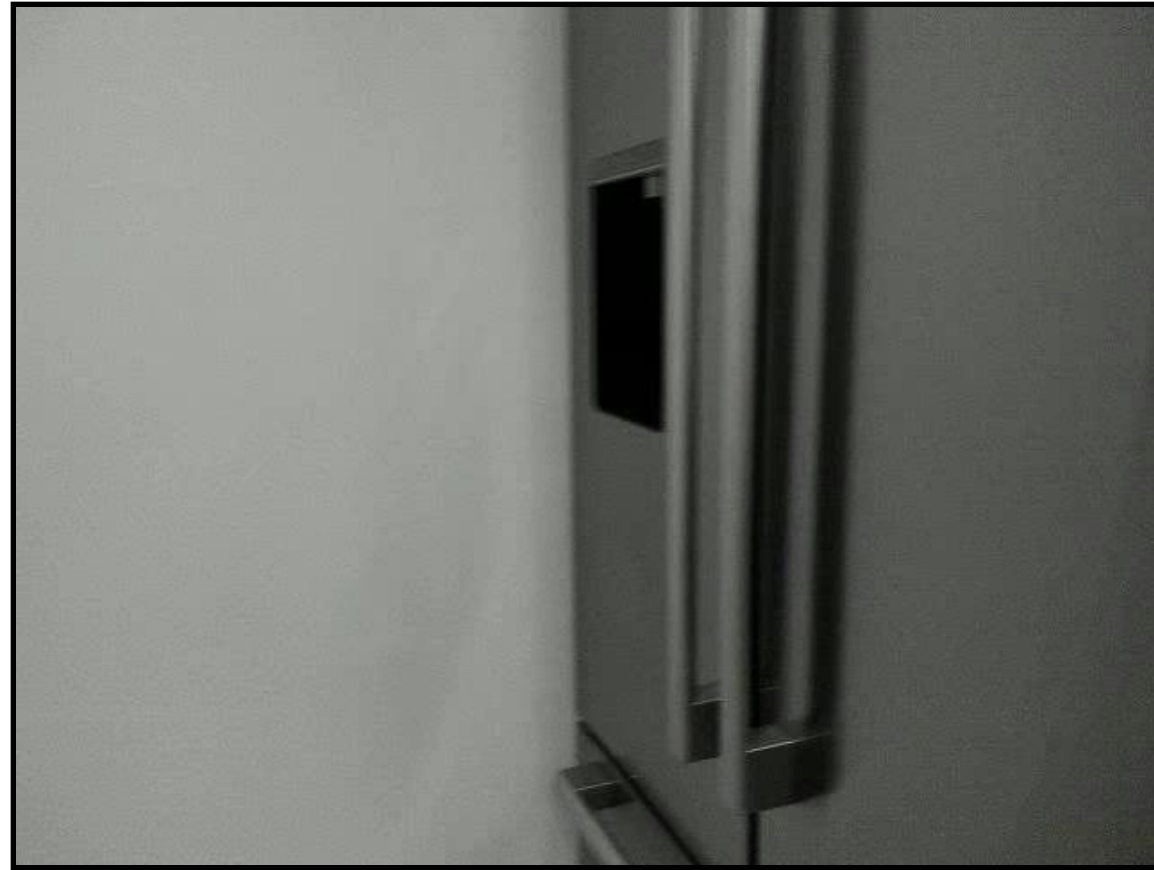
Semantic Map



Robot plans around the dynamic obstacle (person) to go to the refrigerator

Social Navigation

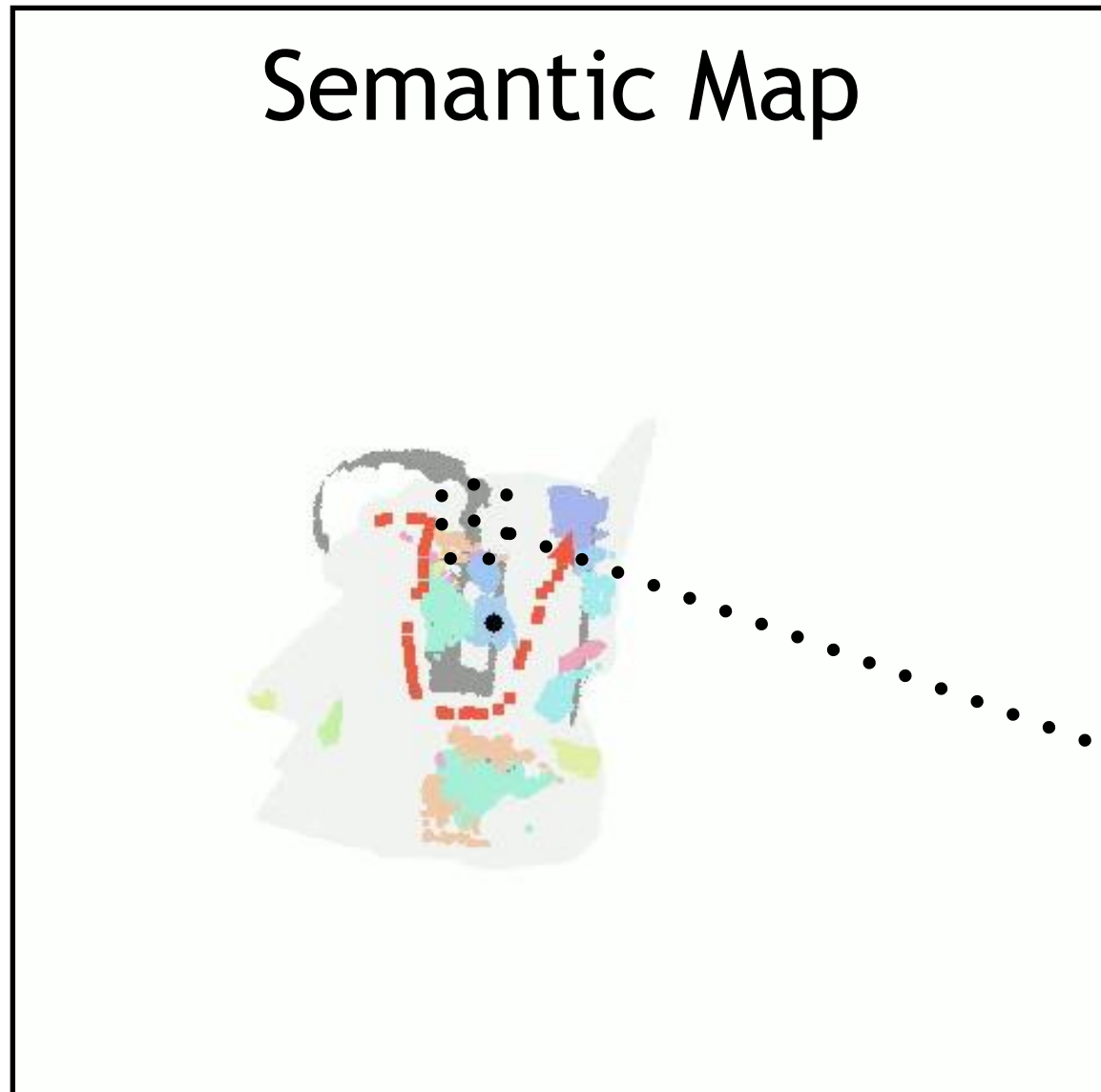
Observation



Third-person view



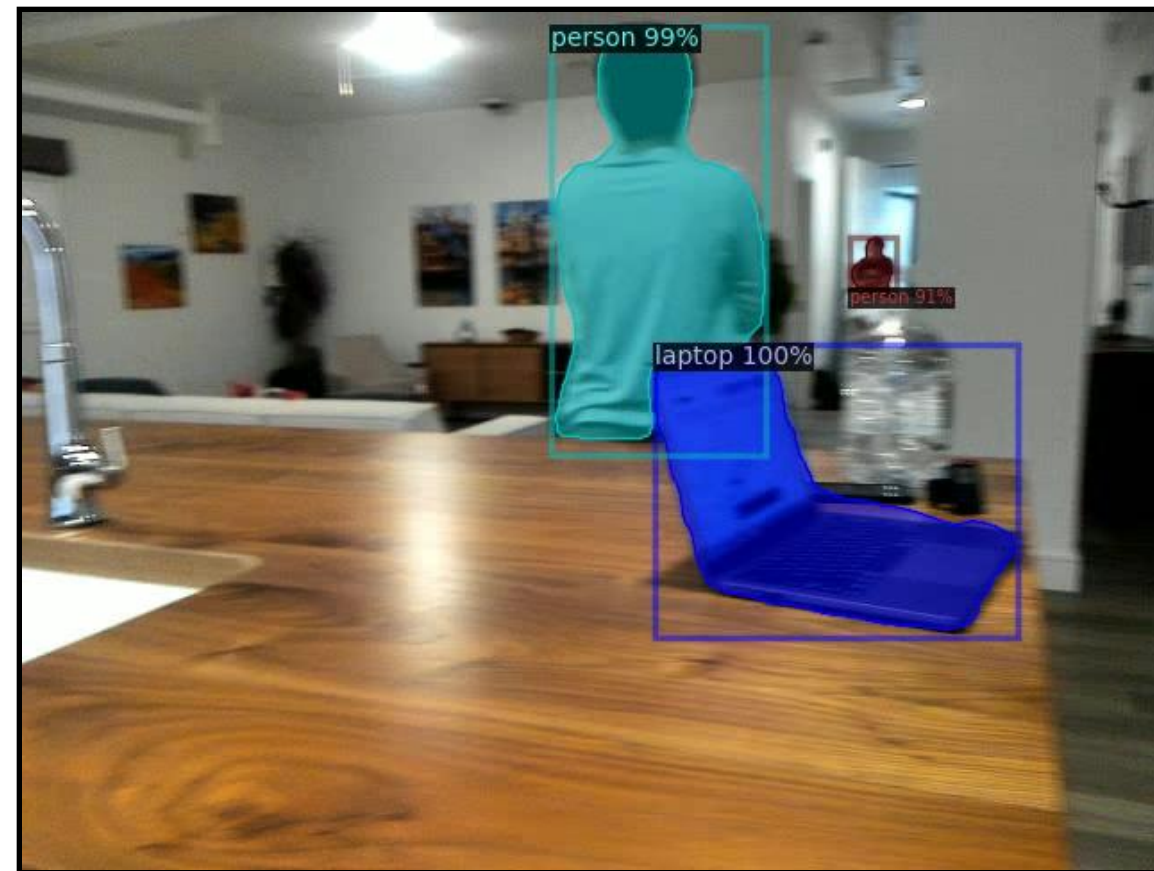
Semantic Map



Robot removes previous location of person from the map

Social Navigation

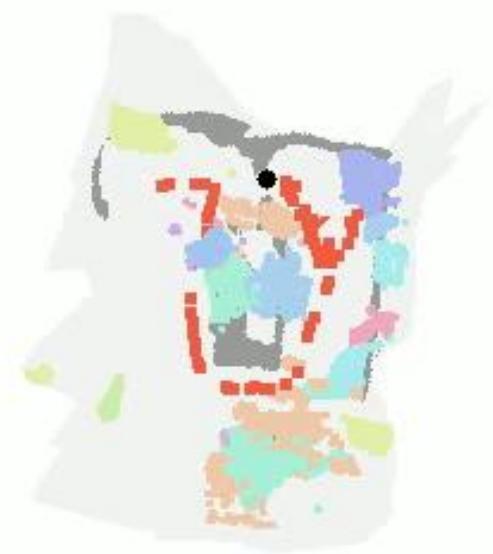
Observation



Third-person view



Semantic Map



Robot follows the person while updating their location

Platform Agnostic

Third-person view

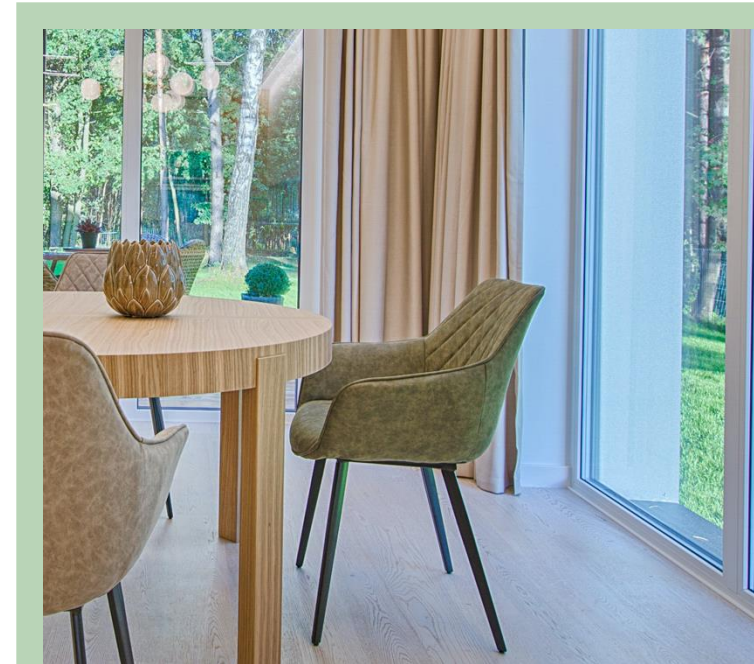


Summary

Universal navigation

- Multimodal

Image



Language

*Find **the fruit basket** on the kitchen counter*

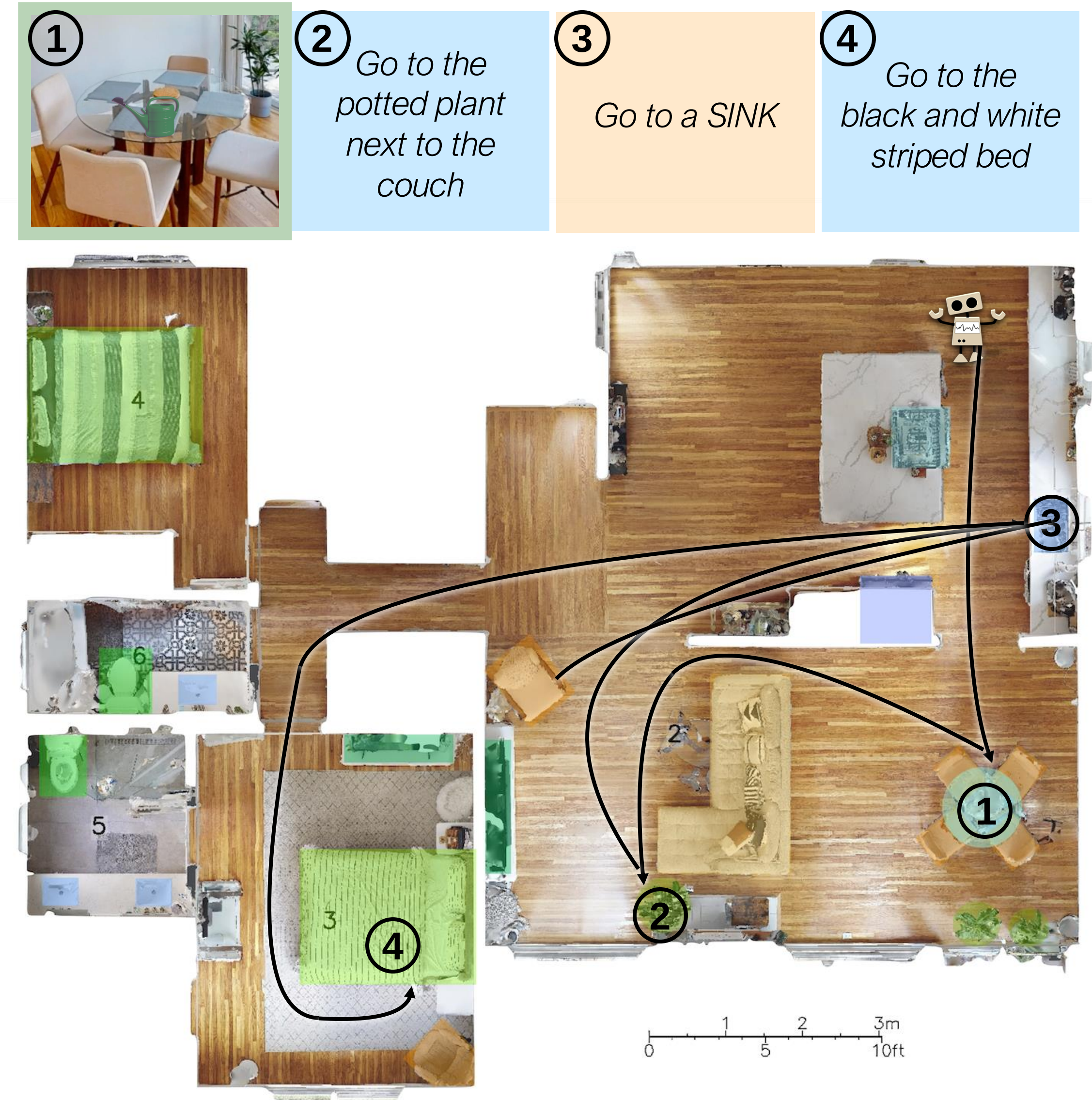
Category

*Bring me a **CUP***

Summary

Universal navigation

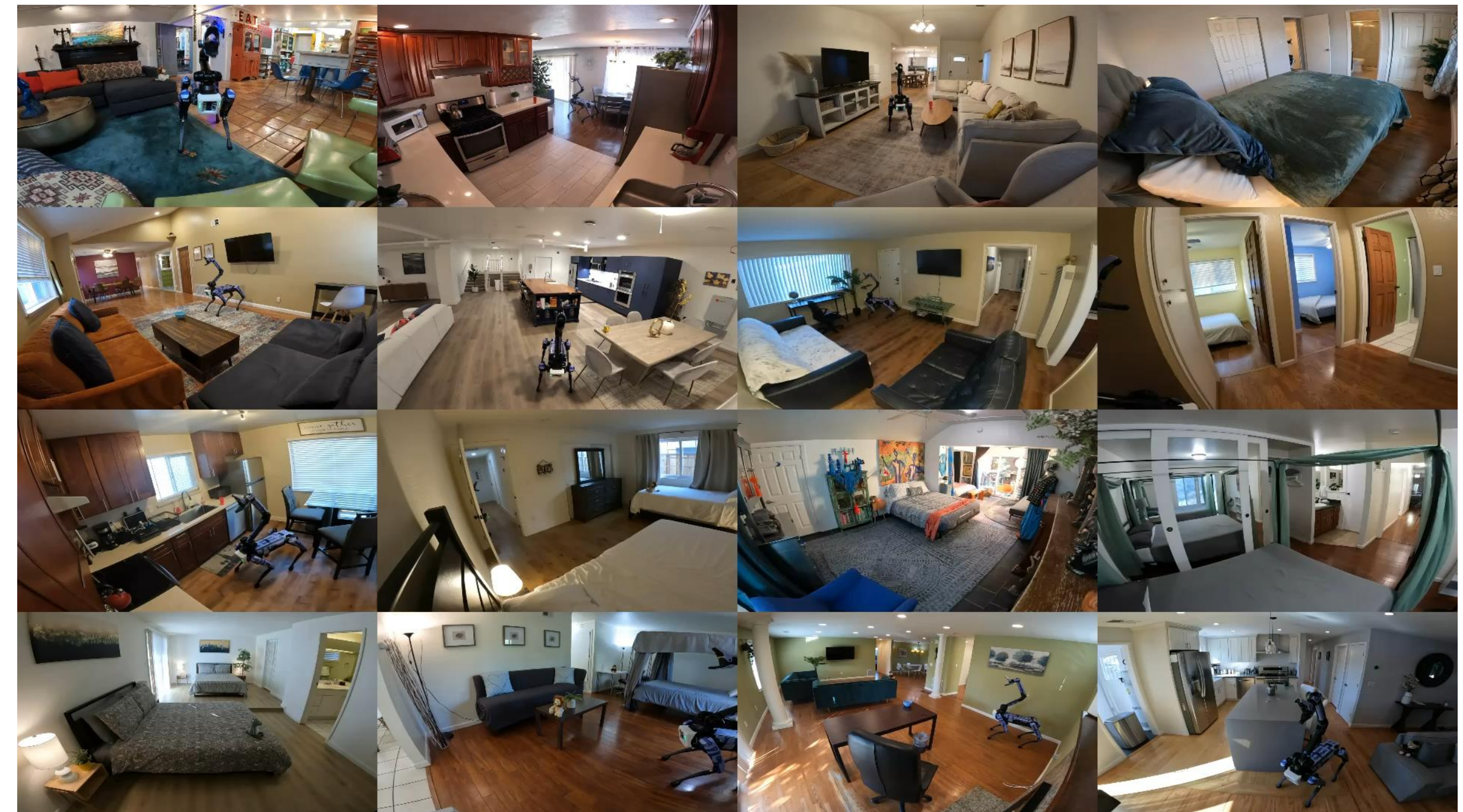
- Multimodal
- Lifelong



Summary

Universal navigation

- Multimodal
- Lifelong
- Unseen environments



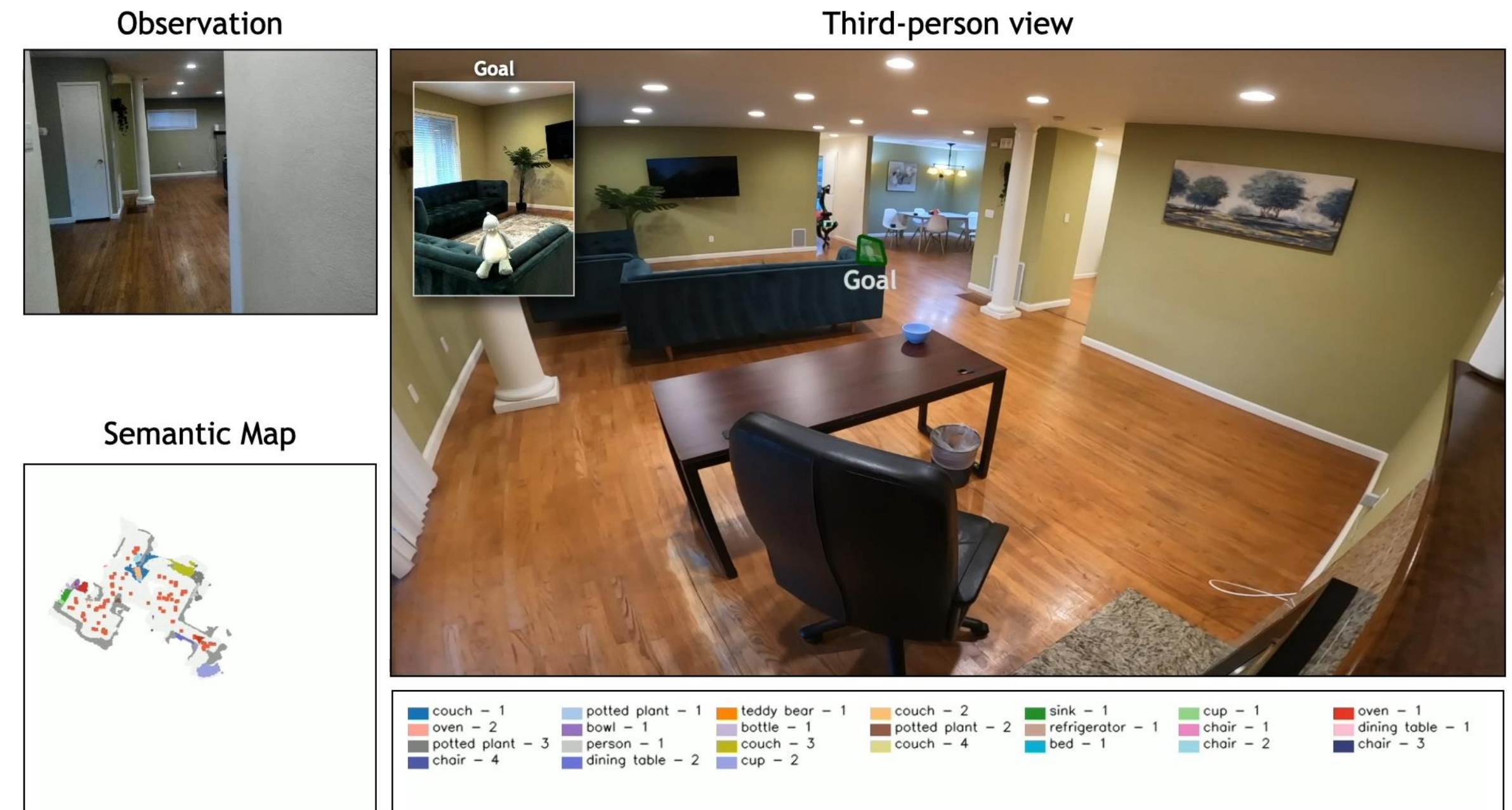
Summary

Universal navigation

- Multimodal
- Lifelong
- Unseen environments

Applications

- Pick & Place



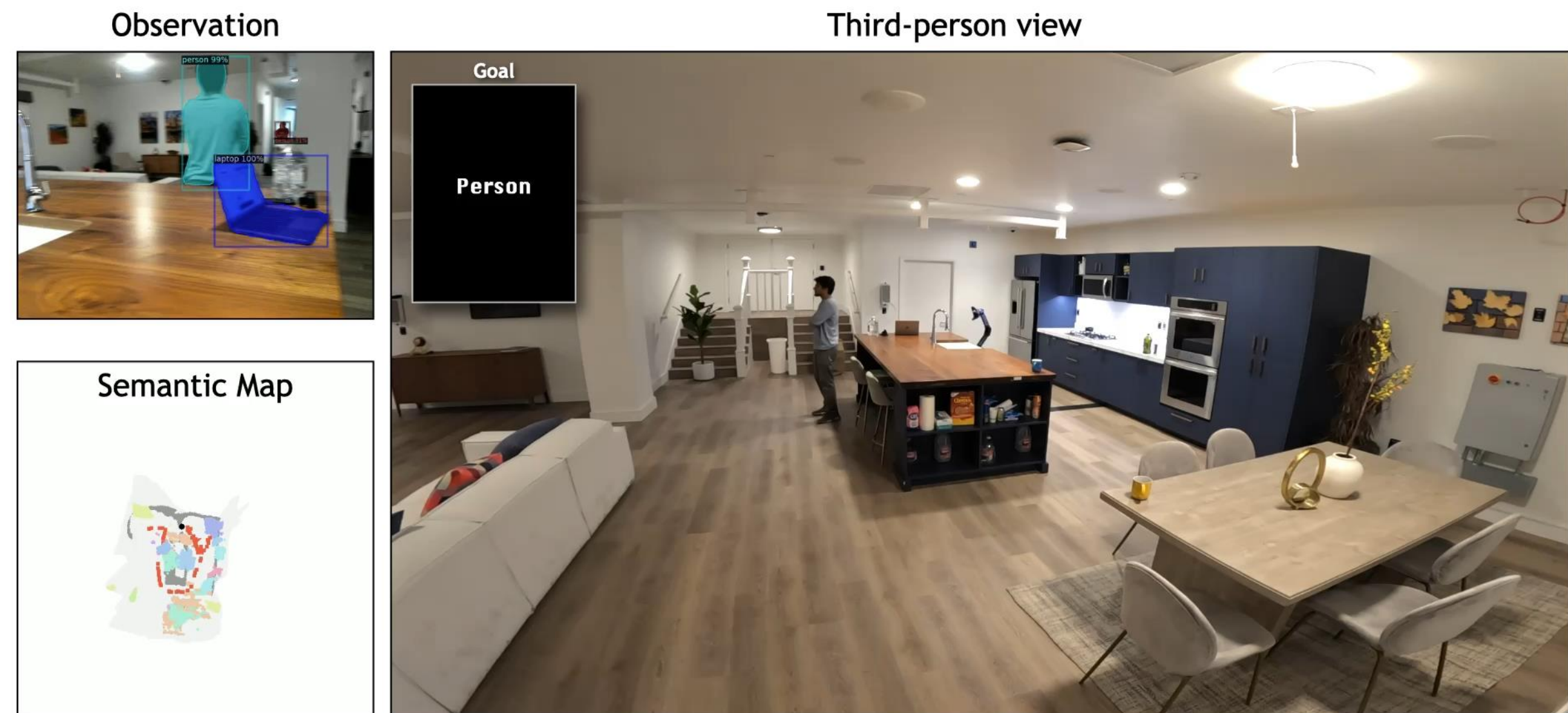
Summary

Universal navigation

- Multimodal
- Lifelong
- Unseen environments

Applications

- Pick & Place
- Social Navigation



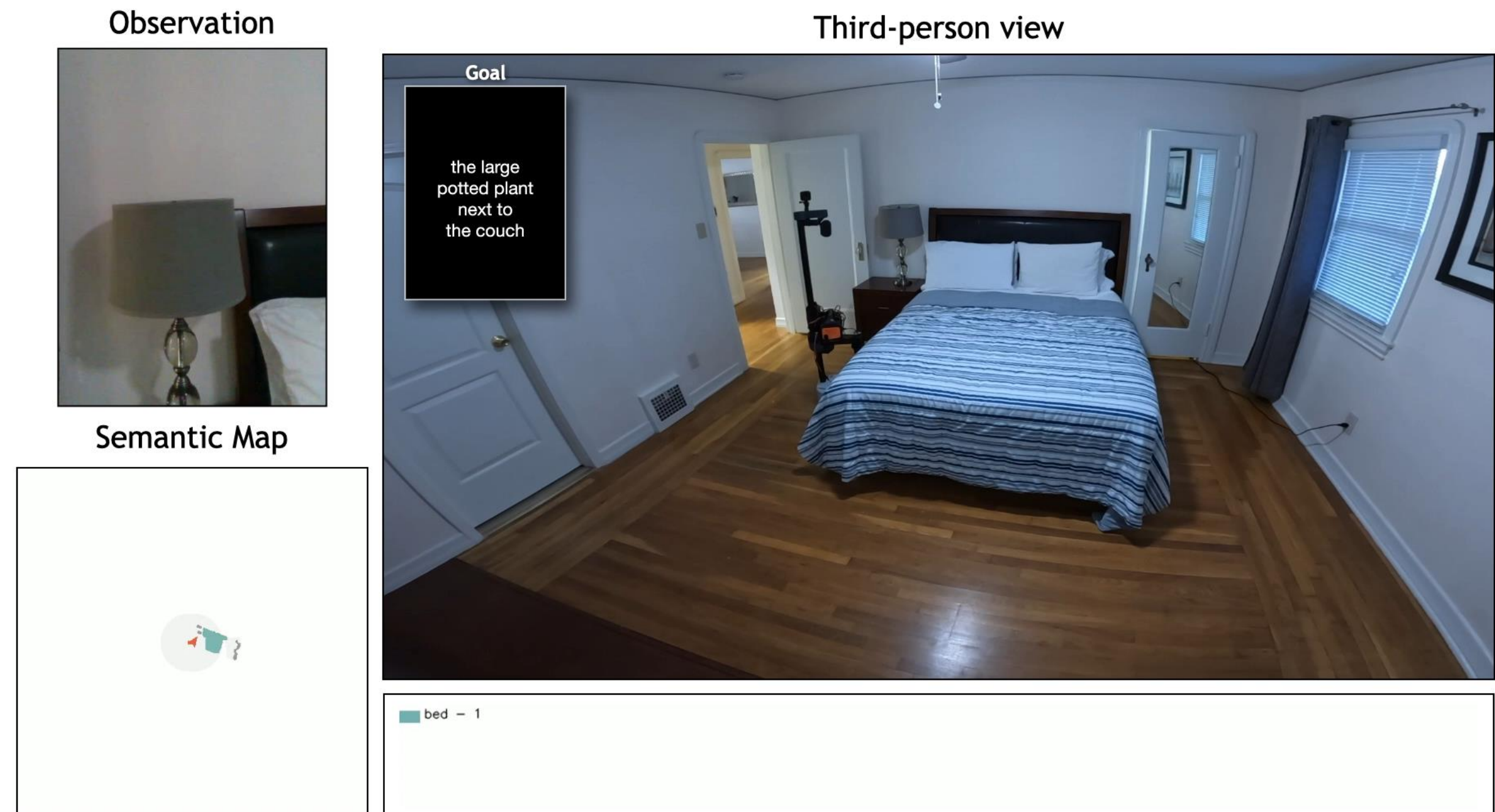
Summary

Universal navigation

- Multimodal
- Lifelong
- Unseen environments

Applications

- Pick & Place
- Social Navigation
- Platform Agnostic





Thank you!

Webpage: <https://theophilegervet.github.io/projects/goat>



Thank you!

